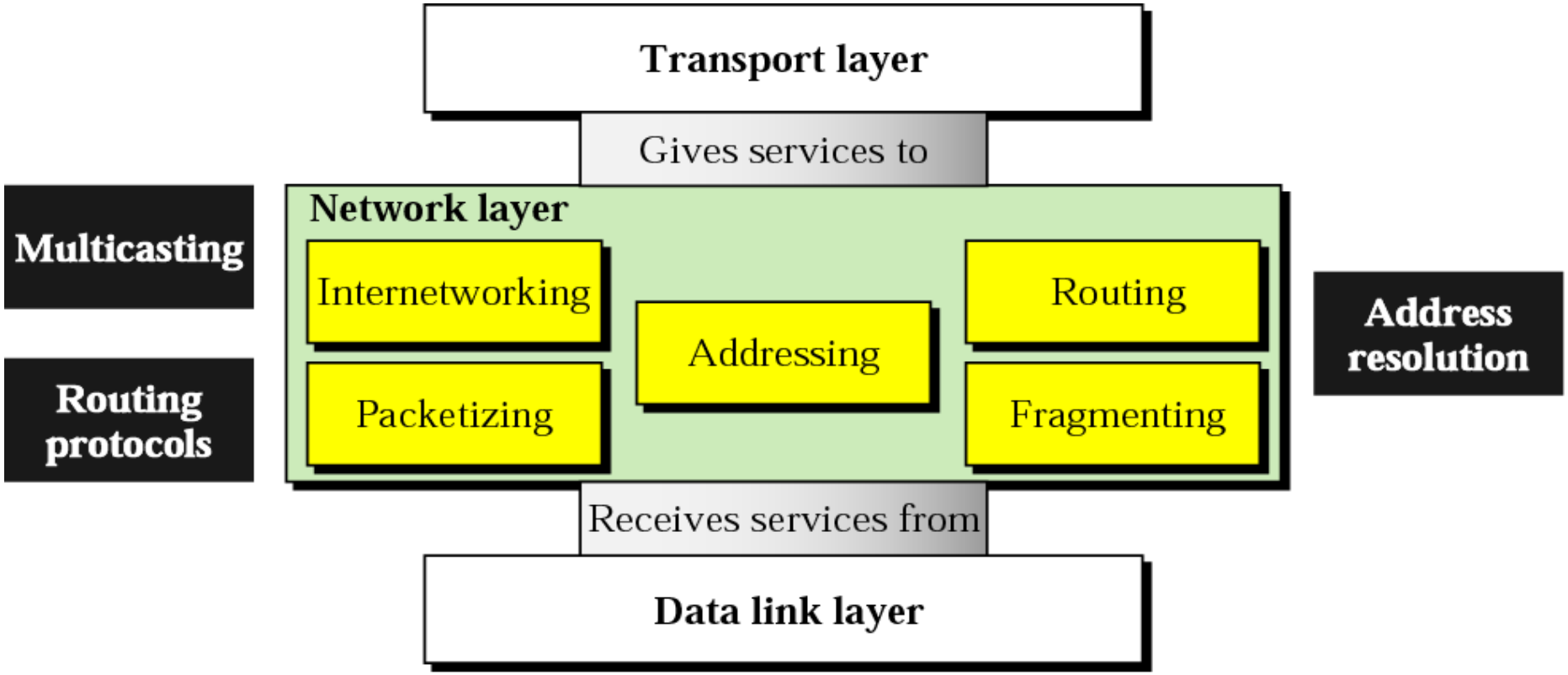
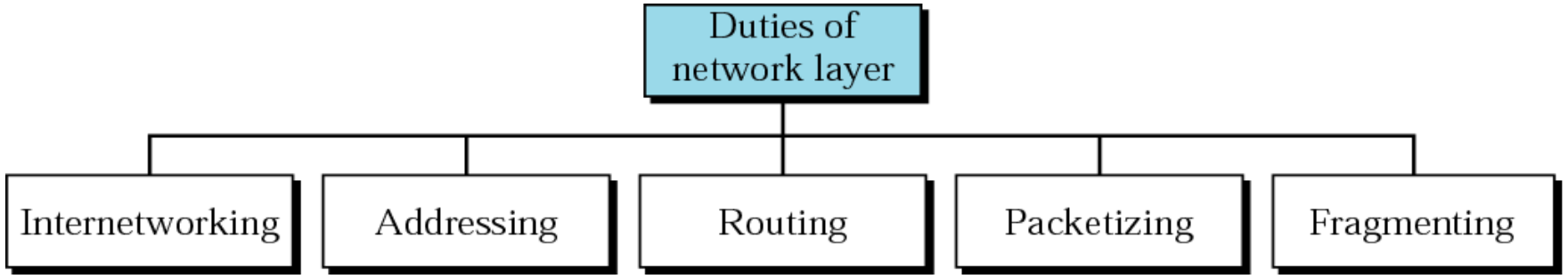


Network Layer

Position of network layer



Network layer duties



Host-to-Host Delivery:

Internetworking, Addressing, and Routing

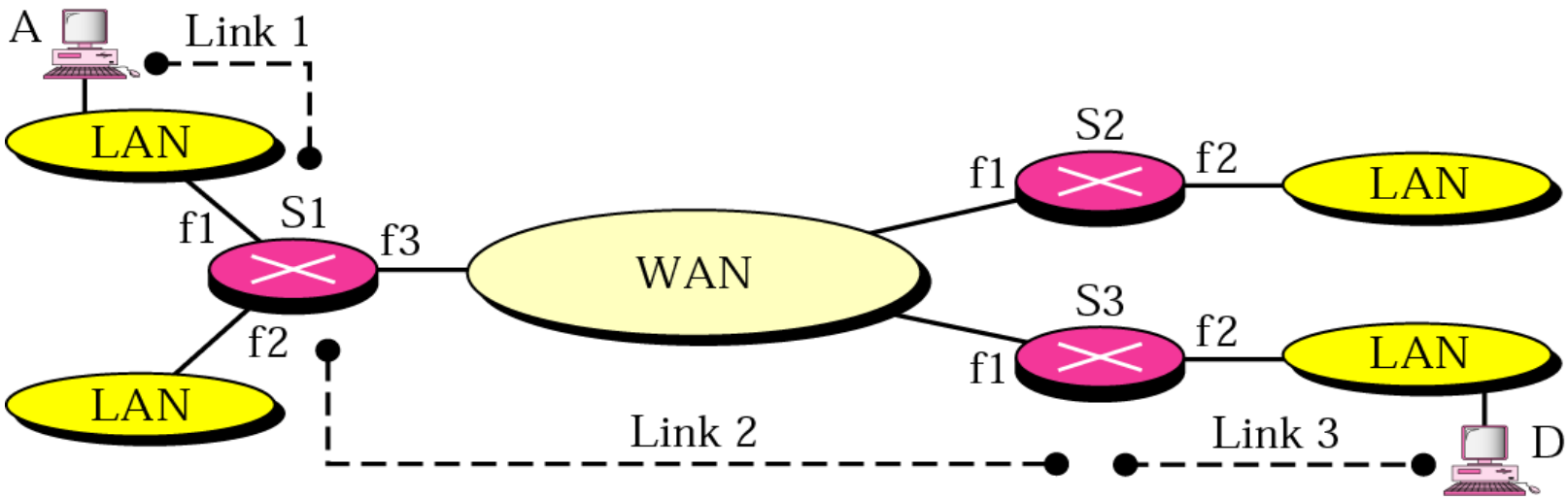
Internetworks

Need For Network Layer

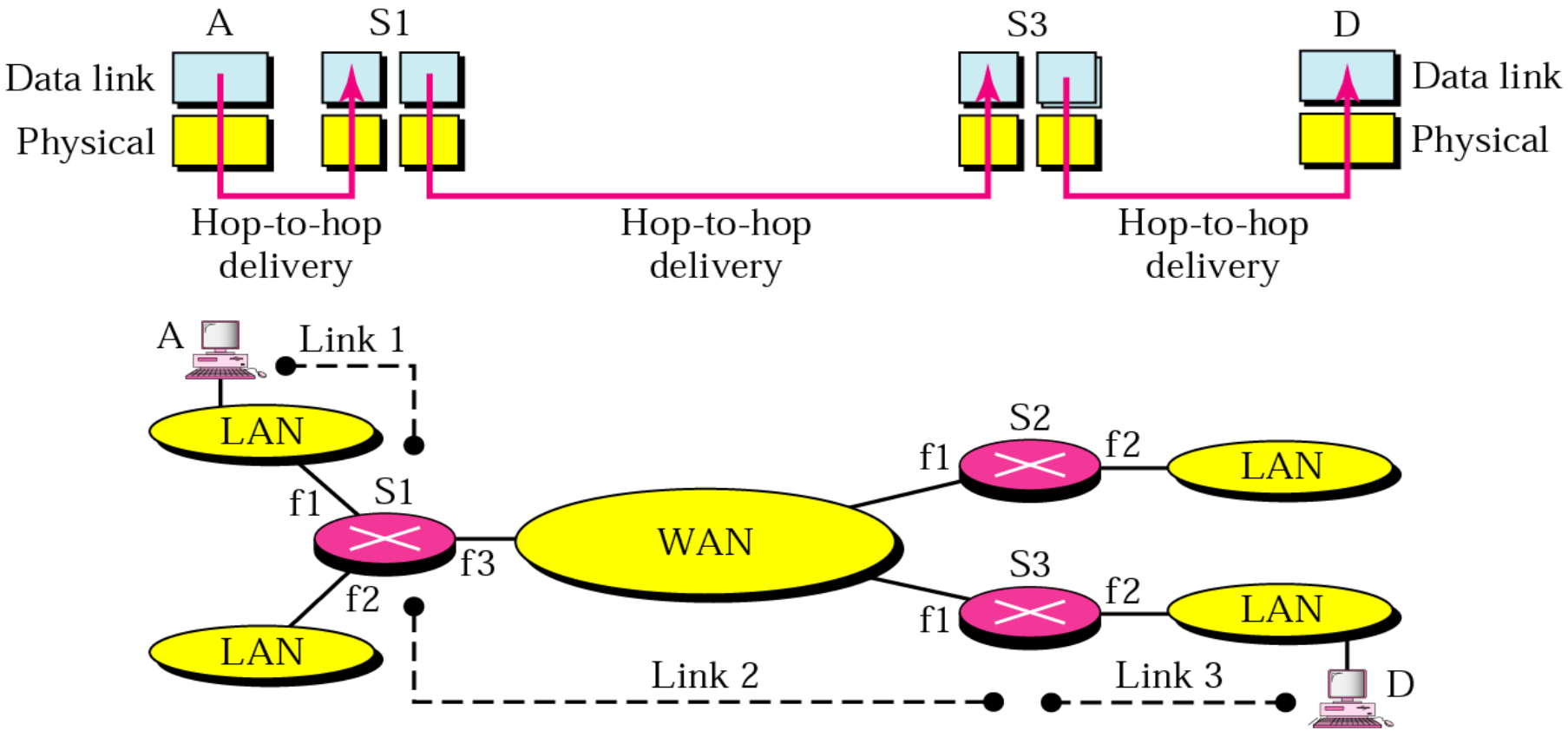
Internet As A Packet-Switched Network

Internet As A Connectionless Network

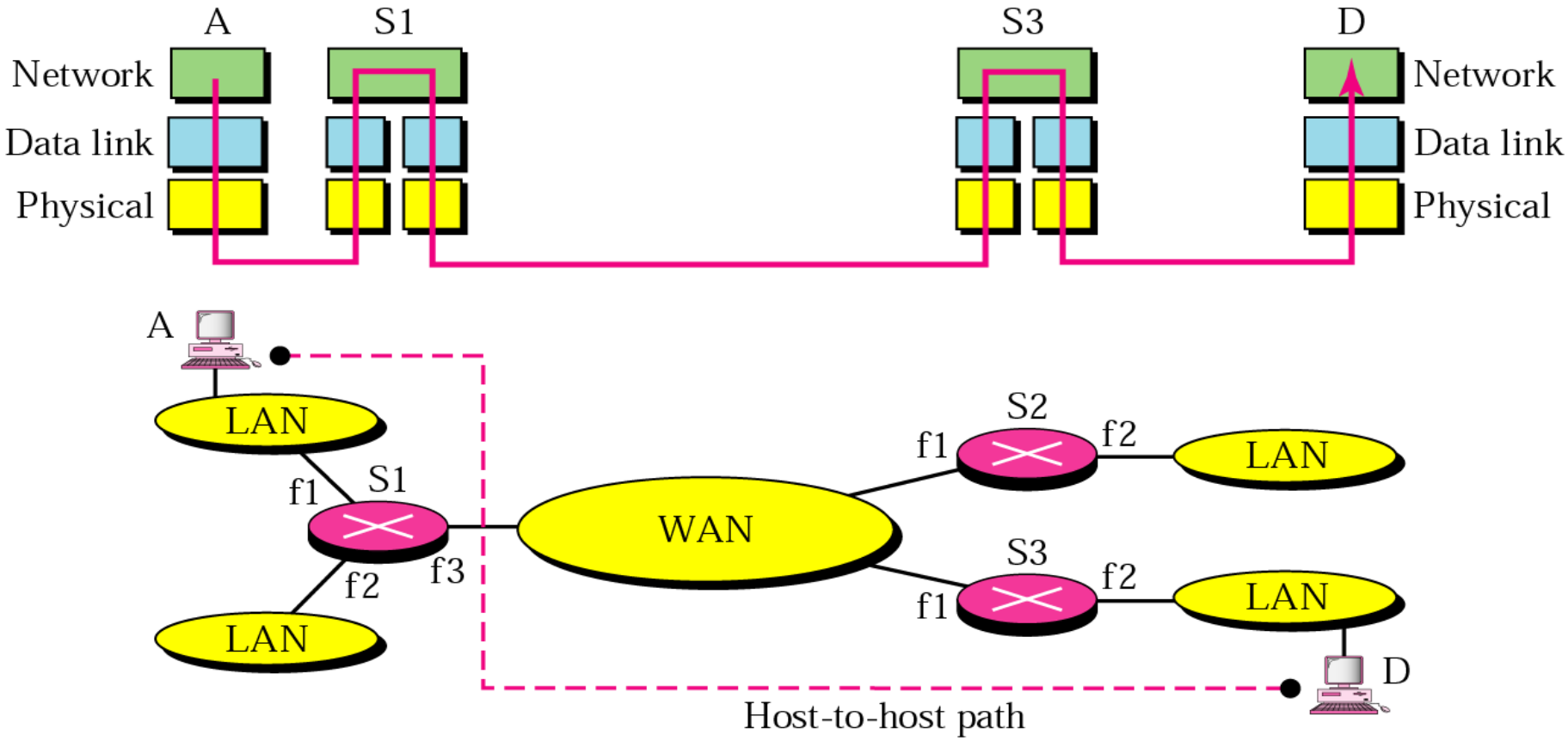
Internetwork



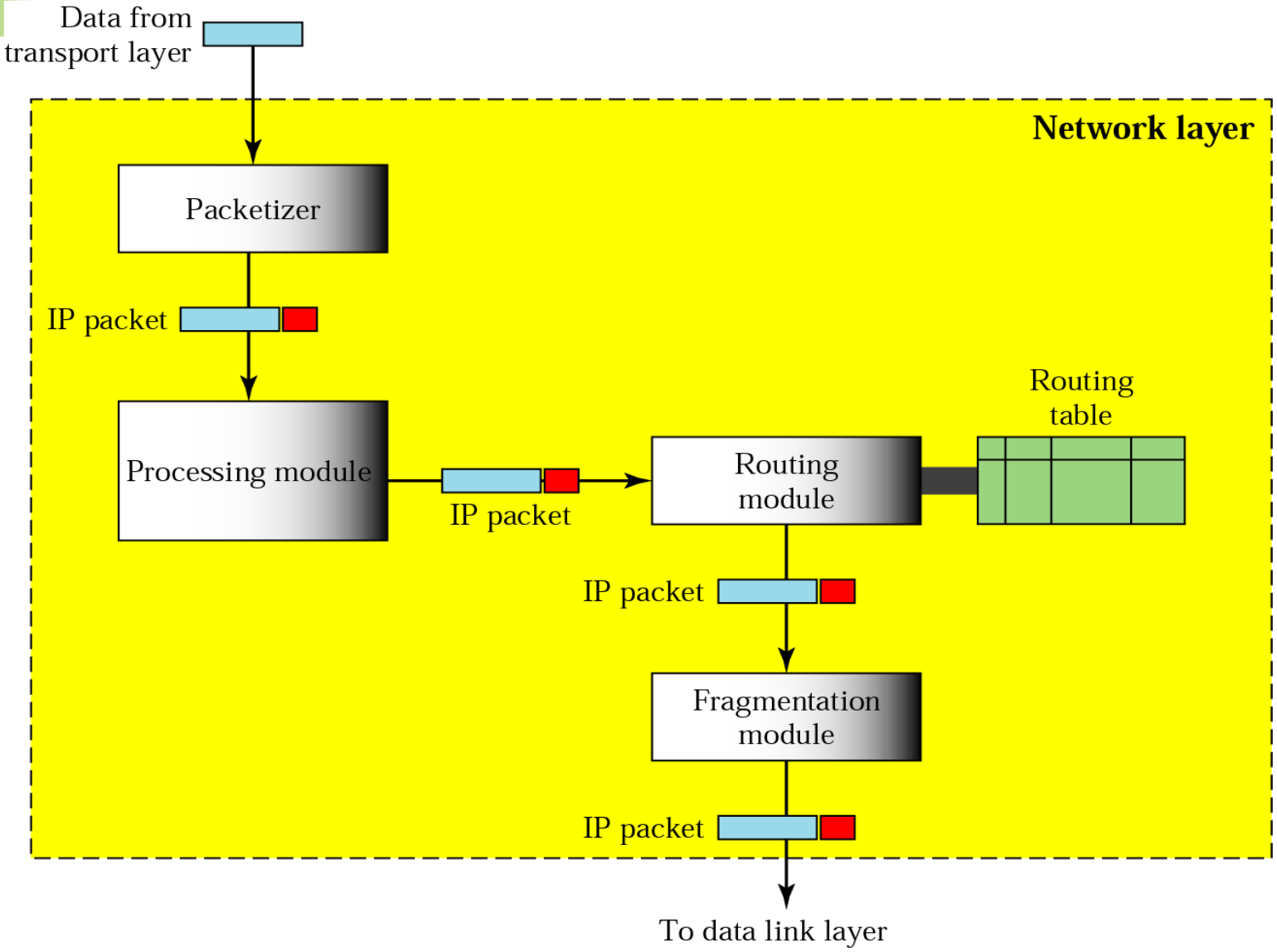
Links in an internetwork



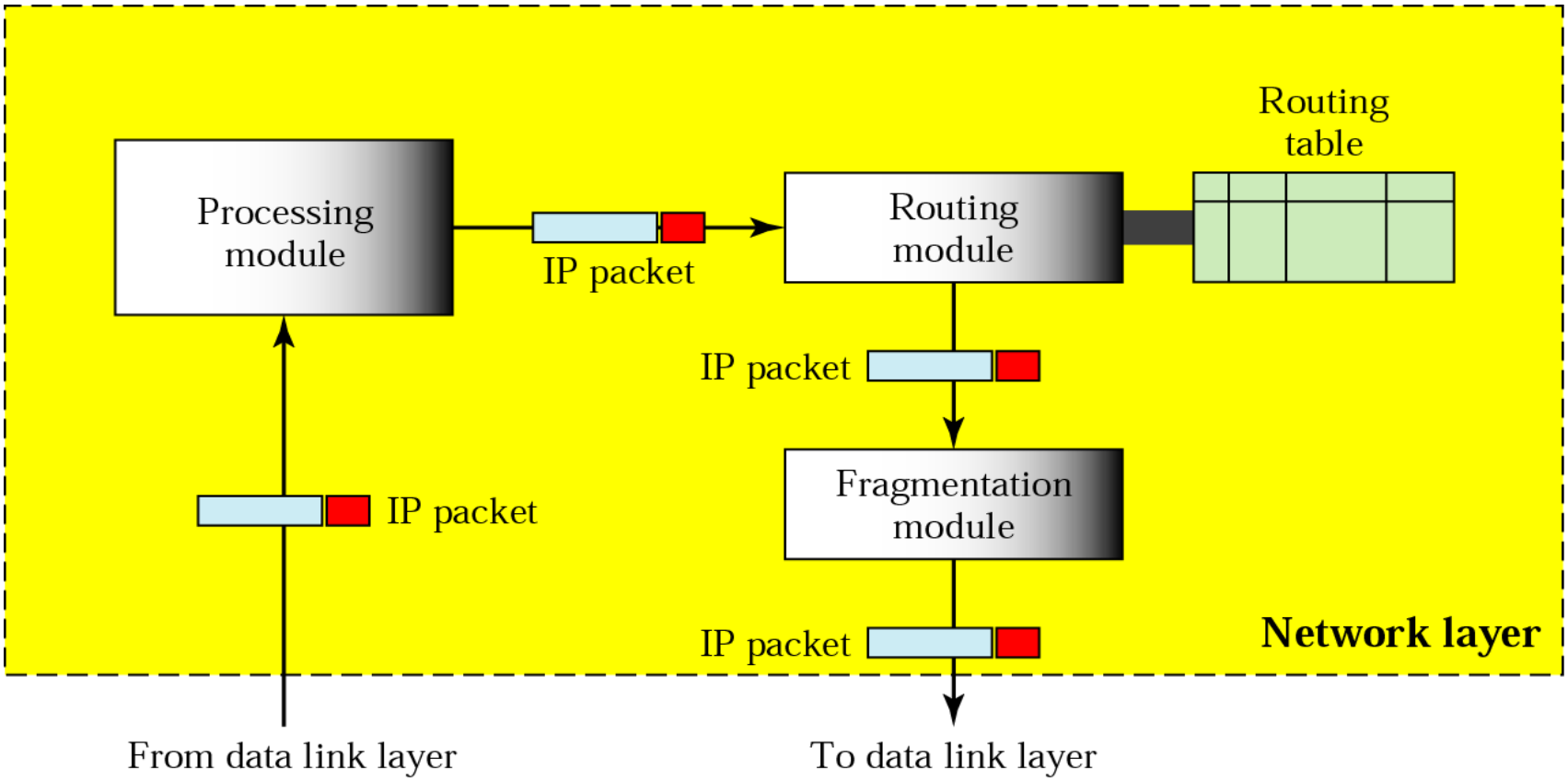
Network layer in an internetwork



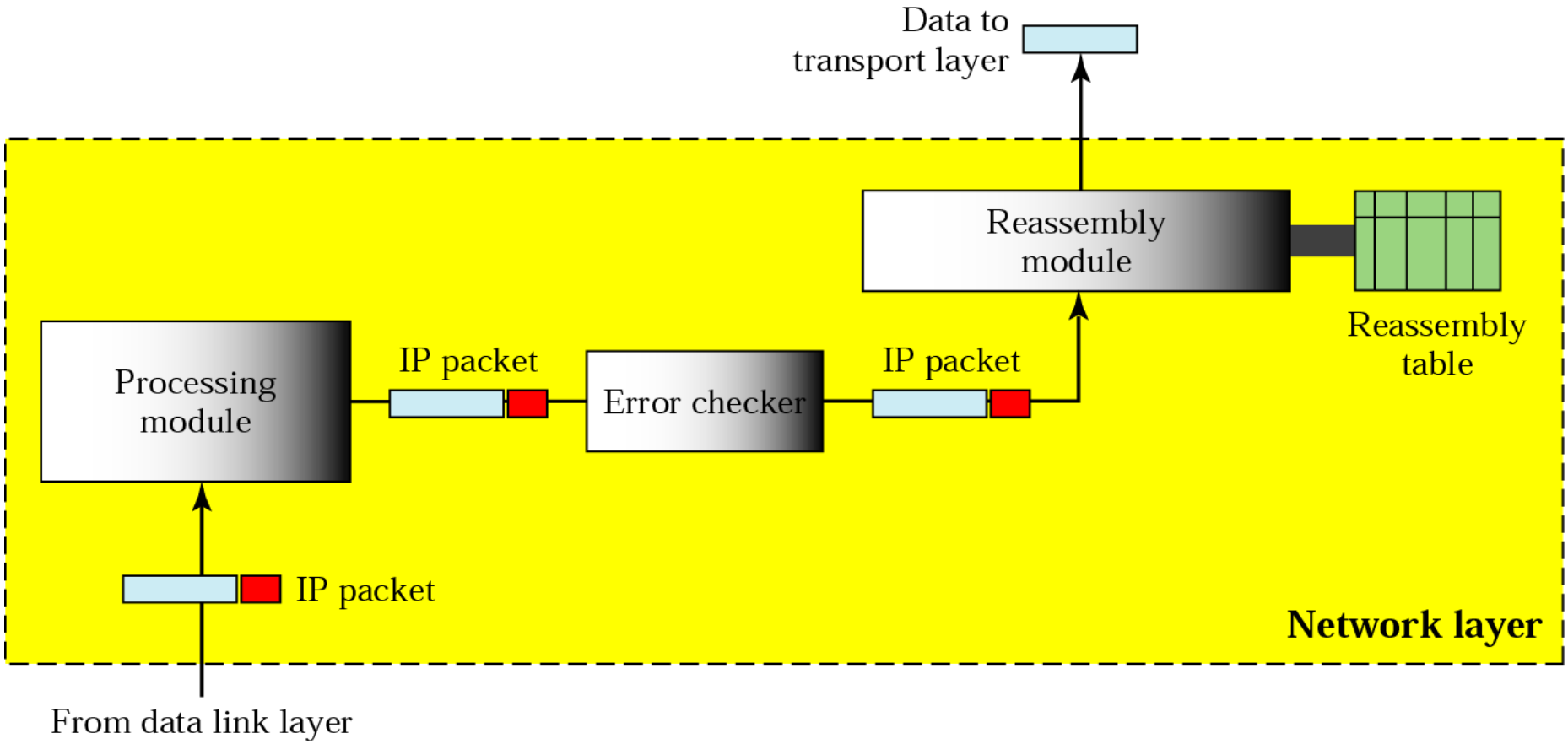
Network layer at the source



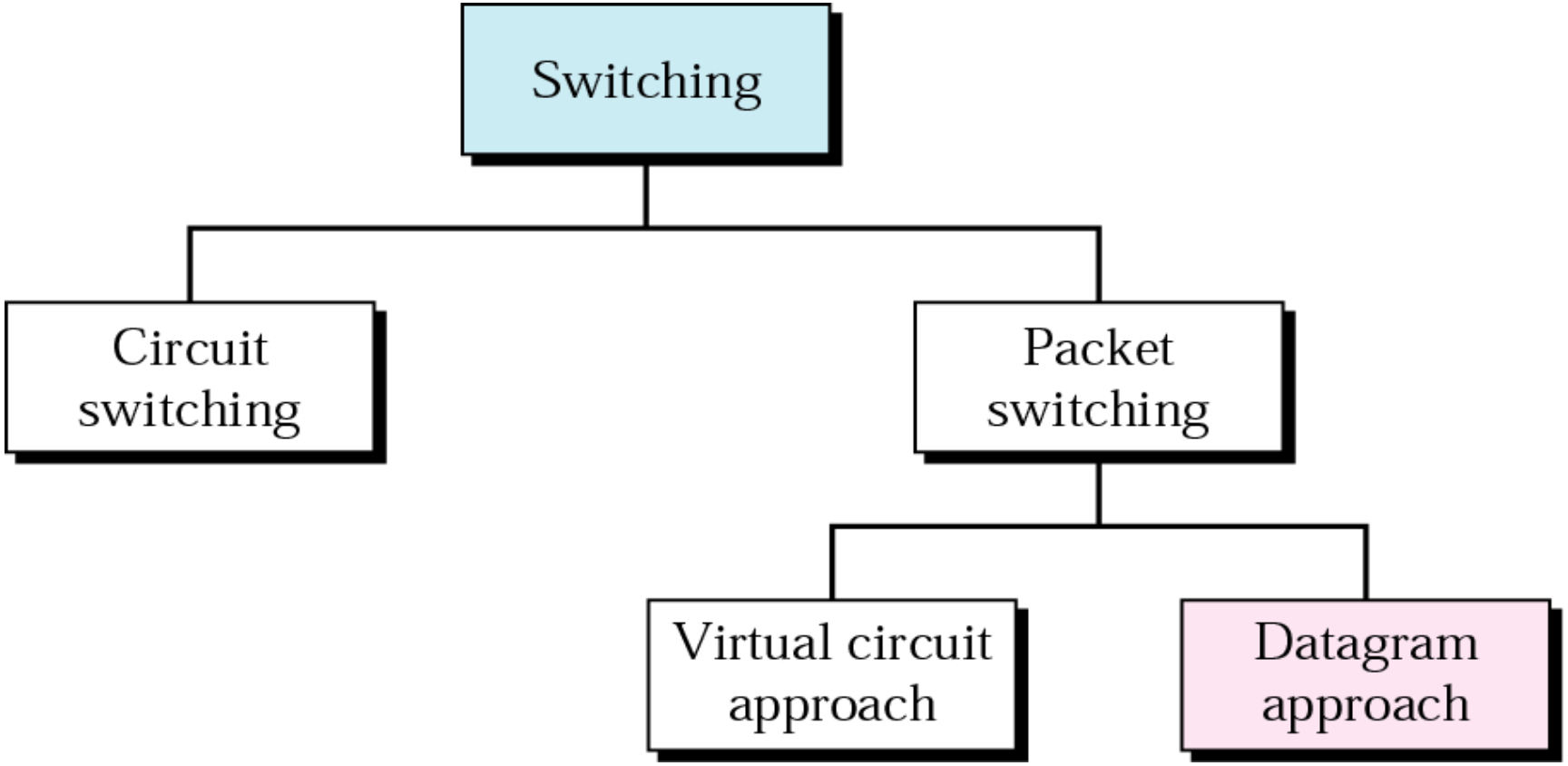
Network layer at a router



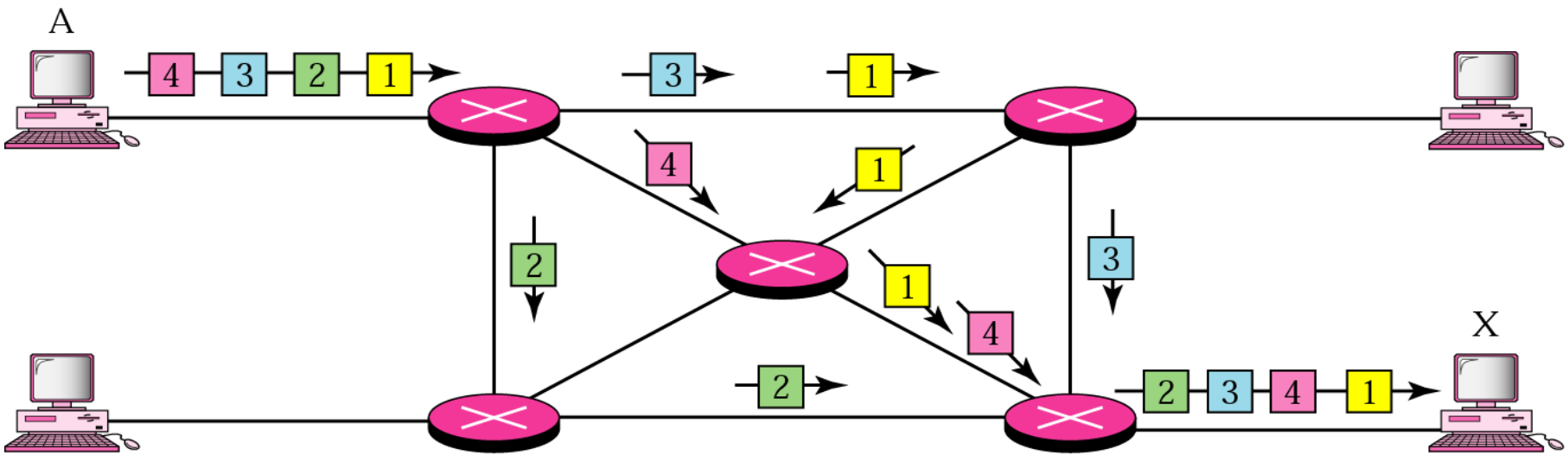
Network layer at the destination



Switching


Switching
Circuit switching
Packet switching
Virtual circuit approach
Datagram approach

Datagram approach





Note:

Switching at the network layer in the Internet is done using the datagram approach to packet switching.



Note:

Communication at the network layer in the Internet is connectionless.

Addressing

Internet Address

Classful Addressing

Subnetting

Supernetting

Classless Addressing

Dynamic Address Configuration

Network Address Translation



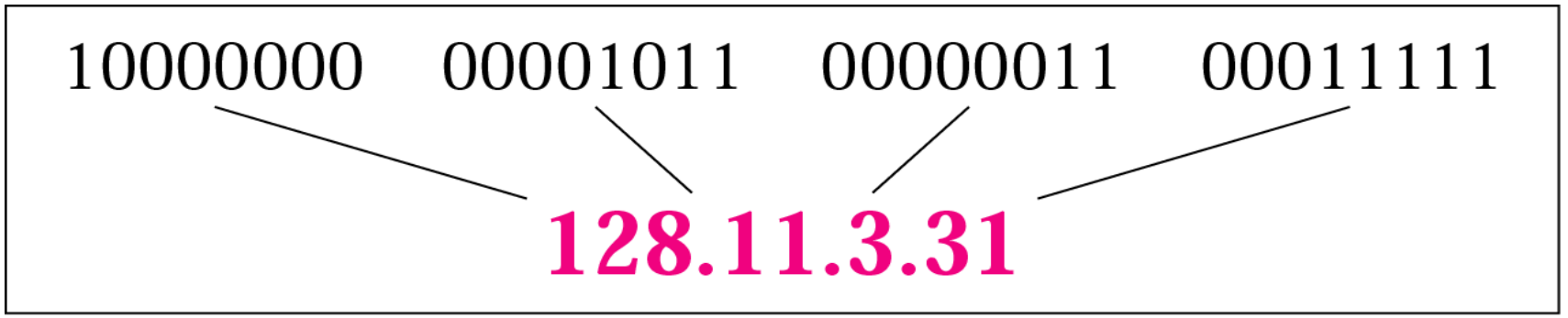
An IP address is a 32-bit address.



Note:

The IP addresses are unique and universal.

Dotted-decimal notation



Example 1

Change the following IP addresses from binary notation to dotted-decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11111001 10011011 11111011 00001111

Solution

We replace each group of 8 bits with its equivalent decimal number (see Appendix B) and add dots for separation:

a. 129.11.11.239

b. 249.155.251.15

Example 2

Change the following IP addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 75.45.34.78

Solution

We replace each decimal number with its binary equivalent (see Appendix B):

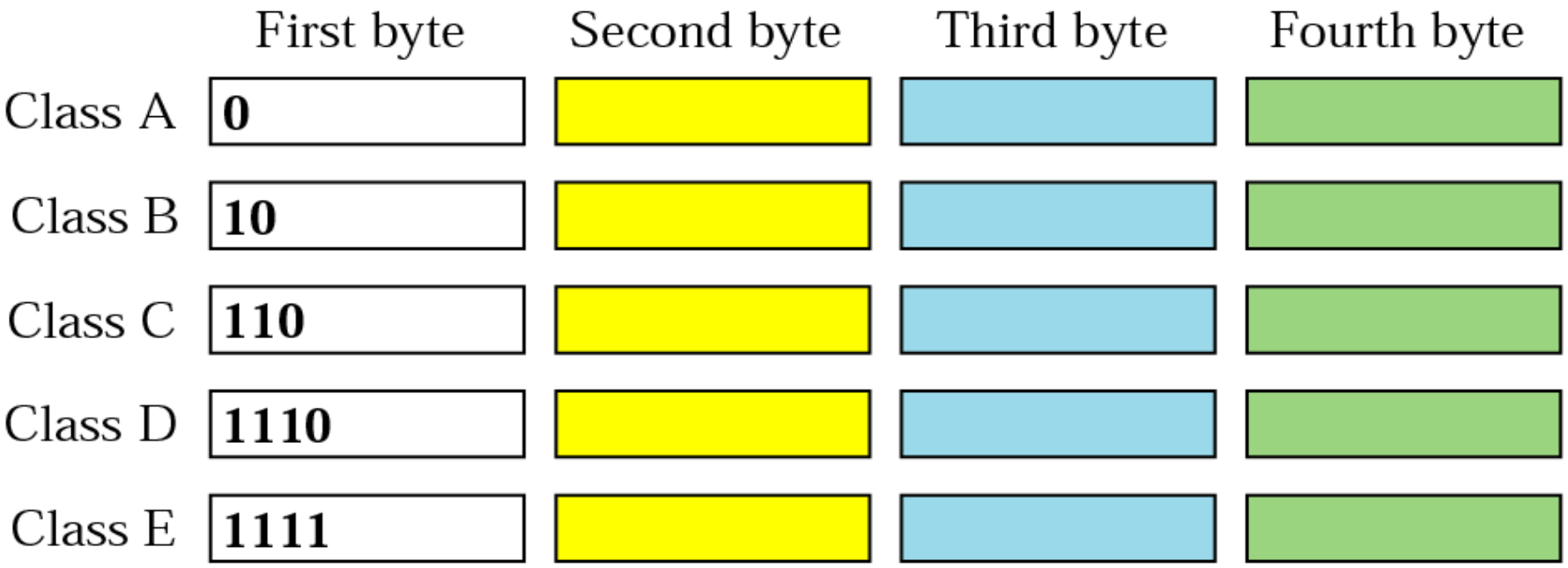
a. 01101111 00111000 00101101 01001110
b. 01001011 00101101 00100010 01001110



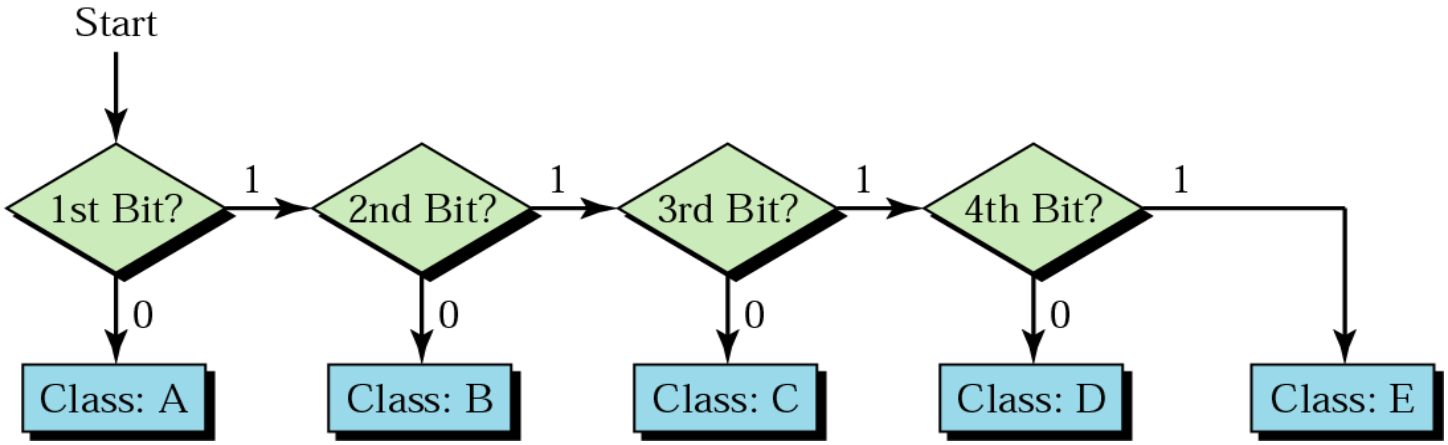
Note:

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

Finding the class in binary notation



Finding the address class



Example 3

Find the class of each address:

a. **0**0000001 00001011 00001011 11101111

b. **1111**0011 10011011 11111011 00001111

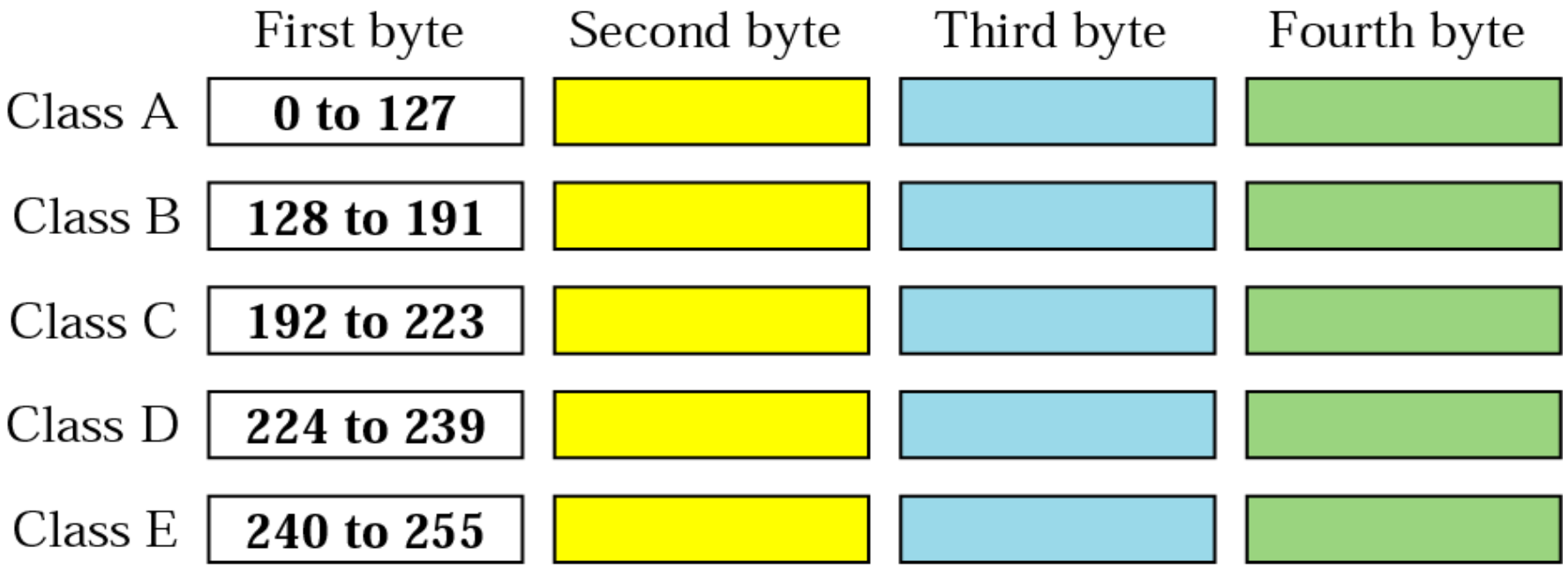
Solution

See the procedure in Figure 19.11.

a. The first bit is 0; this is a class A address.

b. The first 4 bits are 1s; this is a class E address.

Finding the class in decimal notation



Example 4

Find the class of each address:

a. **227**.12.14.87

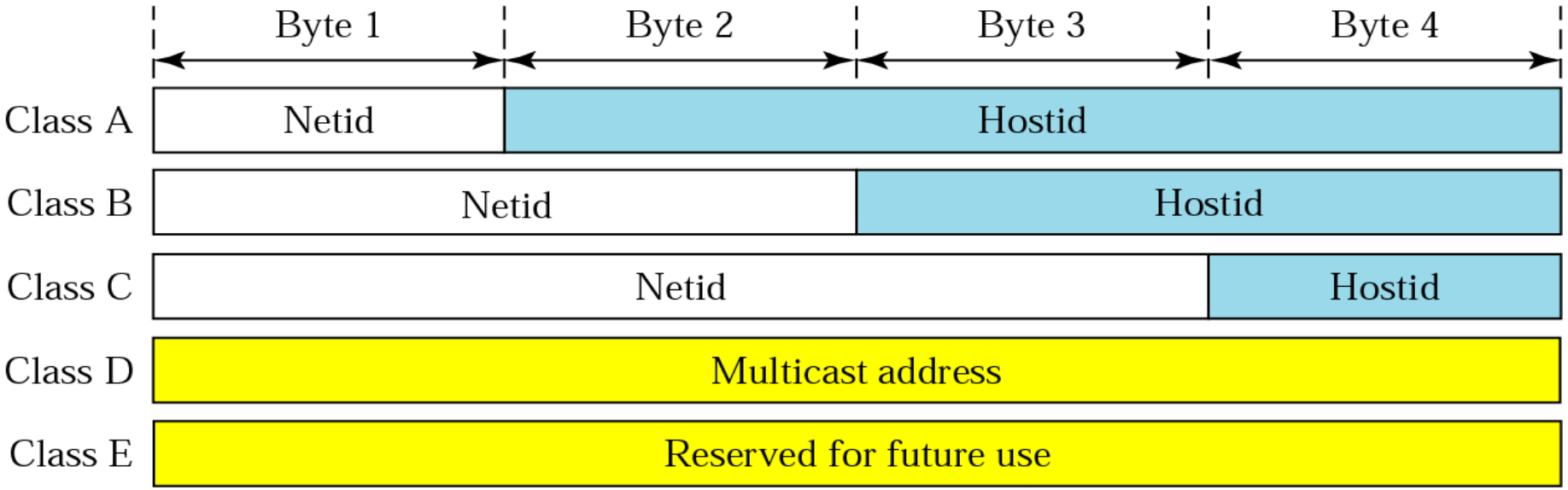
b. **252**.5.15.111

c. **134**.11.78.56

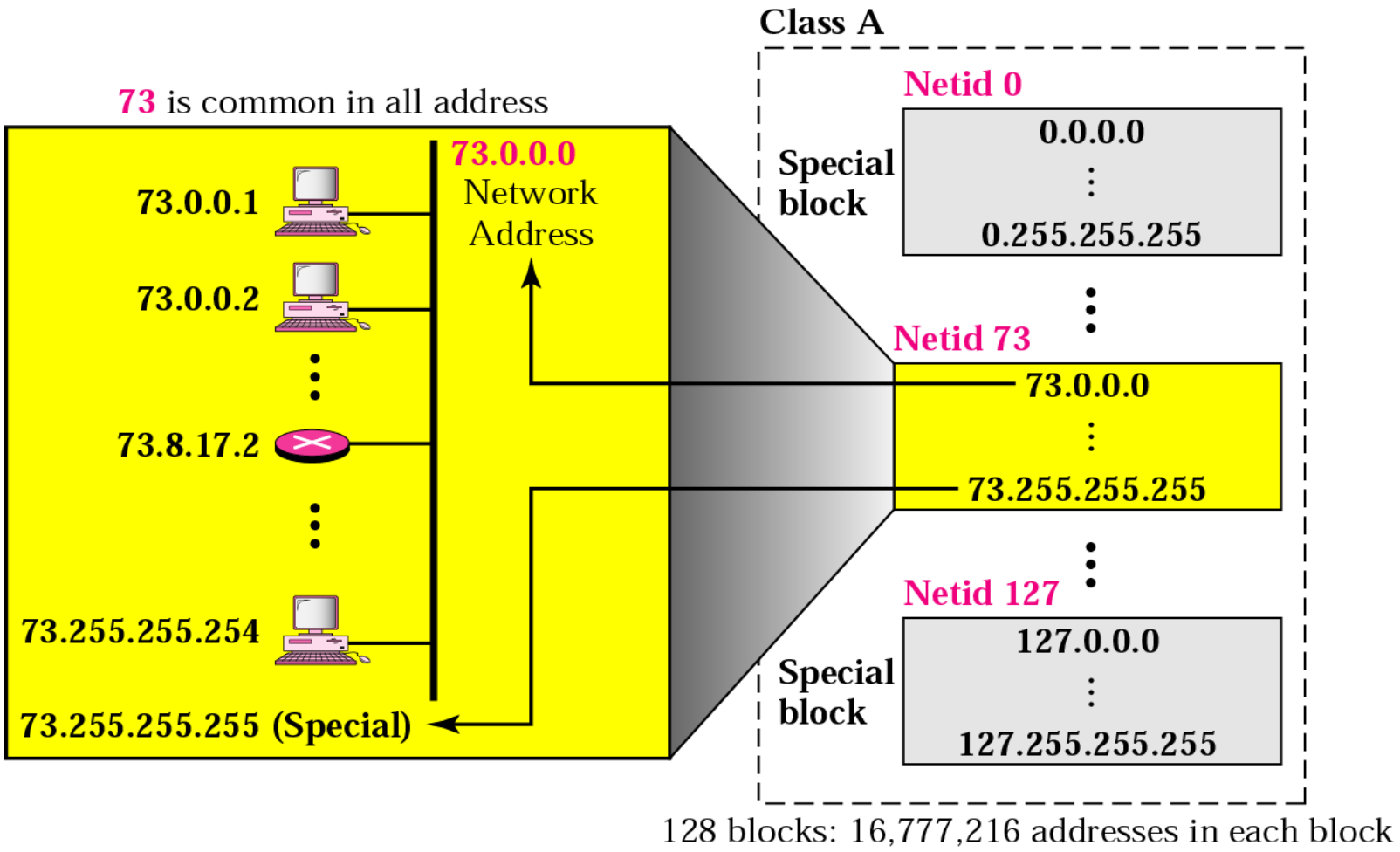
Solution

a. The first byte is 227 (between 224 and 239); the class is D.
b. The first byte is 252 (between 240 and 255); the class is E.
c. The first byte is 134 (between 128 and 191); the class is B.

Netid and hostid



Blocks in class A

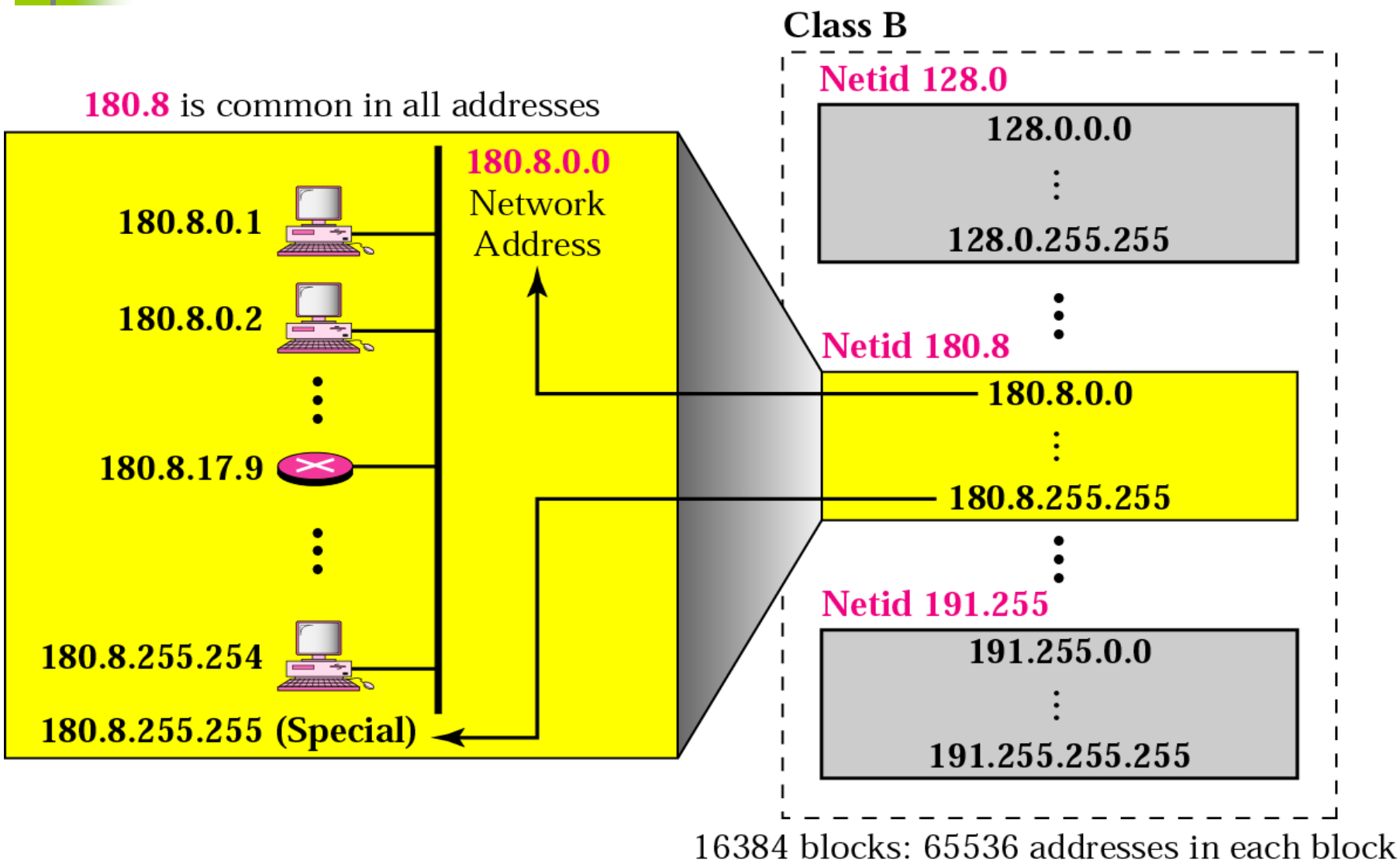




Note:

Millions of class A addresses are wasted.

Blocks in class B





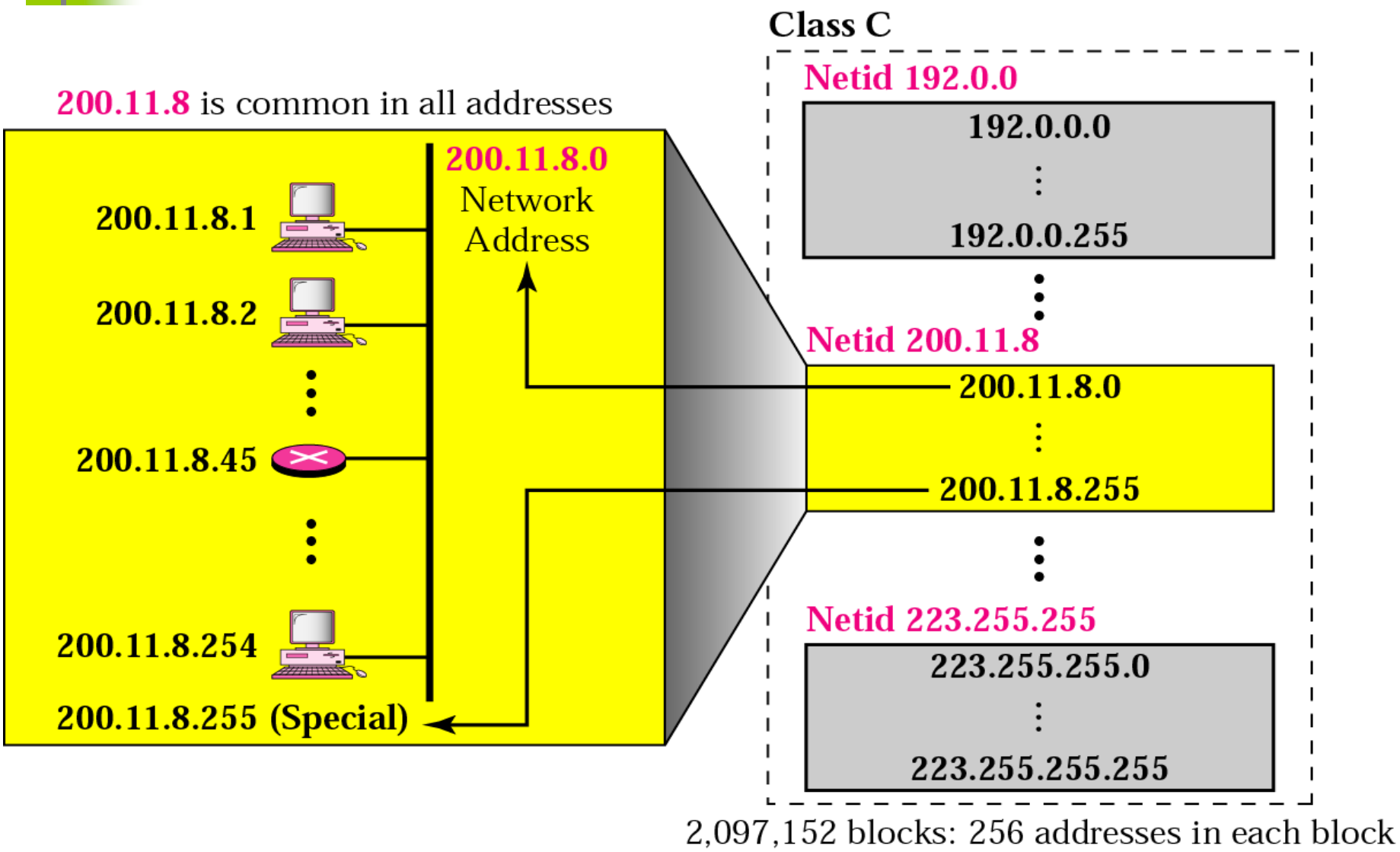
Many class B addresses are wasted.



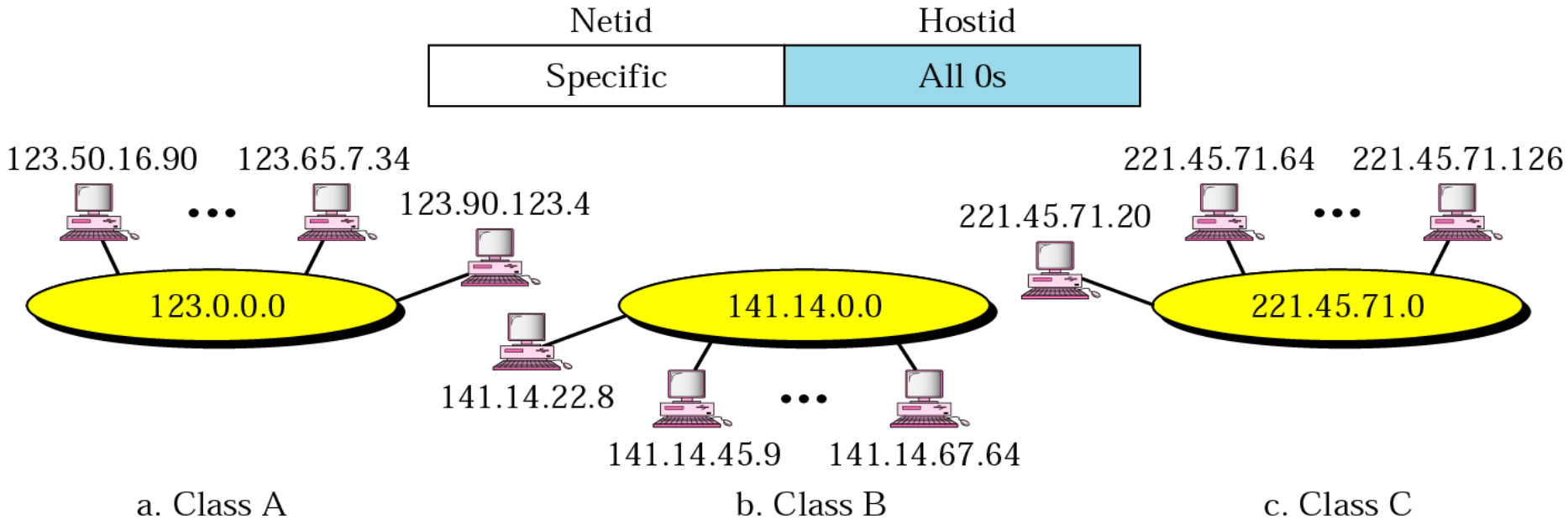
Note:

The number of addresses in class C is smaller than the needs of most organizations.

Blocks in class C



Network address





Note:

In classful addressing, the network address is the one that is assigned to the organization.

Example 5

Given the address 23.56.7.91, find the network address.

Solution

The class is A. Only the first byte defines the netid. We can find the network address by replacing the hostid bytes (56.7.91) with 0s. Therefore, the network address is 23.0.0.0.

Example 6

Given the address 132.6.17.85, find the network address.

Solution

The class is B. The first 2 bytes defines the netid. We can find the network address by replacing the hostid bytes (17.85) with 0s. Therefore, the network address is 132.6.0.0.

Example 7

Given the network address 17.0.0.0, find the class.

Solution

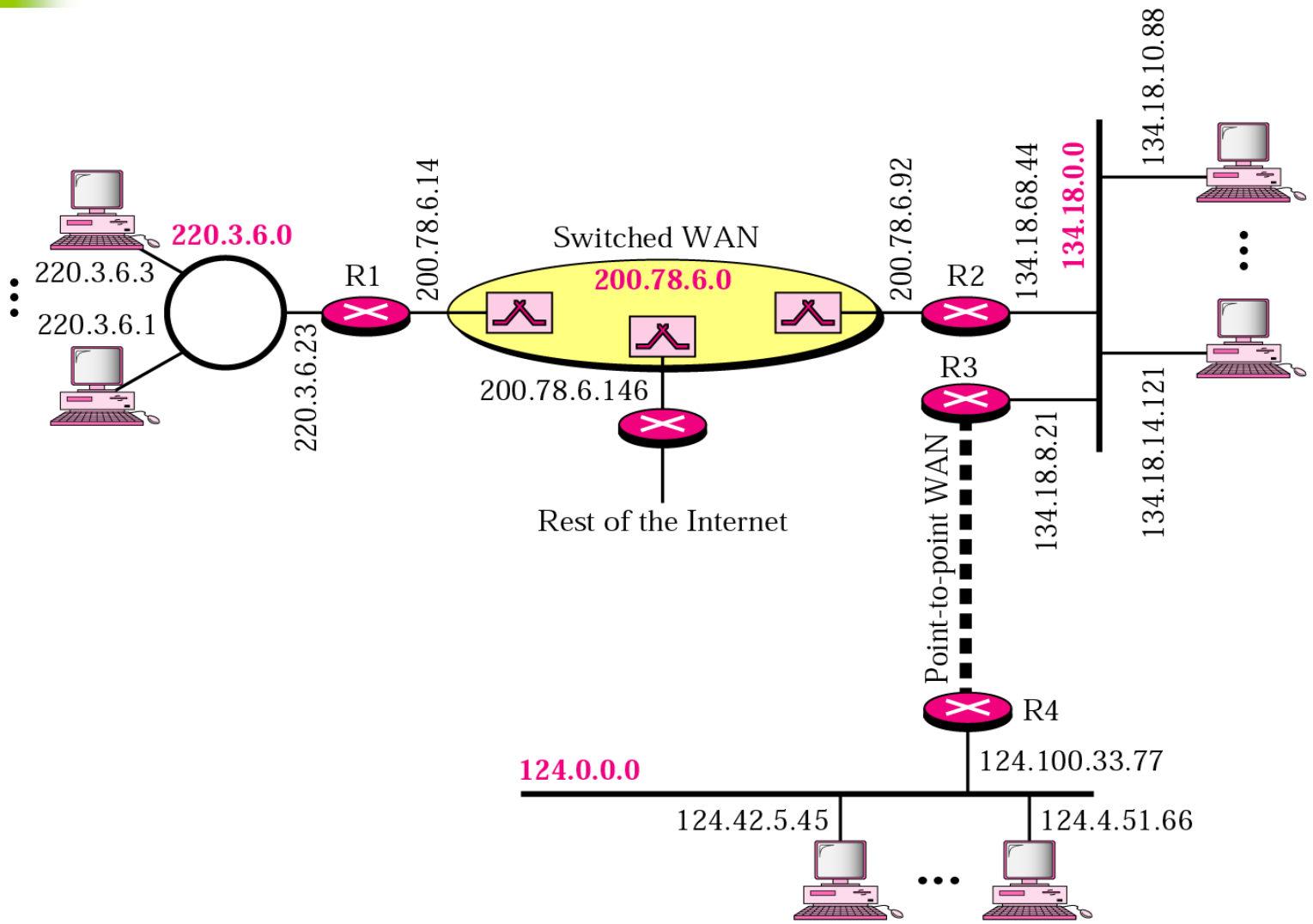
The class is A because the netid is only 1 byte.



Note:

A network address is different from a netid. A network address has both netid and hostid, with 0s for the hostid.

Sample internet

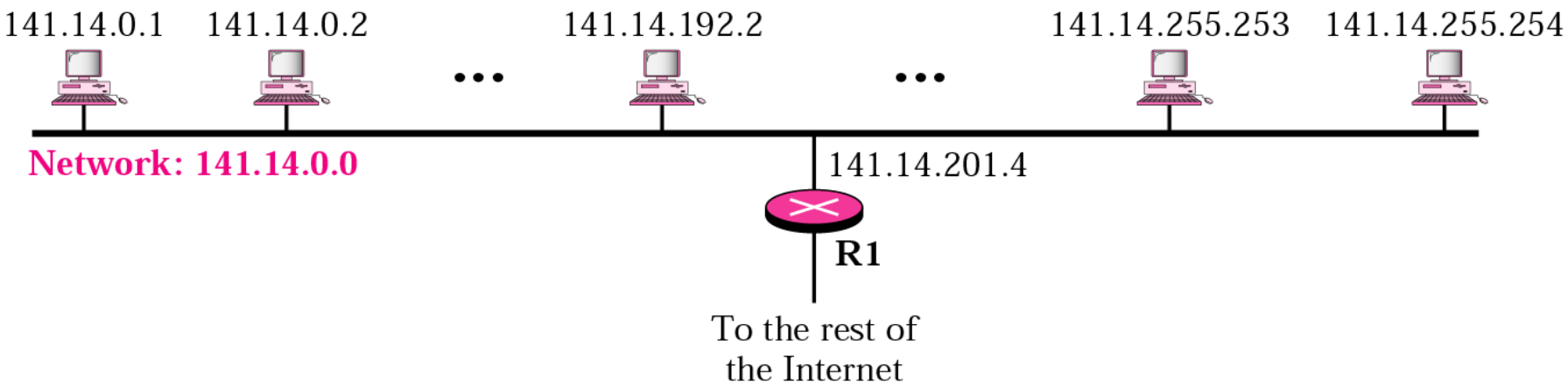




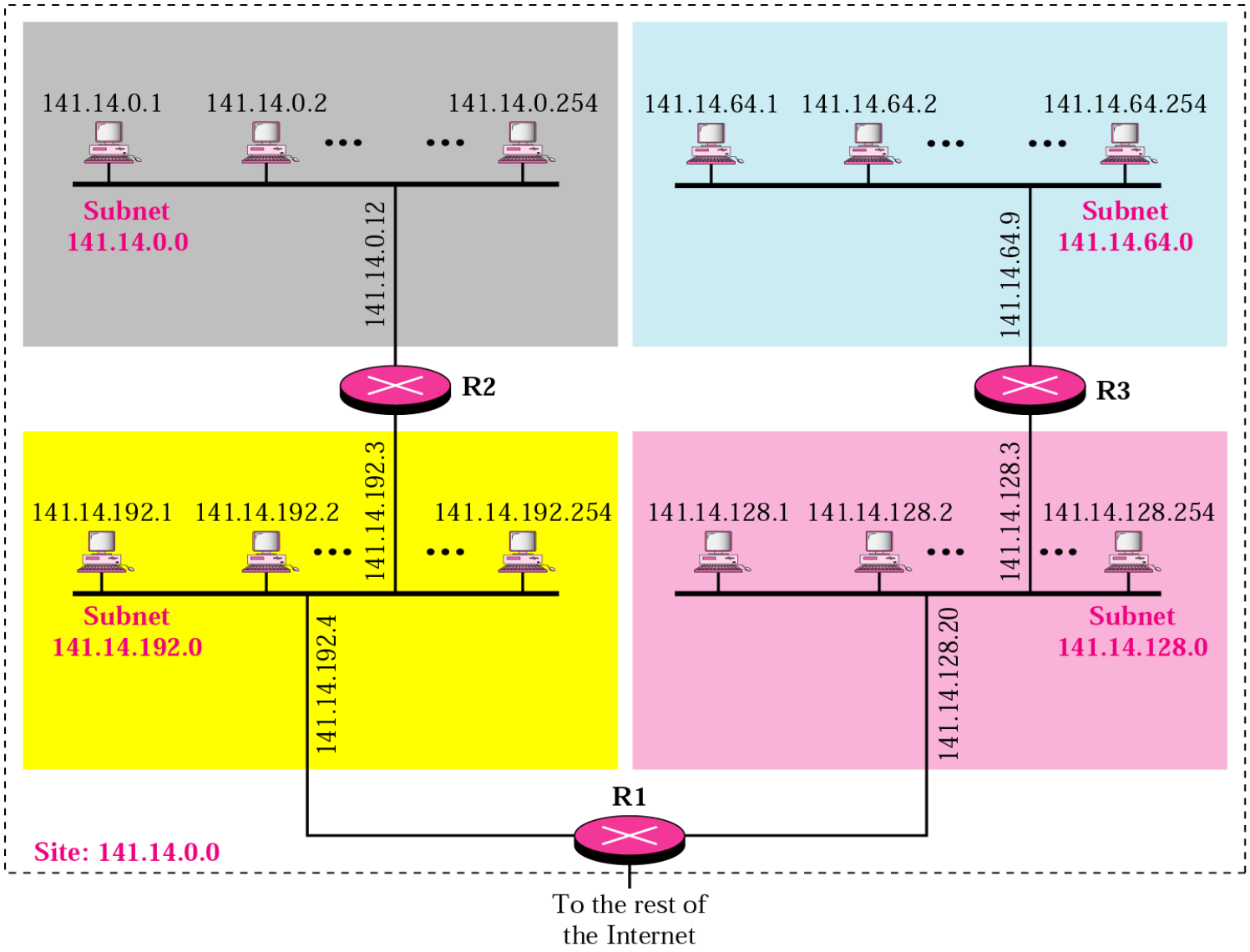
Note:

IP addresses are designed with two levels of hierarchy.

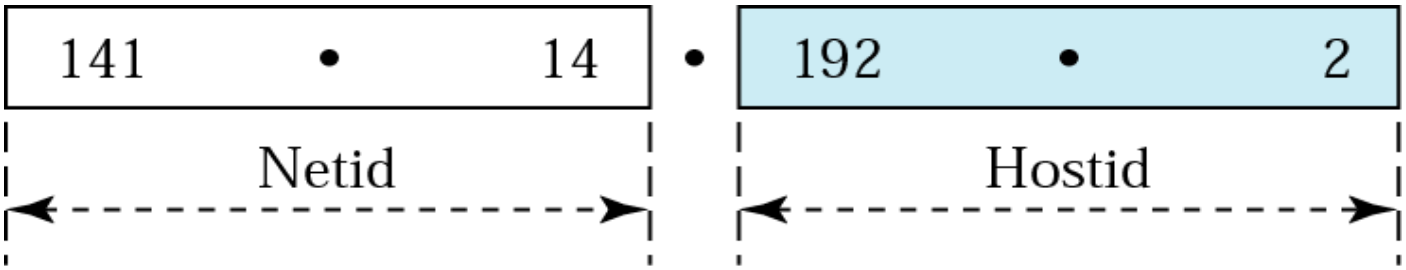
A network with two levels of hierarchy



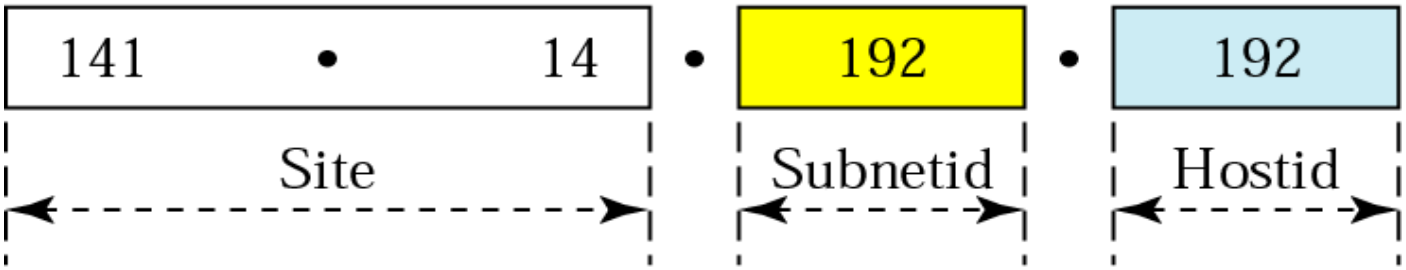
A network with three levels of hierarchy (subnetted)



Addresses in a network with and without subnetting



a. Without subnetting



b. With subnetting

Hierarchy concept in a telephone number



Default masks

Class	*In Binary*	*In Dotted-Decimal*	*Using Slash*
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 111111111 11111111 00000000	255.255.255.0	/24



Note:

The network address can be found by applying the default mask to any address in the block (including itself). It retains the netid of the block and sets the hostid to 0s.

Example 8

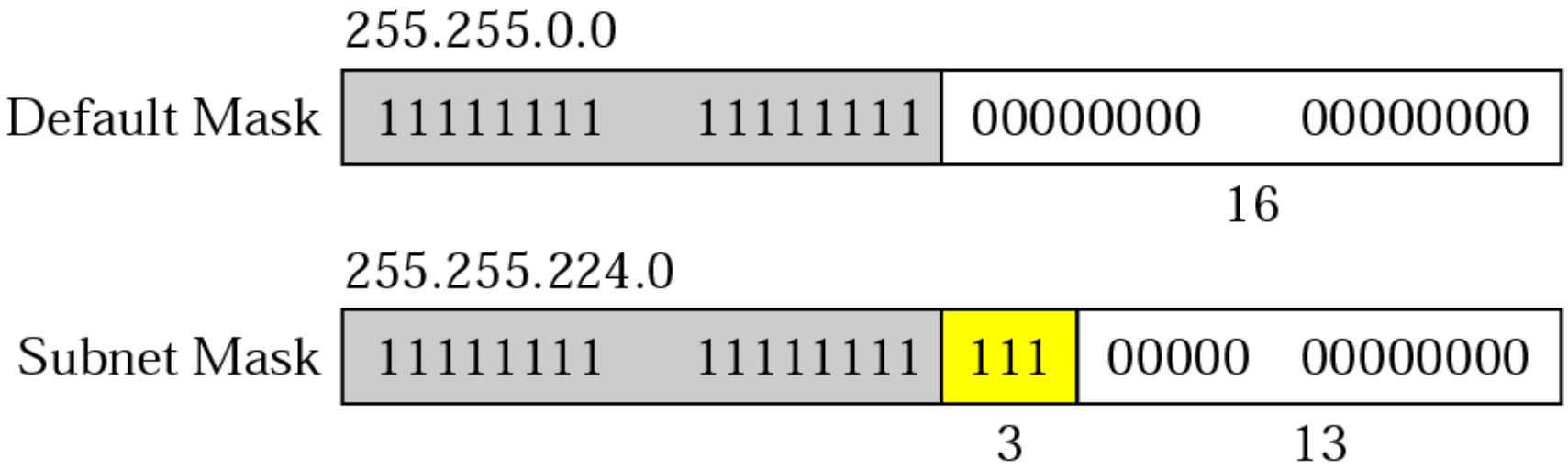
A router outside the organization receives a packet with destination address 190.240.7.91. Show how it finds the network address to route the packet.

Solution

The router follows three steps:

1. The router looks at the first byte of the address to find the class. It is class B.
2. The default mask for class B is 255.255.0.0. The router ANDs this mask with the address to get 190.240.0.0.
3. The router looks in its routing table to find out how to route the packet to this destination. Later, we will see what happens if this destination does not exist.

Subnet mask


255.255.0.0
Default Mask
11111111 11111111 00000000 00000000
16
255.255.224.0
Subnet Mask
11111111 11111111 111 00000 00000000
3
13

Example 9

A router inside the organization receives the same packet with destination address 190.240.33.91. Show how it finds the subnetwork address to route the packet.

Solution

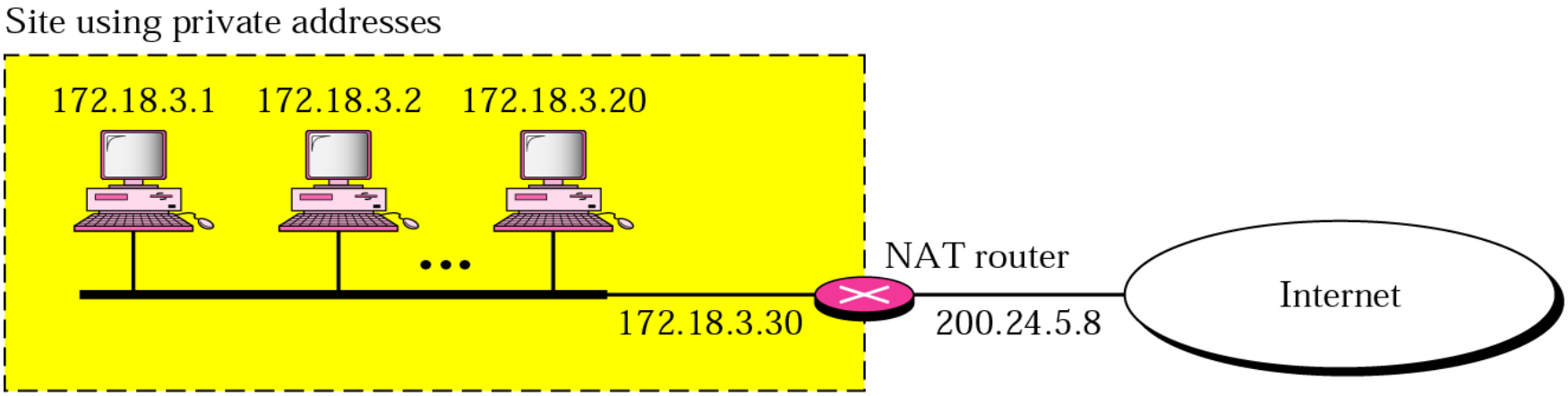
The router follows three steps:

1. **The router must know the mask. We assume it is /19, as shown in Figure 19.23.**
2. **The router applies the mask to the address, 190.240.33.91. The subnet address is 190.240.32.0.**
3. **The router looks in its routing table to find how to route the packet to this destination. Later, we will see what happens if this destination does not exist.**

Default masks

Range			*Total*
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

NAT


Site using private addresses
172.18.3.1
172.18.3.2
172.18.3.20
NAT router
172.18.3.30
200.24.5.8
Internet

20.4 IPv6

IPv6 Addresses

Categories of Addresses

IPv6 Packet Format

Fragmentation

ICMPv6

Transition

Figure 20.15 IPv6 address

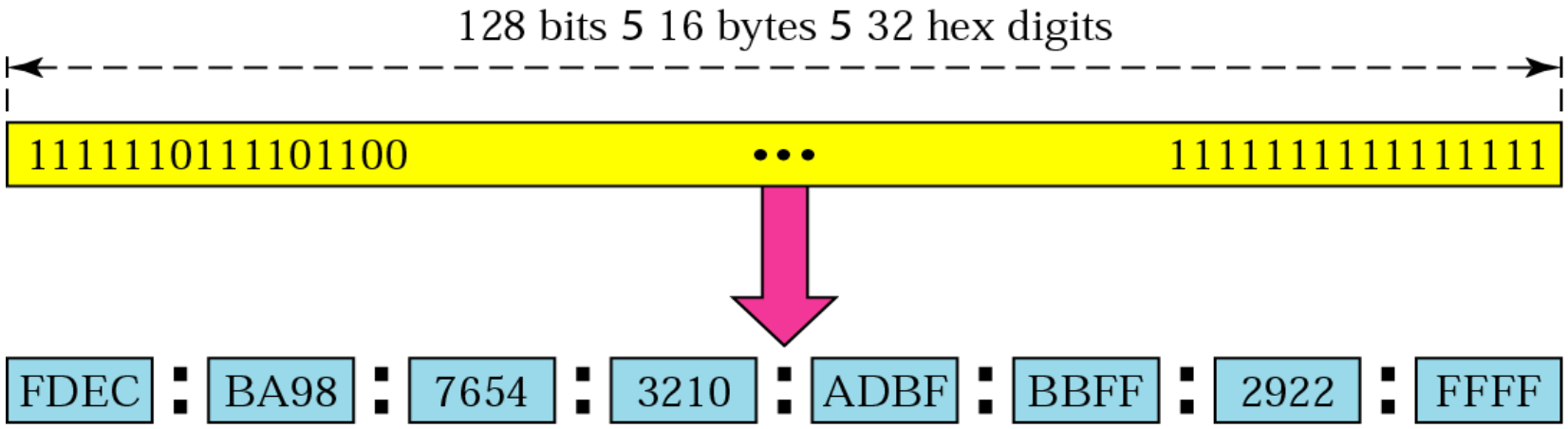


Figure 20.16 Abbreviated address

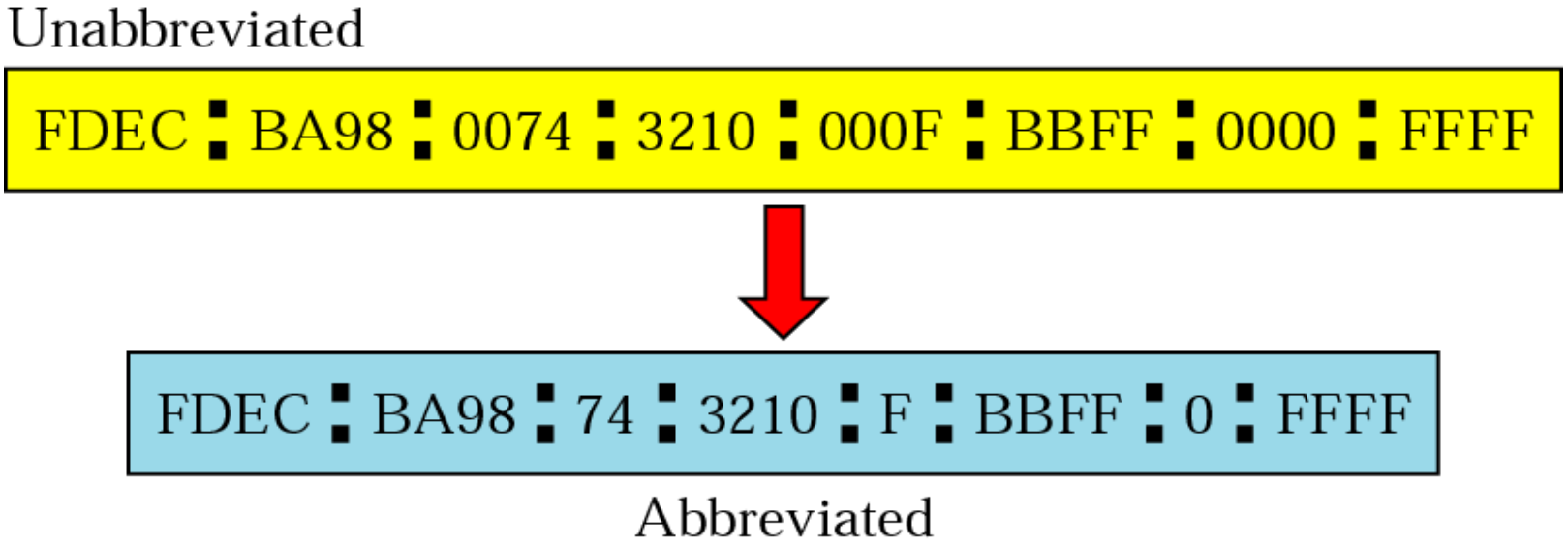


Figure 20.17 Abbreviated address with consecutive zeros

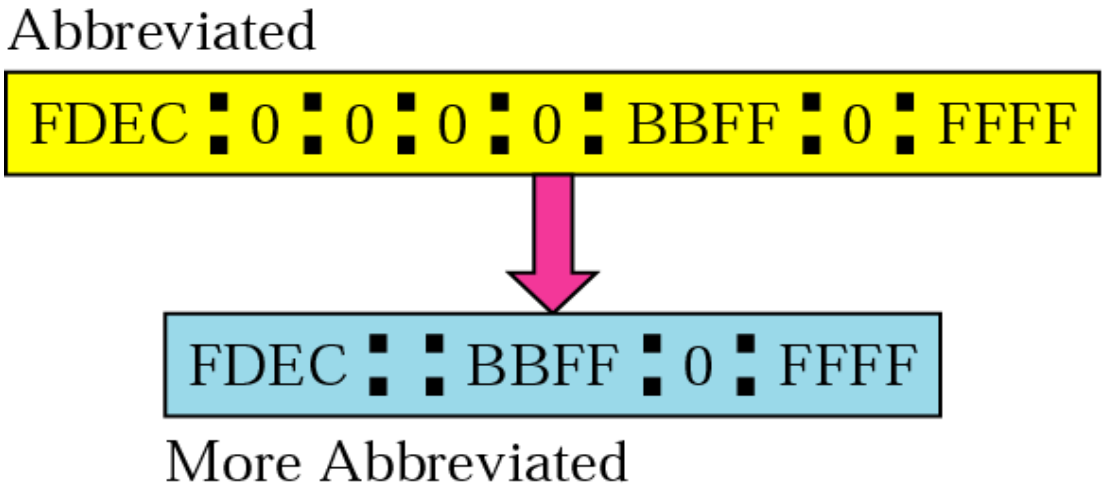


Figure 20.18 CIDR address

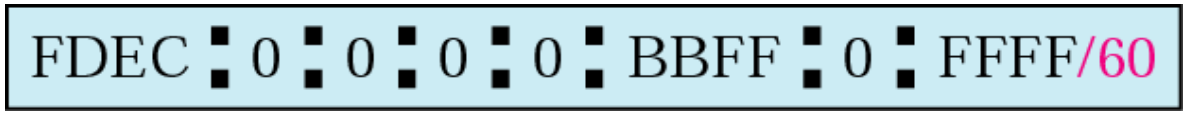


Figure 20.19 **Format of an IPv6 datagram**

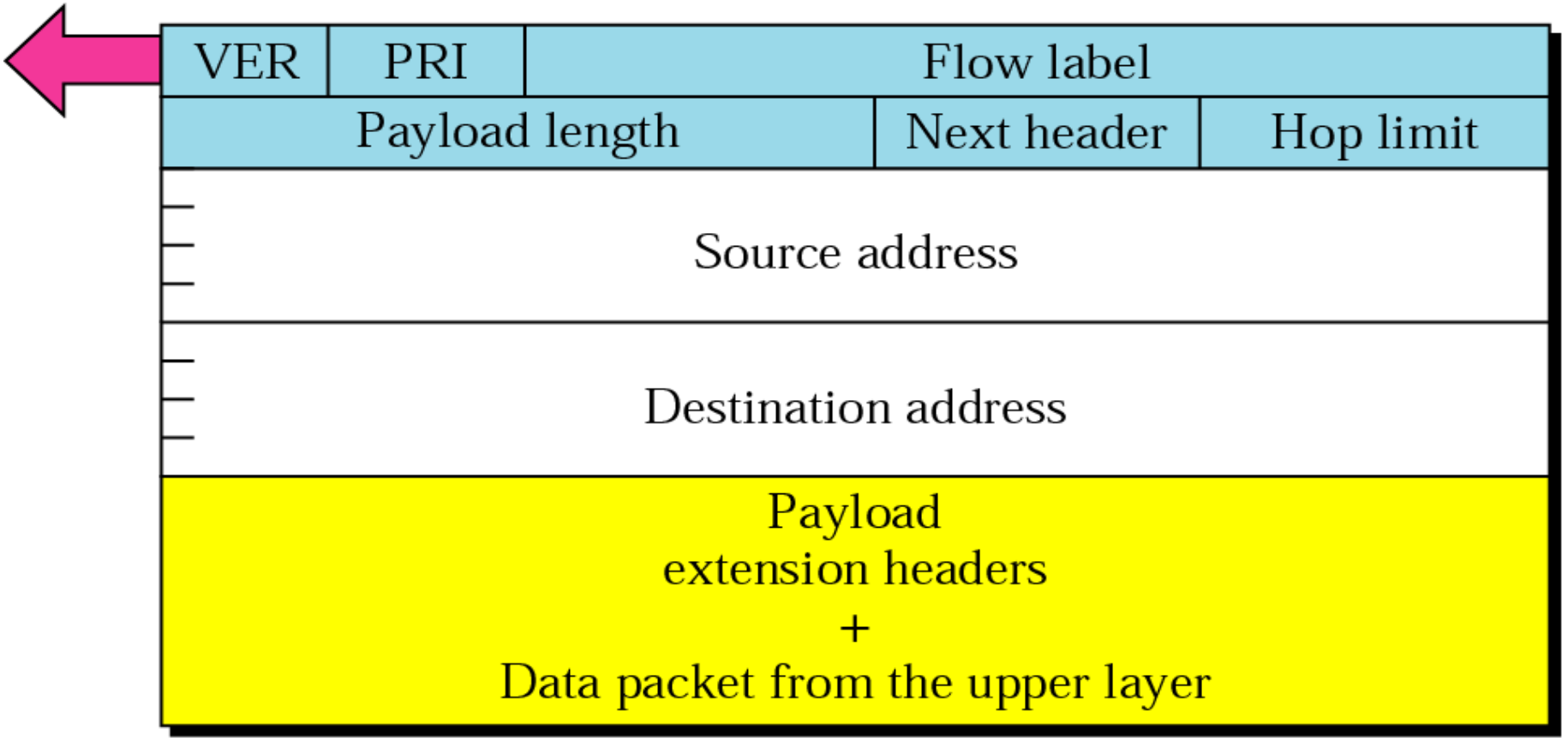
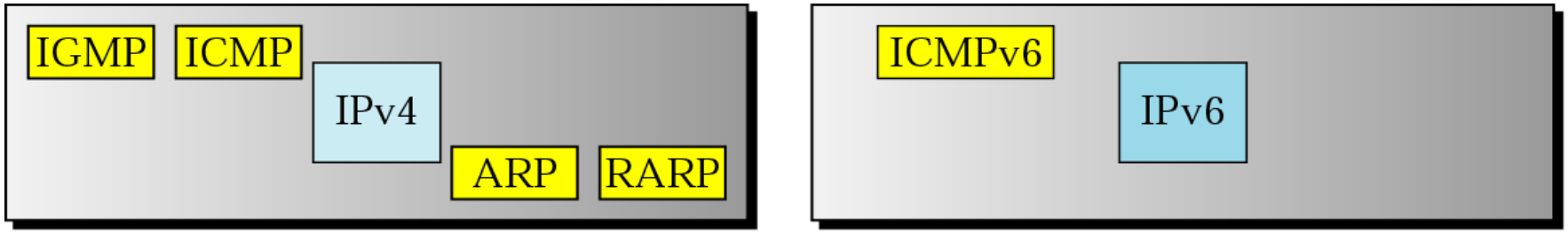


Figure 20.20 Comparison of network layers in version 4 and version 6



Network layer in version 4

Network layer in version 6

Figure 20.21 Three transition strategies

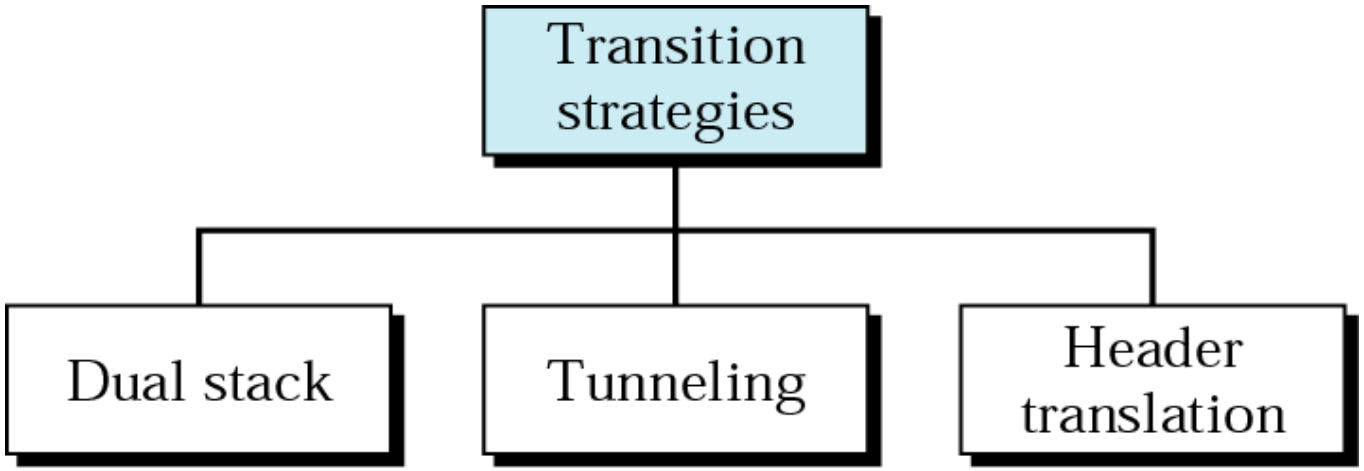


Figure 20.22 Three transition strategies

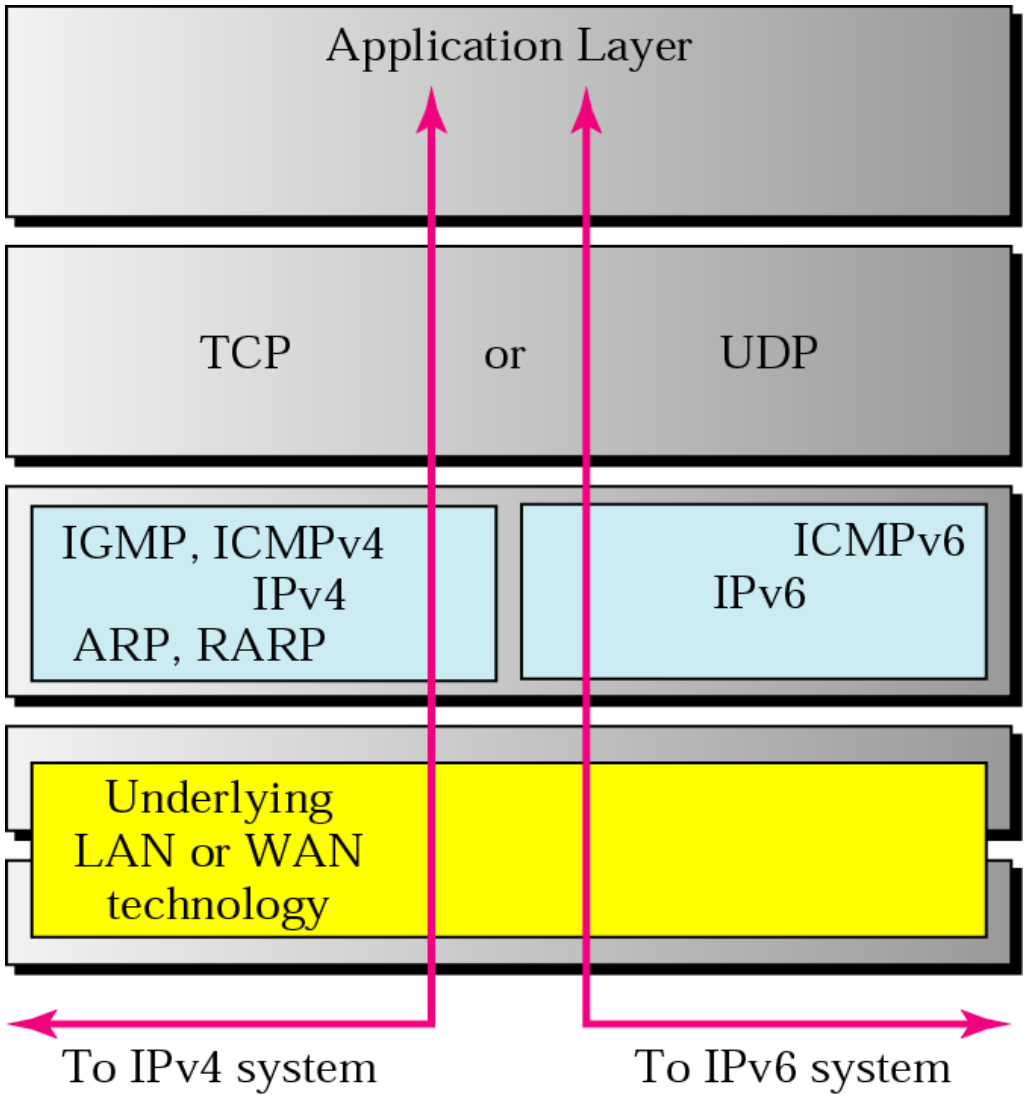


Figure 20.23 Tunneling

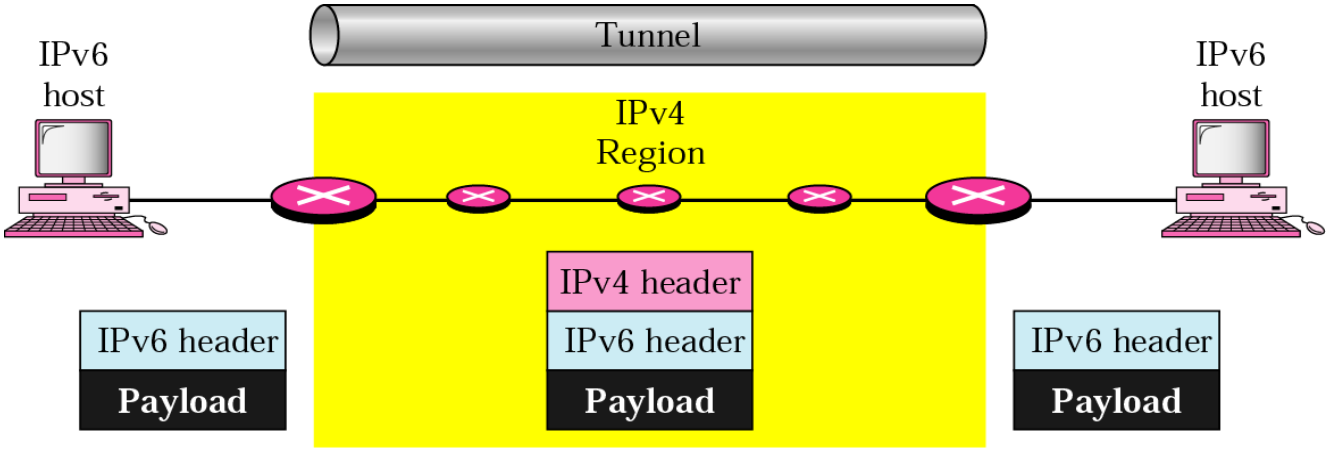
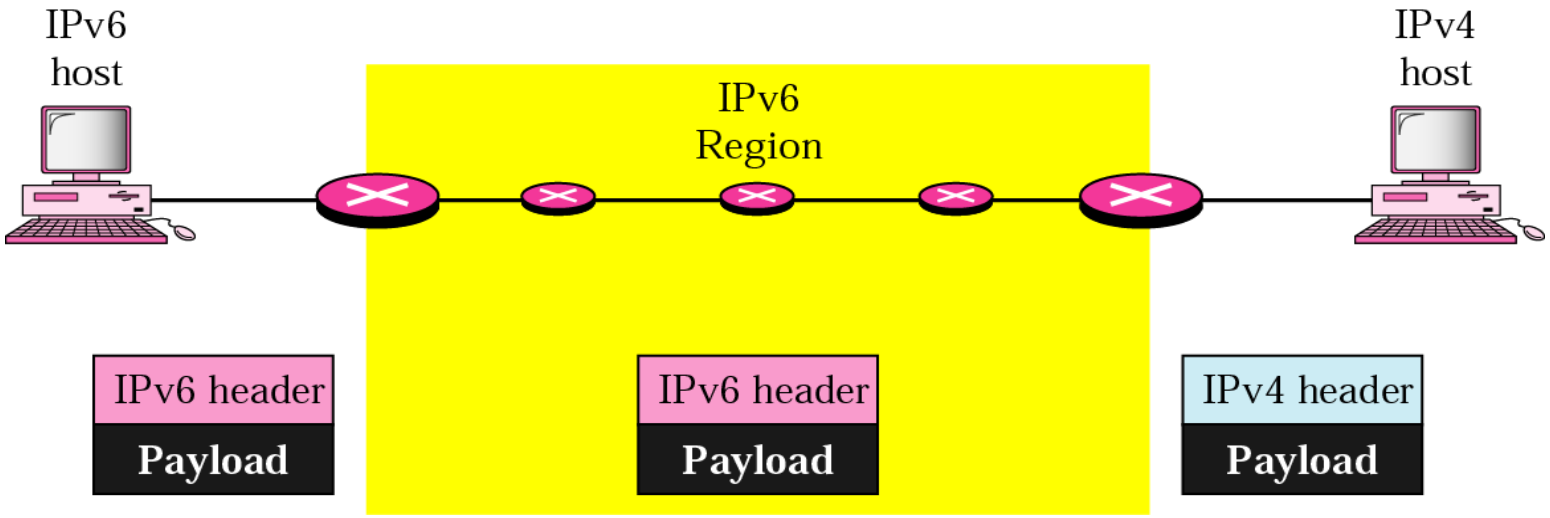


Figure 20.24 Header translation



19.3 Routing

Routing Techniques

Static Versus Dynamic Routing

Routing Table for Classful Addressing

Routing Table for Classless Addressing

Figure 19.28 Next-hop routing

Routing table for host A

Destination	Route
Host B	R1, R2, Host B

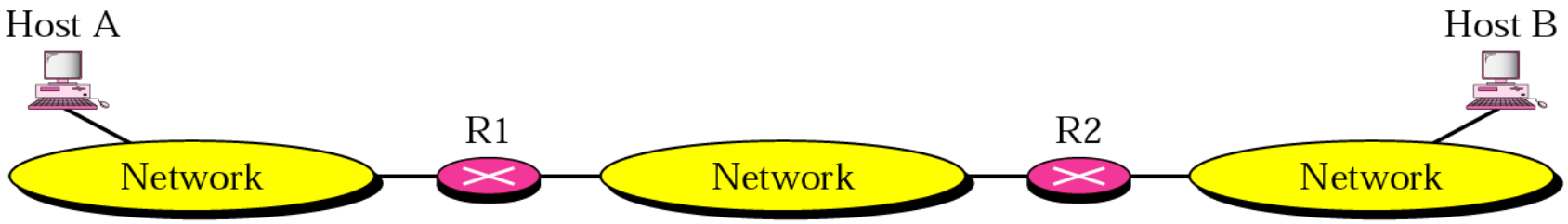
Routing table for R1

Destination	Route
Host B	R2, Host B

Routing table for R2

Destination	Route
Host B	Host B

a. Routing tables based on route



Routing table for host A

Destination	Next Hop
Host B	R1

Routing table for R1

Destination	Next Hop
Host B	R2

Routing table for R2

Destination	Next Hop
Host B	—

b. Routing tables based on next hop

Figure 19.29 Network-specific routing

Routing table for host S based on host-specific routing

Destination	Next Hop
A	R1
B	R1
C	R1
D	R1

Routing table for host S based on network-specific routing

Destination	Next Hop
N2	R1

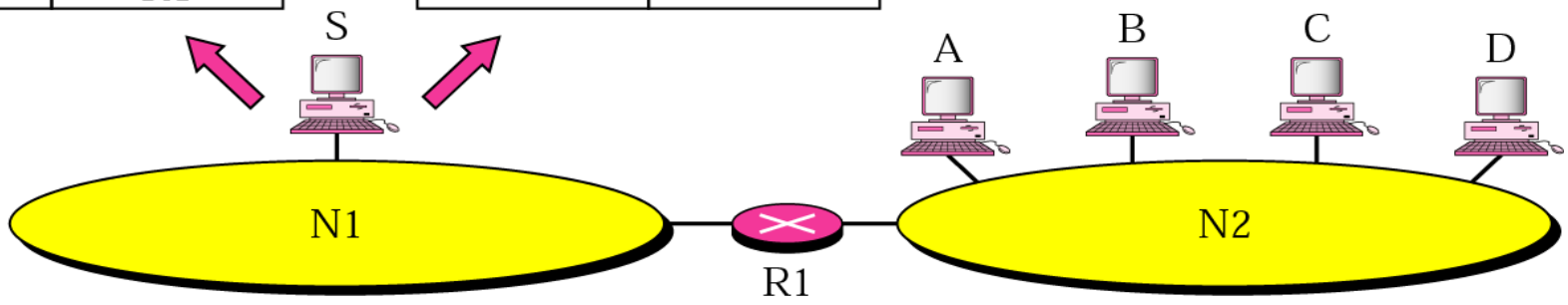


Figure 19.30 Host-specific routing

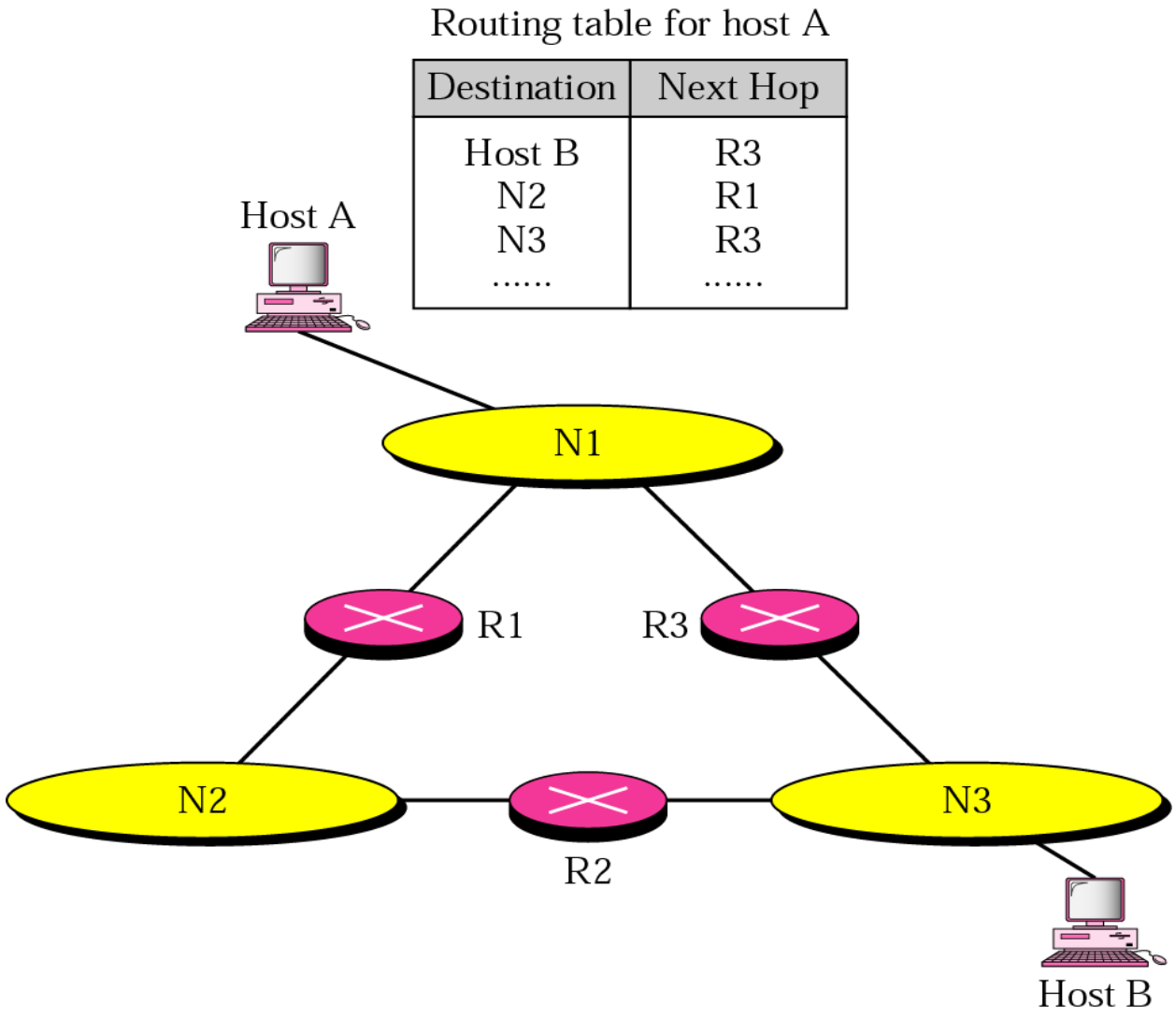


Figure 19.31 Default routing

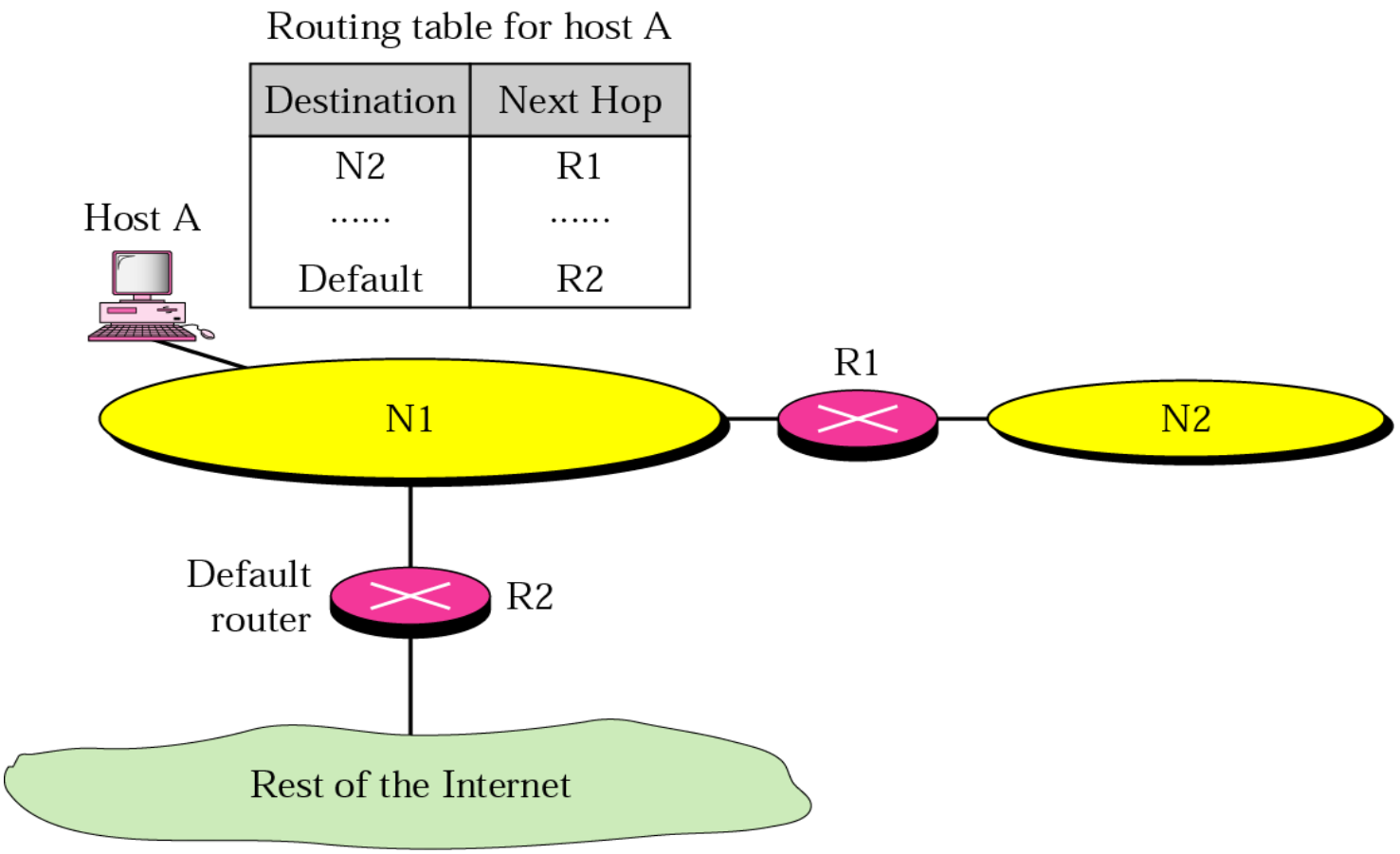


Figure 19.32 Classful addressing routing table

	Mask	Destination address	Next-hop address	Interface
	/8	14.0.0.0	118.45.23.8	m1
Host-specific →	/32	192.16.7.1	202.45.9.3	m0
	/24	193.14.5.0	84.78.4.12	m2
Default →	/0	/0	145.11.10.6	m0

Example 10

Using the table in Figure 19.32, the router receives a packet for destination 192.16.7.1. For each row, the mask is applied to the destination address until a match with the destination address is found. In this example, the router sends the packet through interface m0 (host specific).

Example 11

Using the table in Figure 19.32, the router receives a packet for destination 193.14.5.22. For each row, the mask is applied to the destination address until a match with the next-hop address is found. In this example, the router sends the packet through interface m2 (network specific).

Example 12

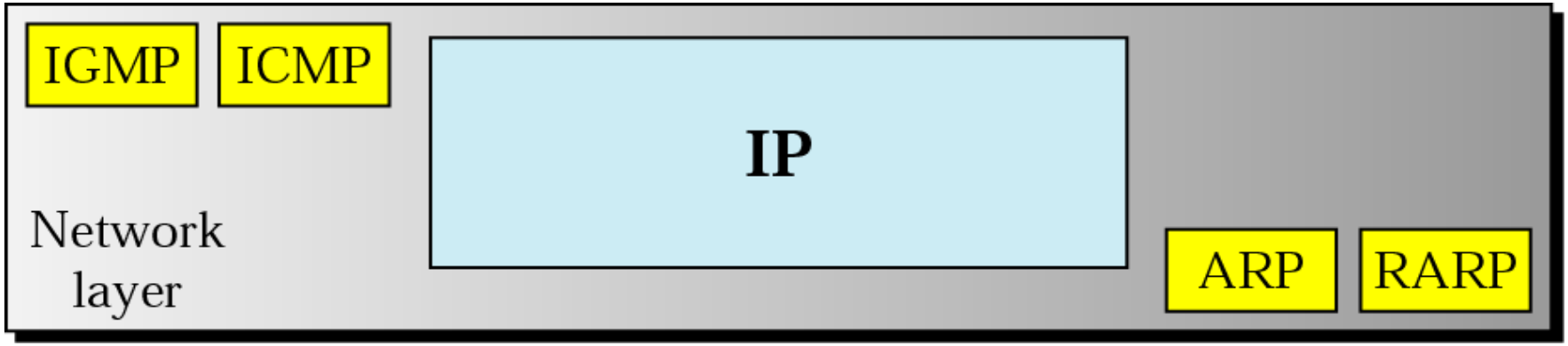
Using the table in Figure 19.32, the router receives a packet for destination 200.34.12.34. For each row, the mask is applied to the destination address, but no match is found. In this example, the router sends the packet through the default interface m0.

Chapter 15

Network Layer Protocols:

ARP, IPv4, ICMPv4, IPv6, and ICMPv6

Figure 20.1 **Protocols at network layer**



20.1 ARP

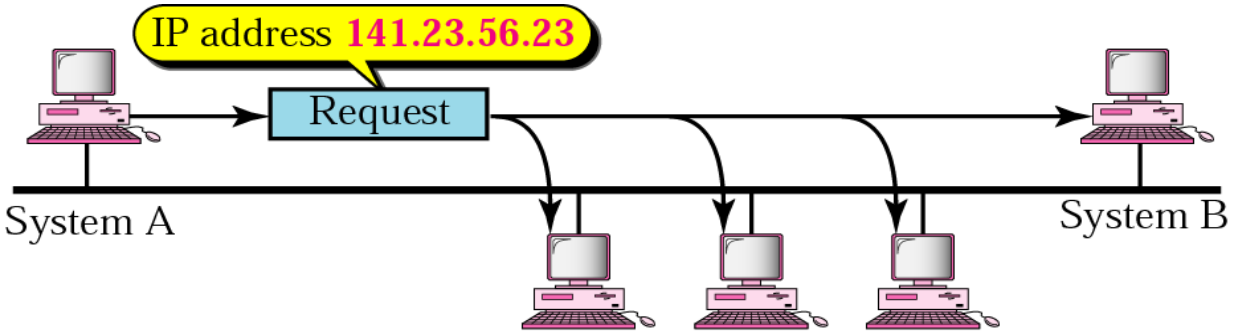
Mapping

Packet Format

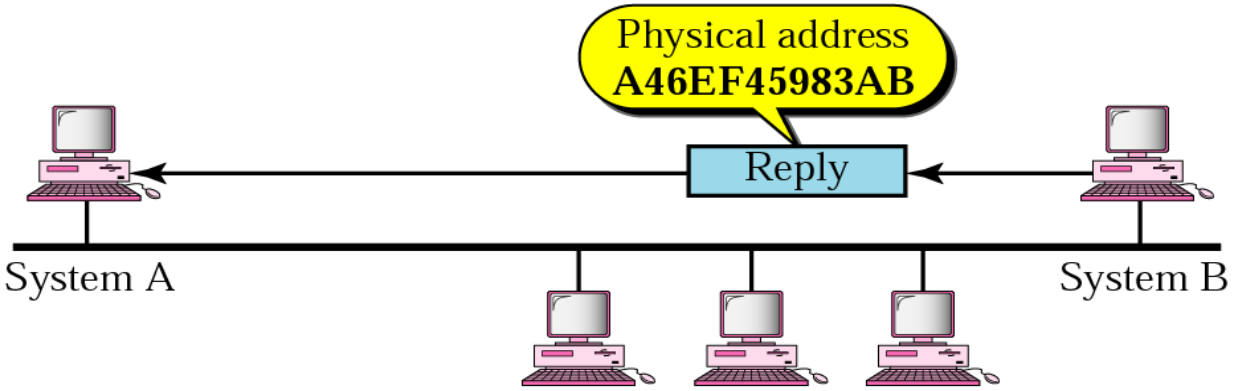
Encapsulation

Operation

Figure 20.2 ARP operation



a. ARP request is broadcast



b. ARP reply is unicast

Figure 20.3 ARP packet

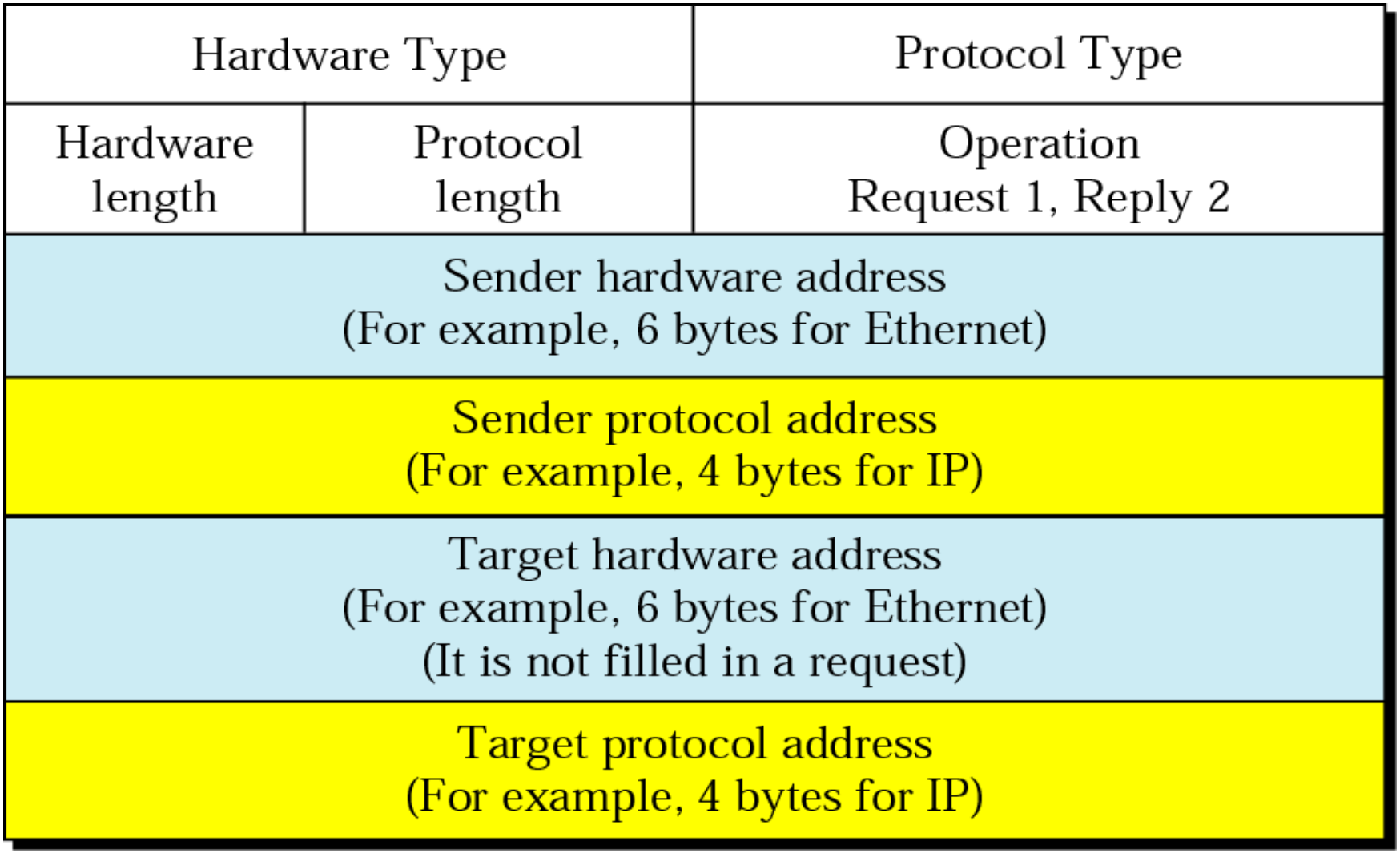


Figure 20.4 Encapsulation of ARP packet

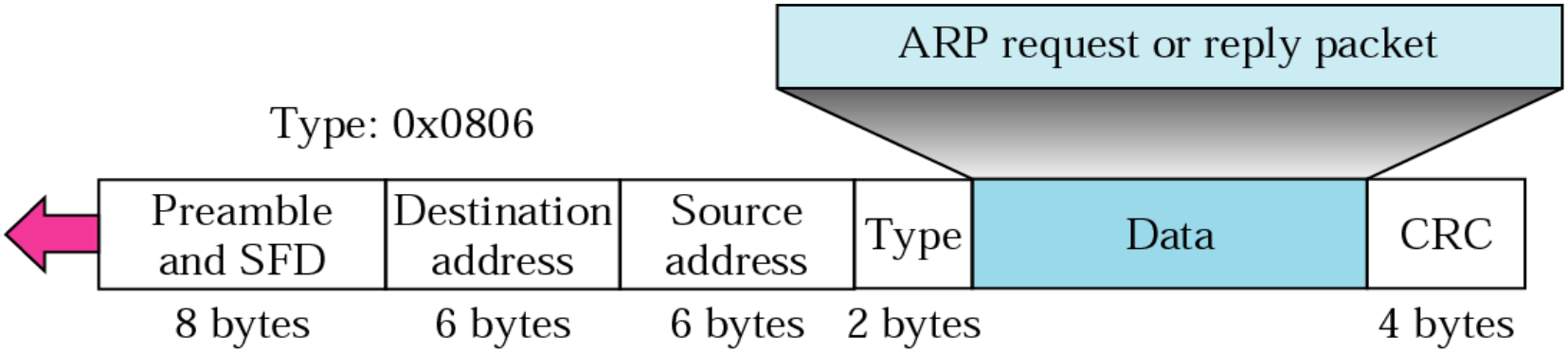
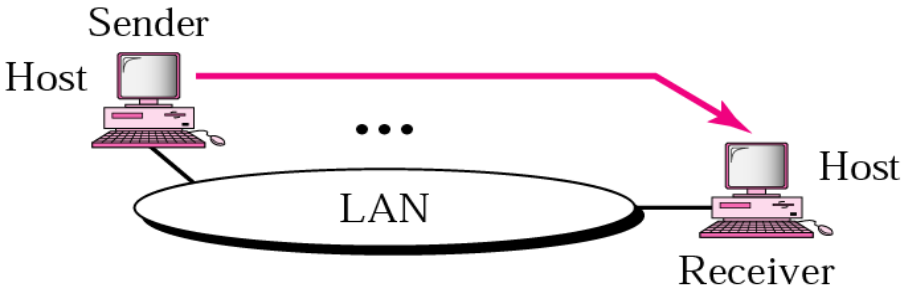
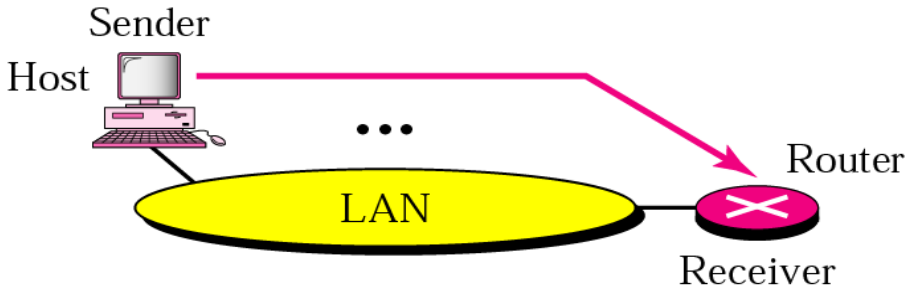


Figure 20.5 Four cases using ARP



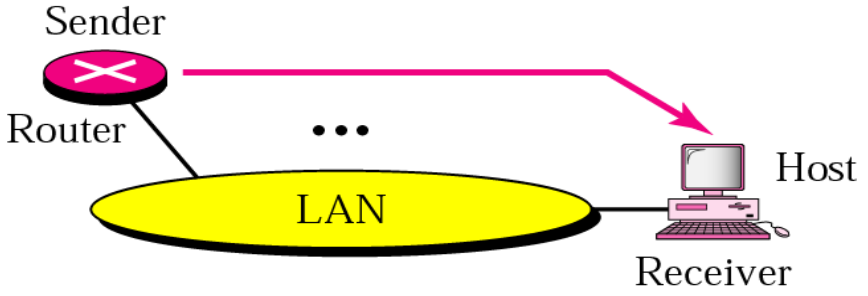
Case 1. A host has a packet to send to another host on the same network.



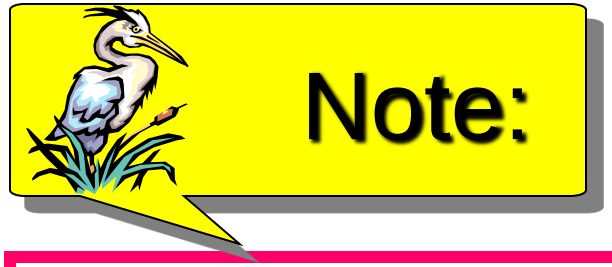
Case 2. A host wants to send a packet to another host on another network.
It must first be delivered to the appropriate router.



Case 3. A router receives a packet to be sent to a host on another network.
It must first be delivered to the appropriate router.



Case 4. A router receives a packet to be sent to a host on the same network.



An ARP request is broadcast; an ARP reply is unicast.

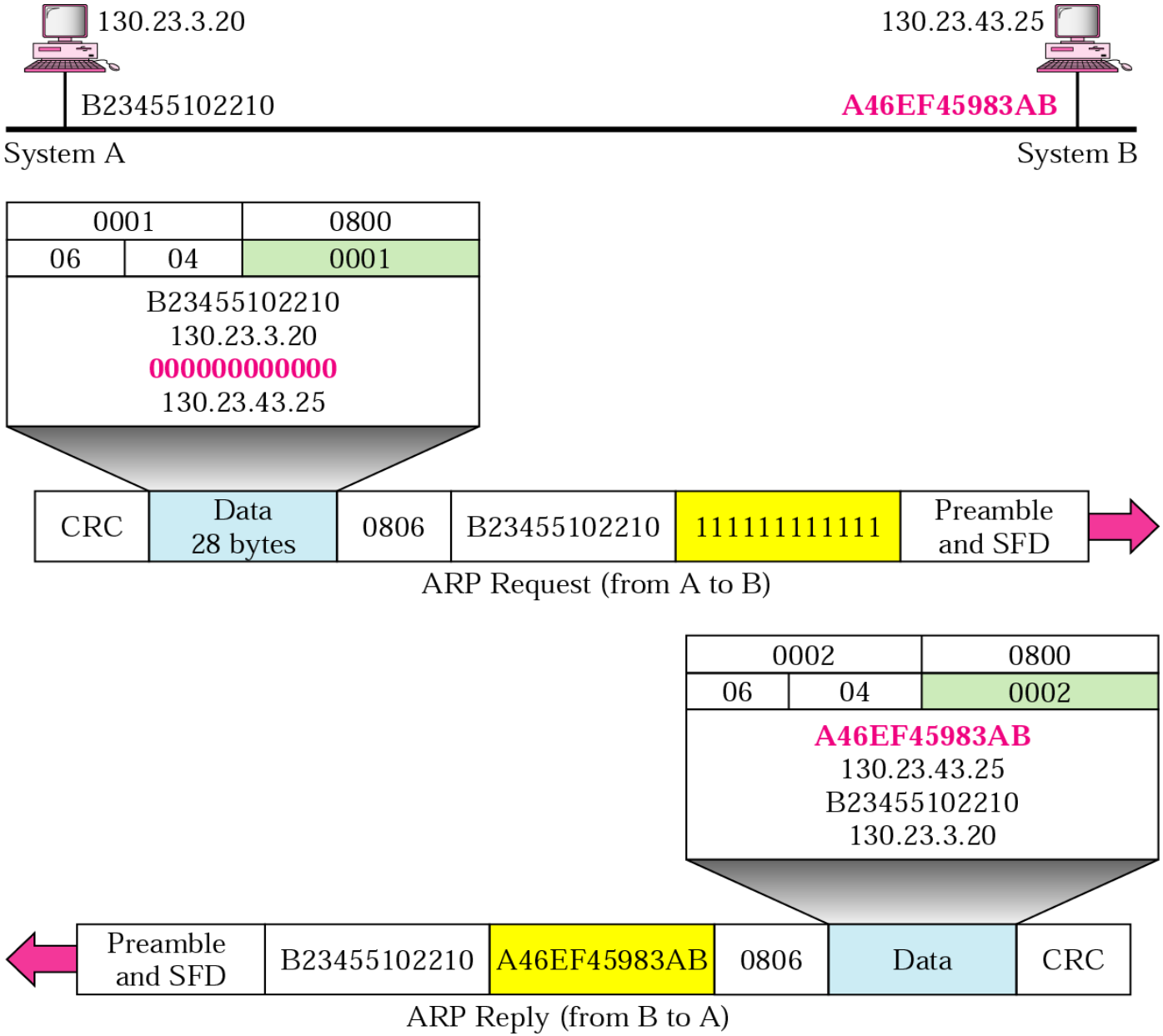
Example 1

A host with IP address 130.23.3.20 and physical address B23455102210 has a packet to send to another host with IP address 130.23.43.25 and physical address A46EF45983AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.

Solution

Figure 20.6 shows the ARP request and reply packets. Note that the ARP data field in this case is 28 bytes, and that the individual addresses do not fit in the 4-byte boundary. That is why we do not show the regular 4-byte boundaries for these addresses. Note that we use hexadecimal for every field except the IP addresses.

Figure 20.6 Example 1

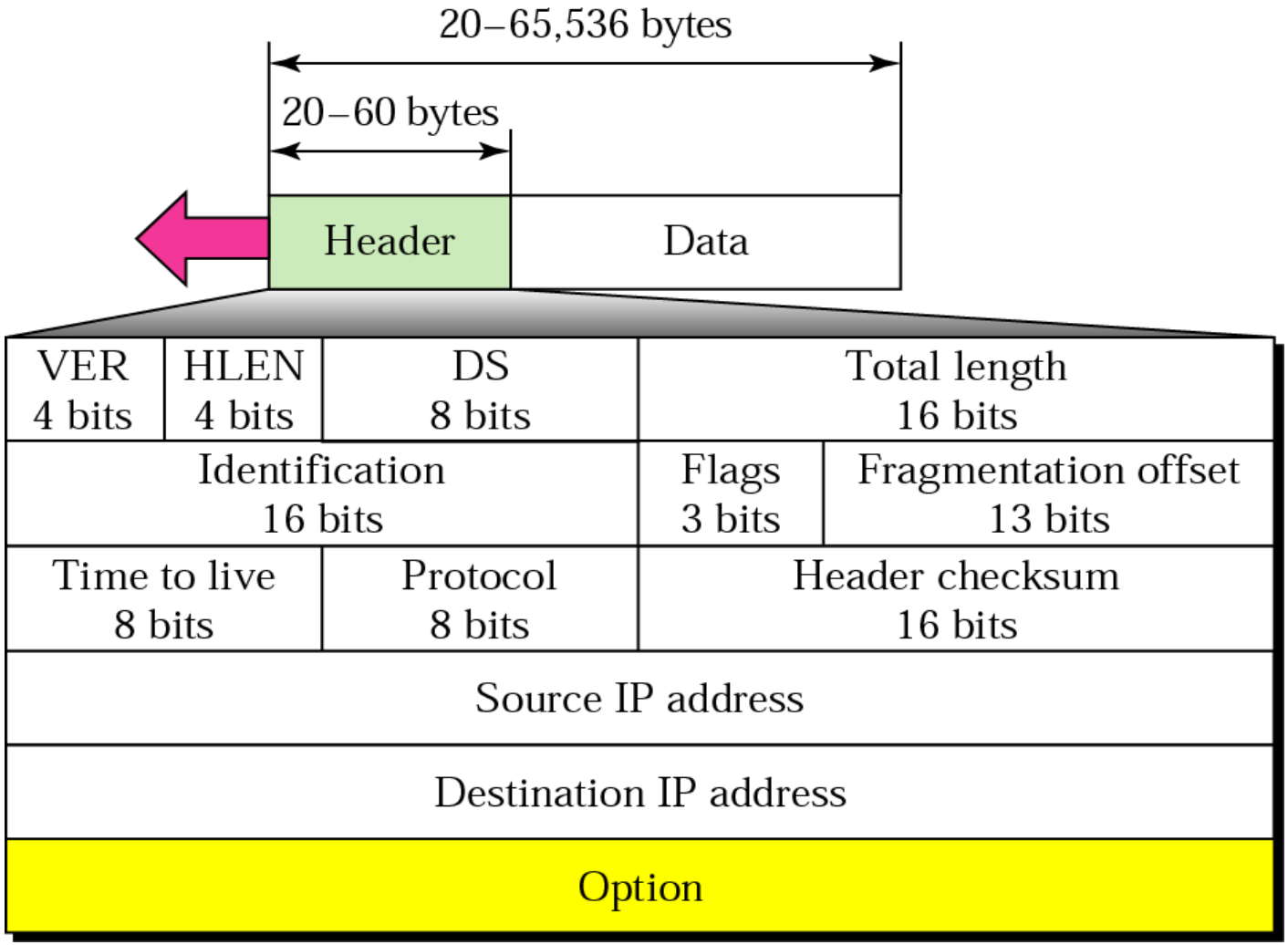


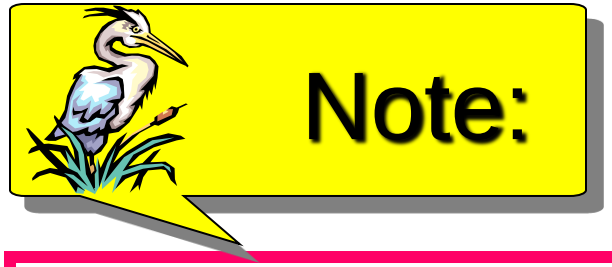
20.2 IP

Datagram

Fragmentation

Figure 20.7 IP datagram





The total length field defines the total length of the datagram including the header.

Figure 20.8 Multiplexing

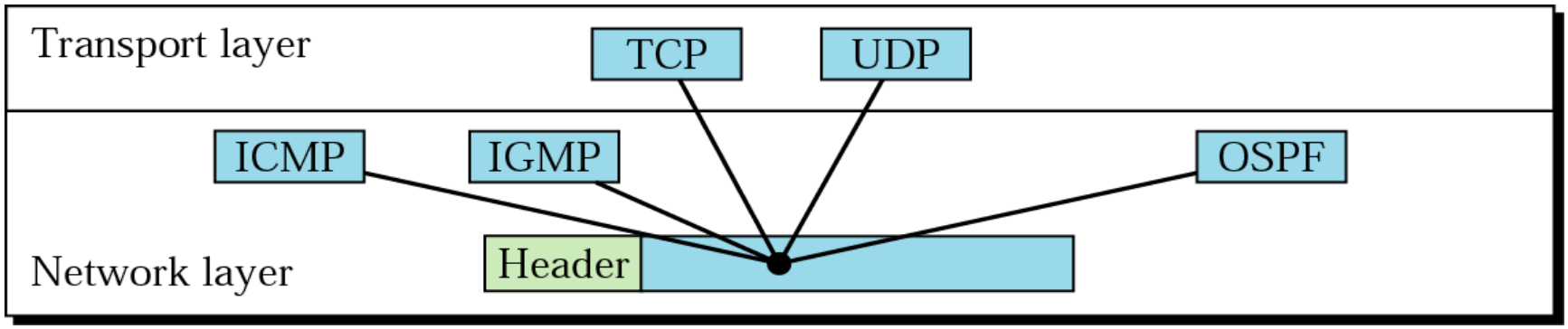


Figure 20.9 Example of checksum calculation

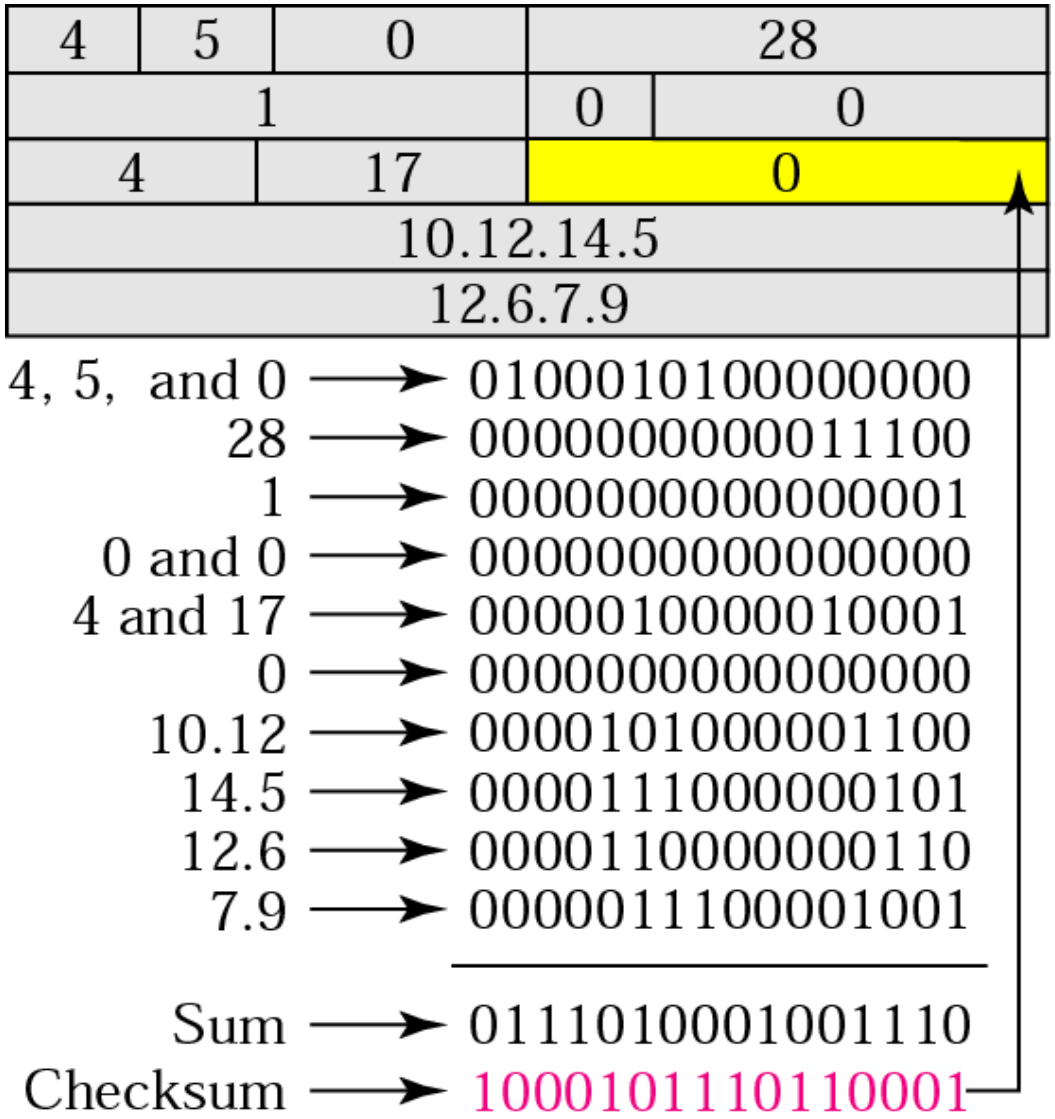


Figure 20.10 MTU

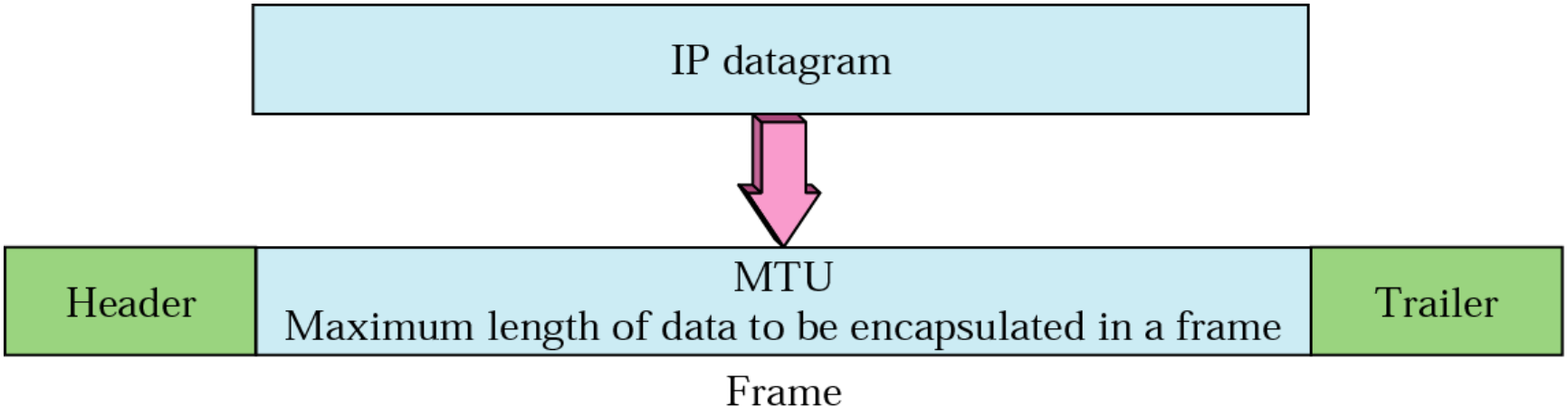
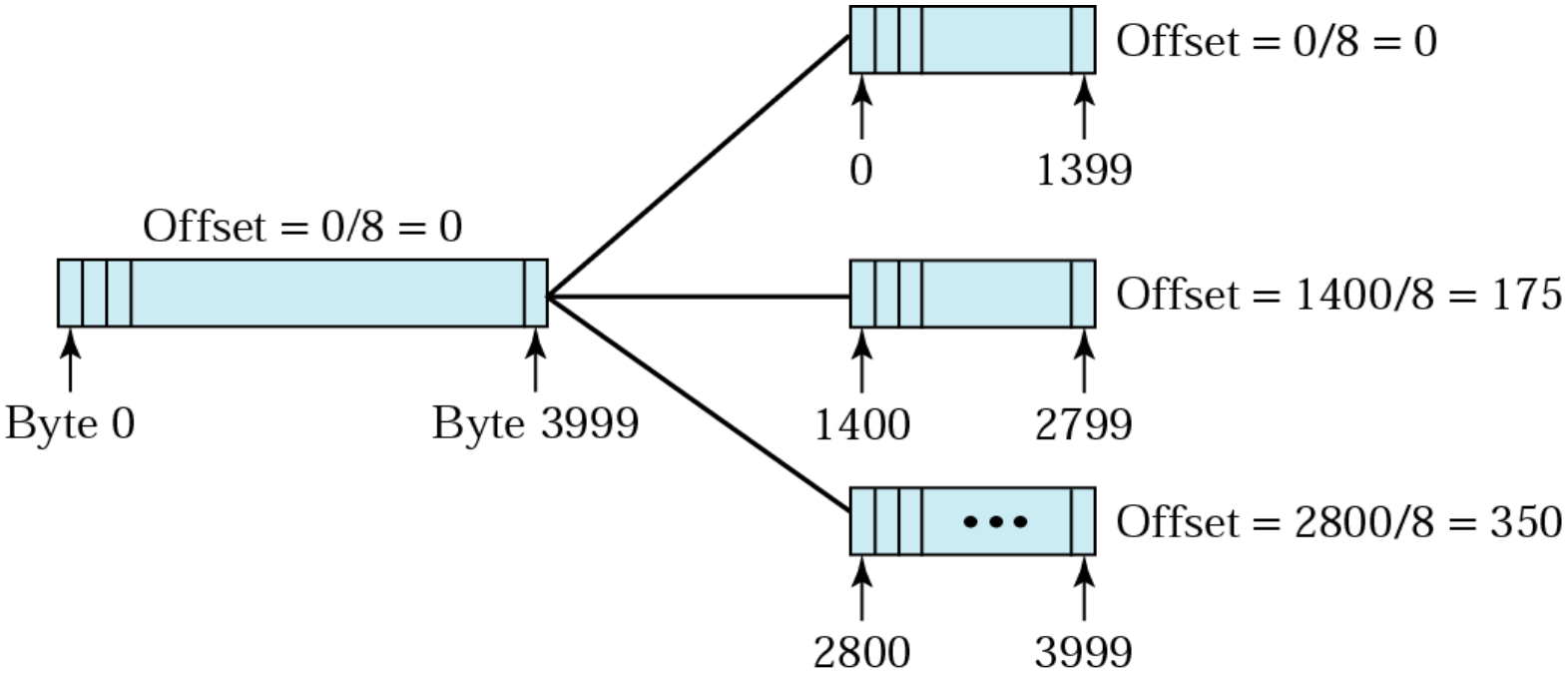


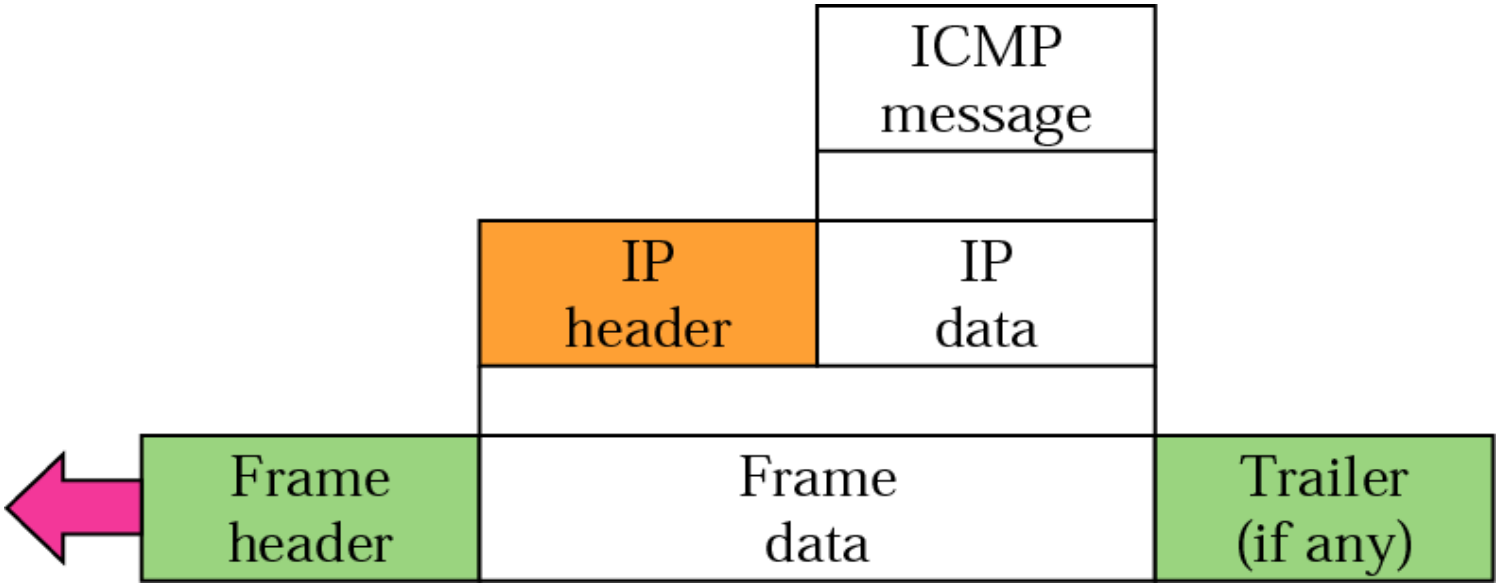
Figure 20.11 Fragmentation example



20.3 ICMP

Types of Messages

Figure 20.12 ICMP encapsulation



Chapter 21

Unicast and Multicast Routing:

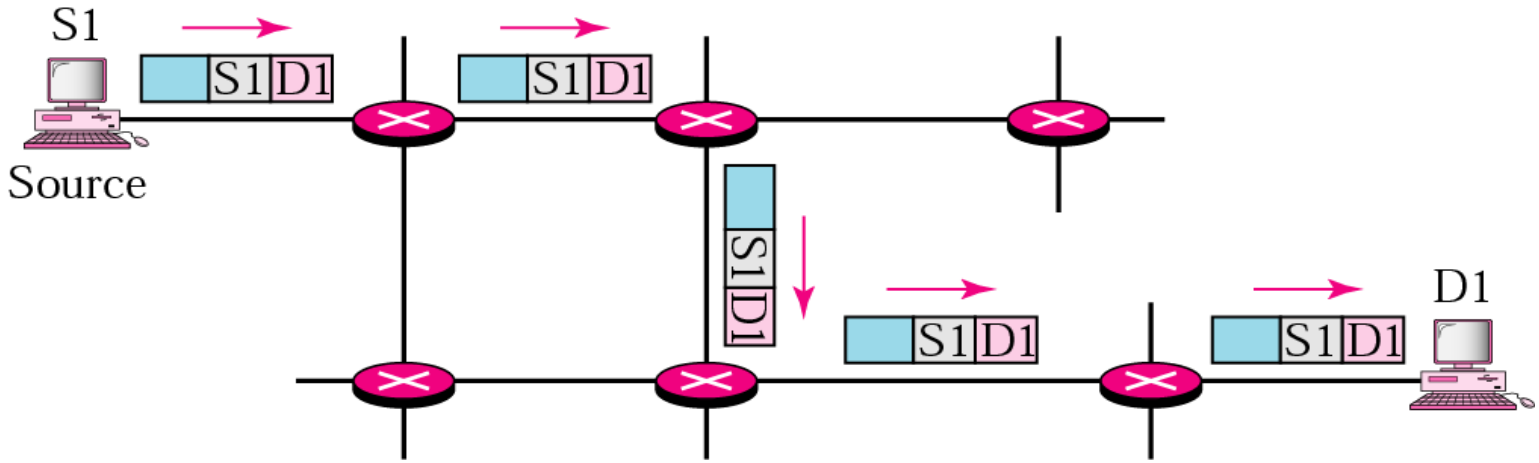
Routing Protocols

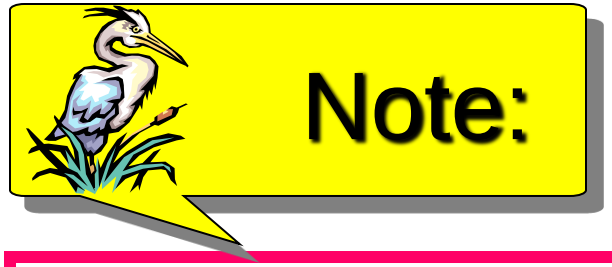
21.1 Unicast Routing

Metric

Interior and Exterior Routing

Figure 21.1 Unicasting





In unicast routing, the router forwards the received packet through only one of its ports.

21.2 Unicast Routing Protocols

RIP

OSPF

BGP

Figure 21.2 *Popular routing protocols*

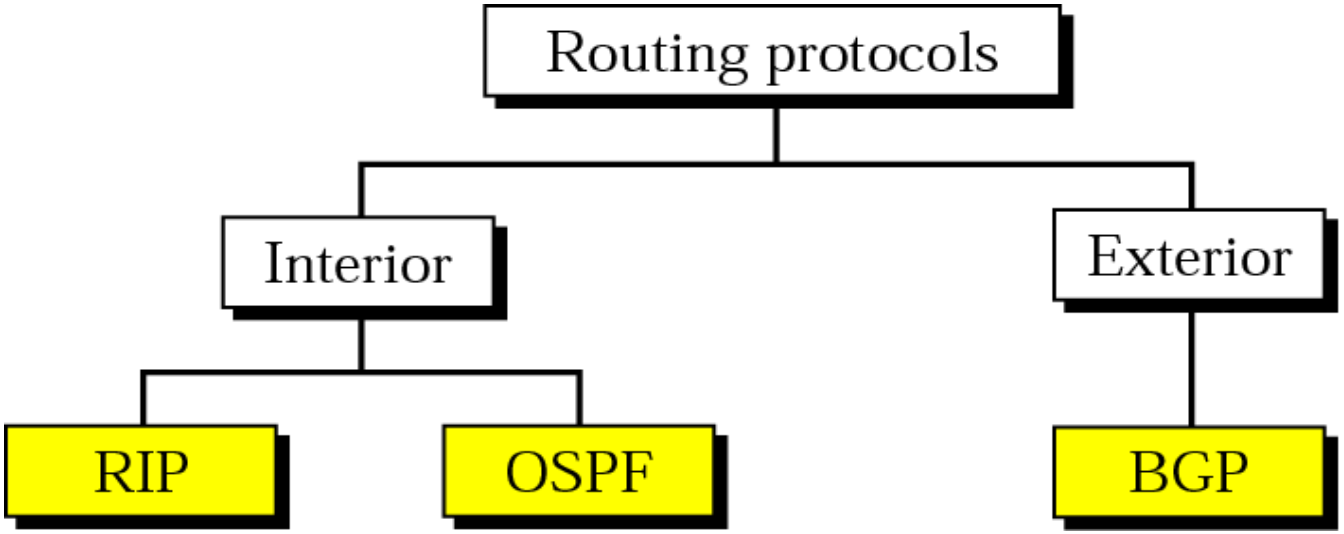


Figure 21.3 Autonomous systems

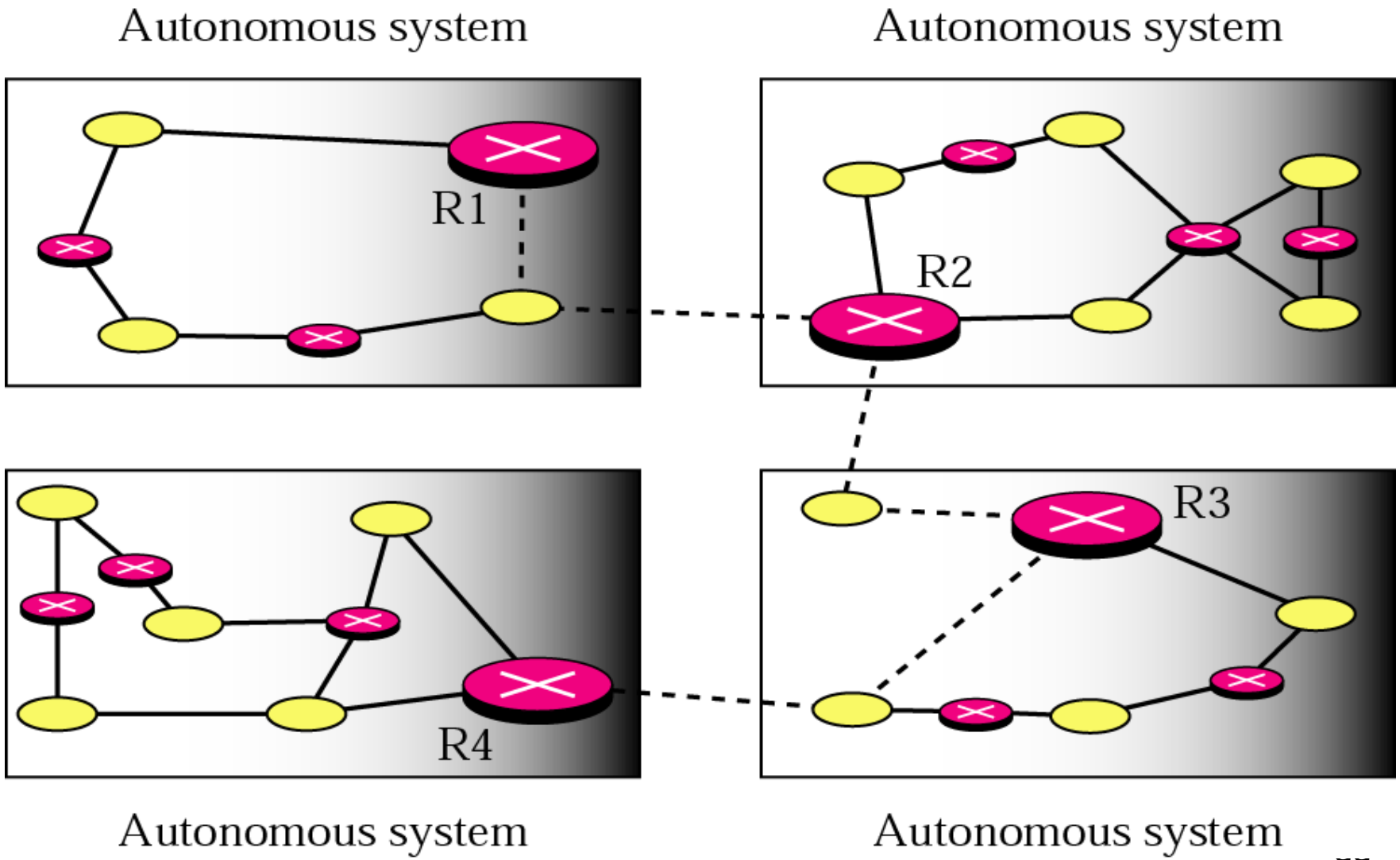


Table 21.1 A distance vector routing table

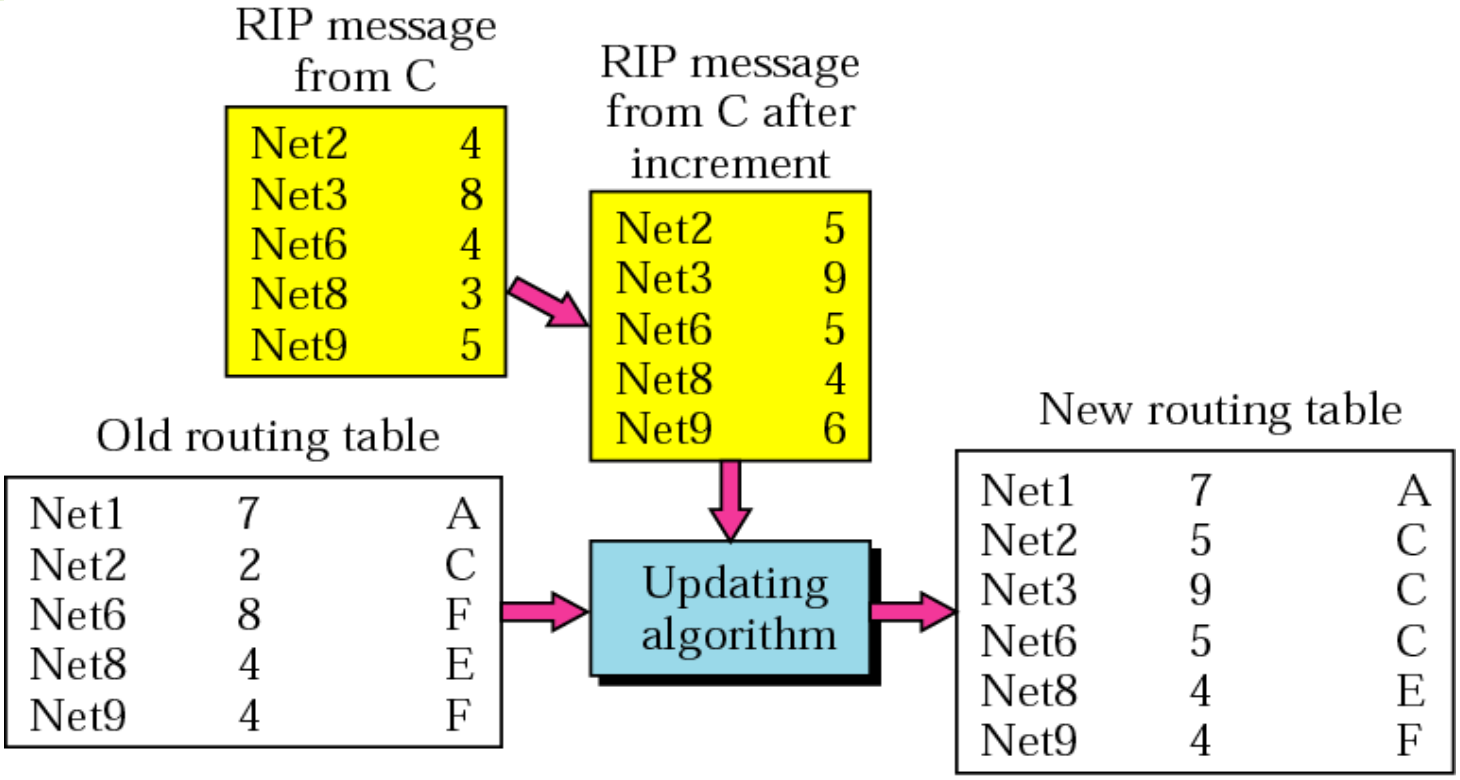
Destination	Hop Count	Next Router	Other information
163.5.0.0	7	172.6.23.4	
197.5.13.0	5	176.3.6.17	
189.45.0.0	4	200.5.1.6	
115.0.0.0	6	131.4.7.19	

RIP Updating Algorithm

Receive: a response RIP message

1. Add one hop to the hop count for each advertised destination.
2. Repeat the following steps for each advertised destination:
  1. If (destination not in the routing table)
    1. Add the advertised information to the table.
  2. Else
    1. If (next-hop field is the same)
      1. Replace entry in the table with the advertised one.
    2. Else
      1. If (advertised hop count smaller than one in the table)
        1. Replace entry in the routing table.
3. Return.

Figure 21.4 Example of updating a routing table



Net1: No news, do not change
Net2: Same next hop, replace
Net3: A new router, add
Net6: Different next hop, new hop count smaller, replace
Net8: Different next hop, new hop count the same, do not change
Net9: Different next hop, new hop count larger, do not change

Figure 21.5 Initial routing tables in a small autonomous system

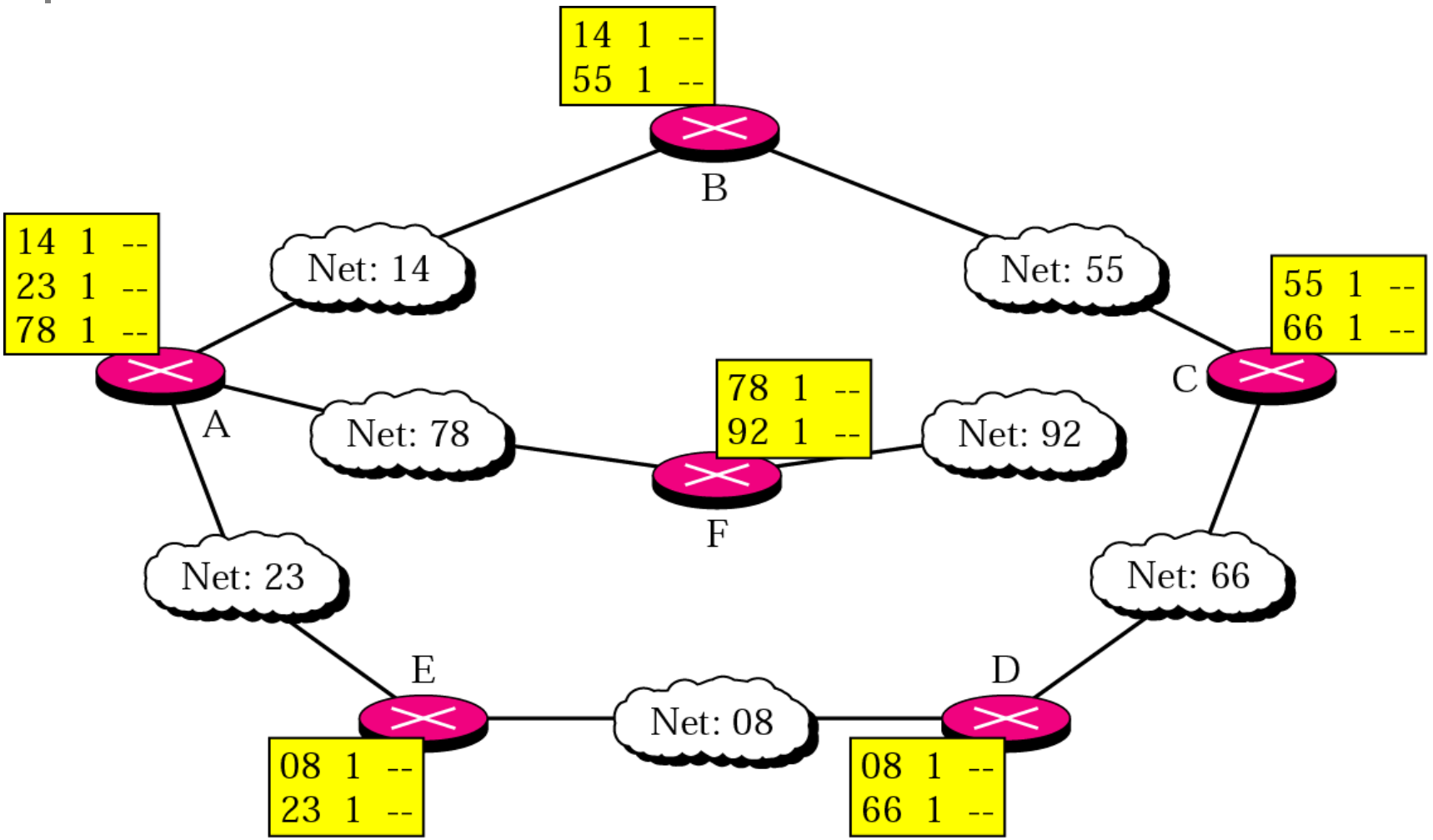


Figure 21.6 Final routing tables for Figure 21.5

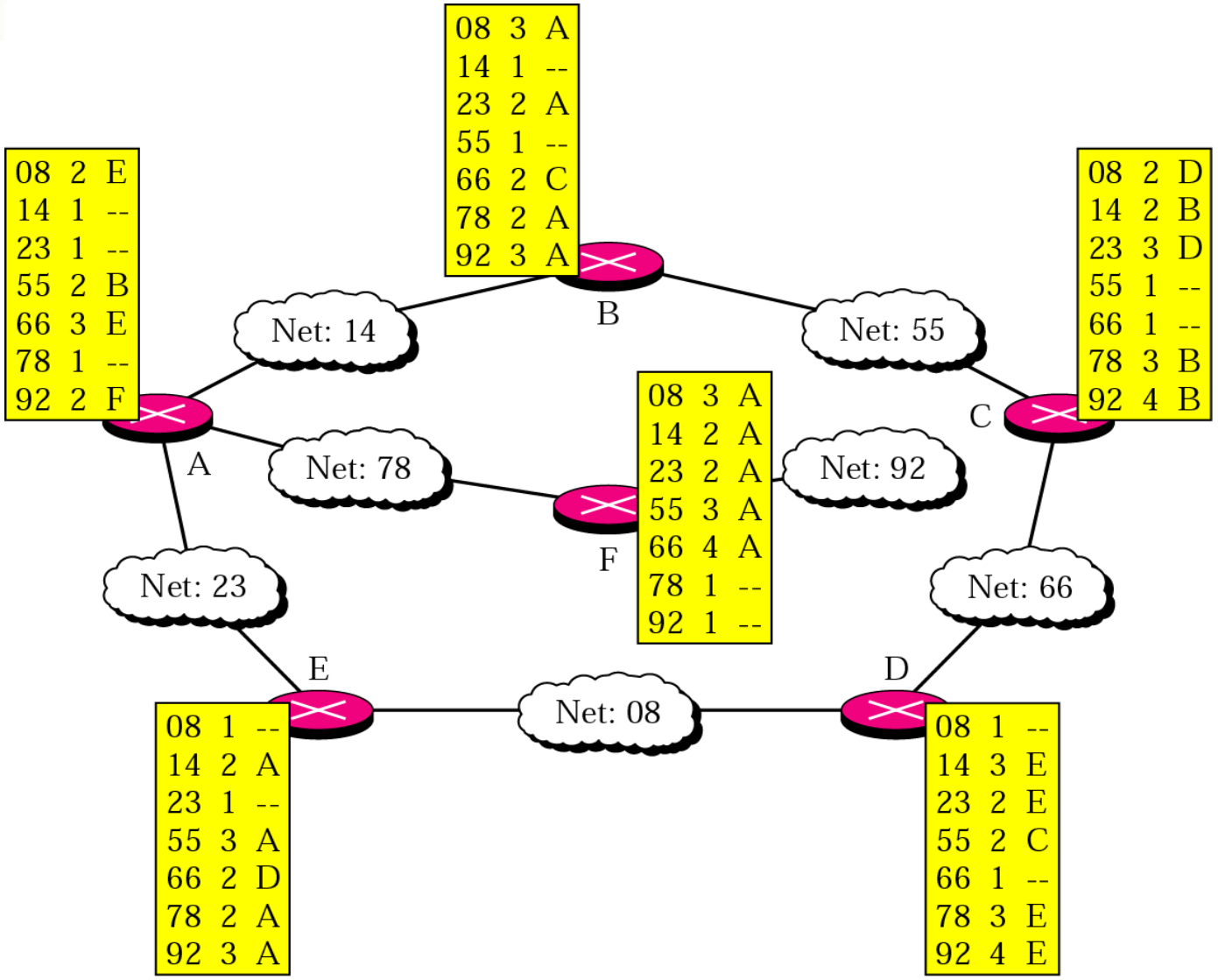


Figure 21.7 Areas in an autonomous system

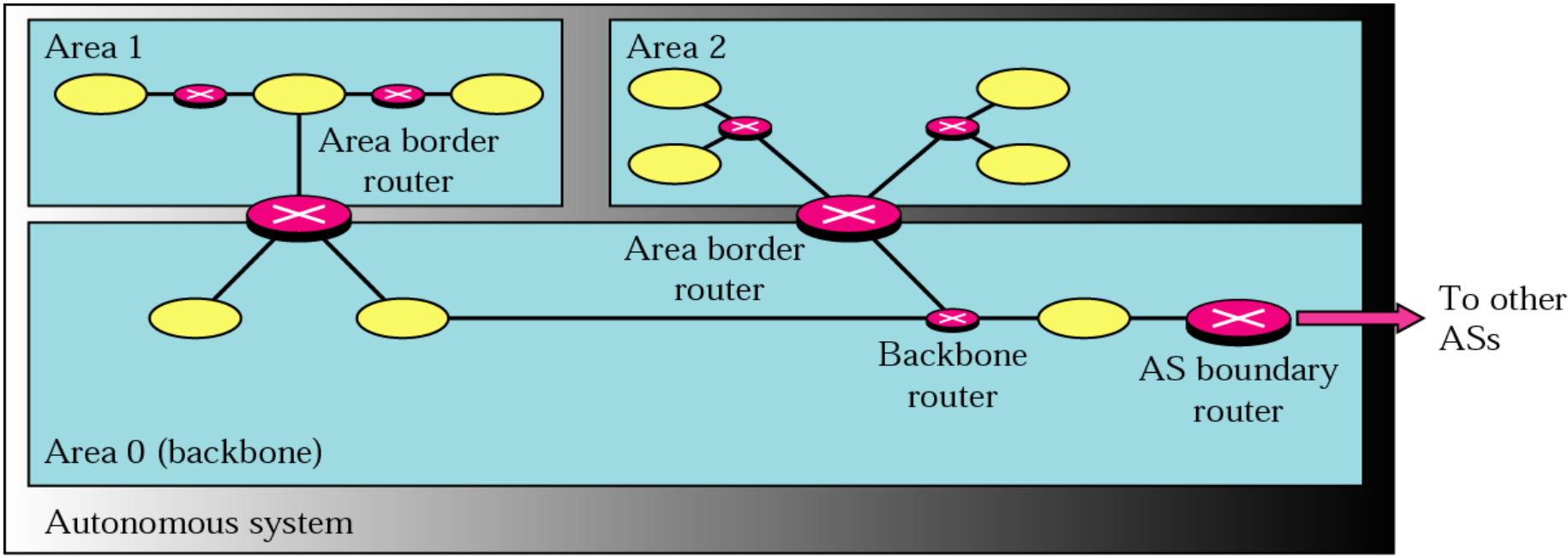


Figure 21.8 **Types of links**

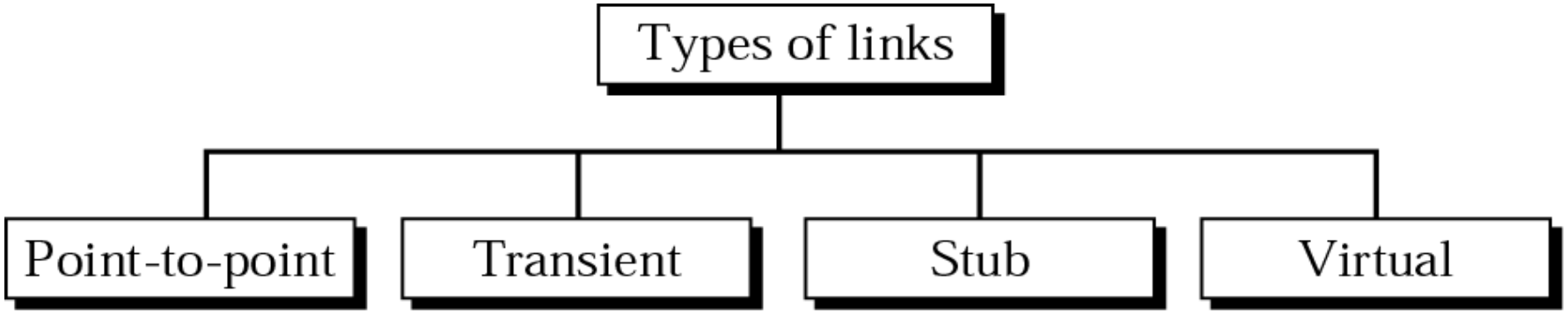


Figure 21.9 Point-to-point link

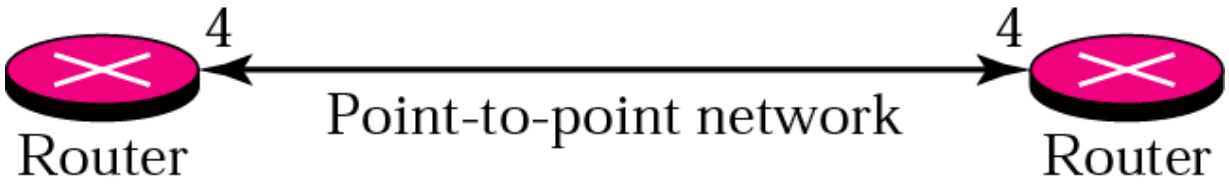
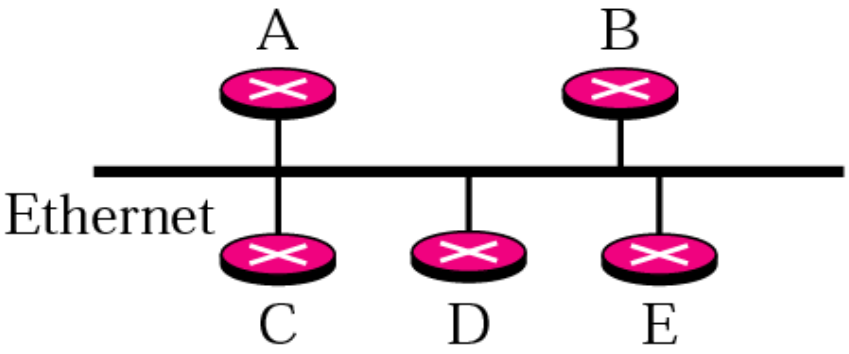
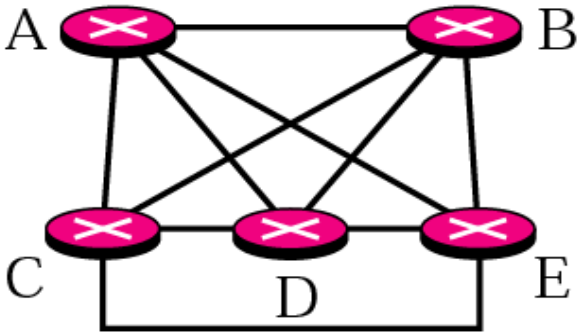


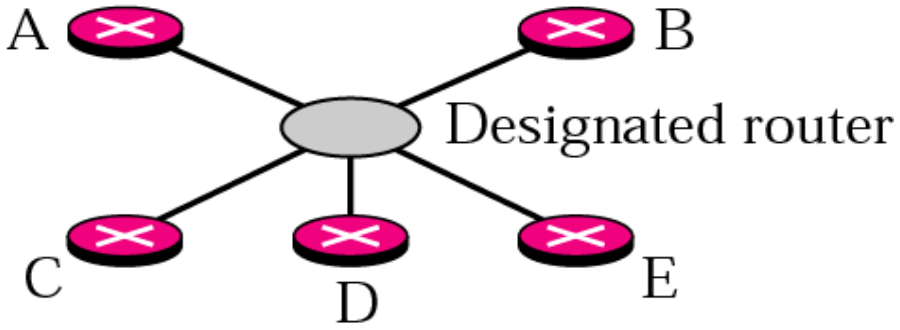
Figure 21.10 Transient link



a. Transient network

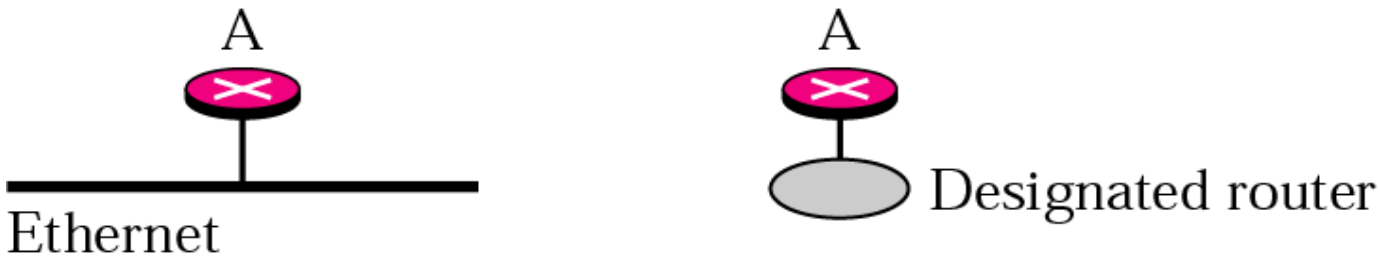


b. Unrealistic representation



c. Realistic representation

Figure 21.11 Stub link



a. Stub network b. Representation

Figure 21.12 Example of an internet

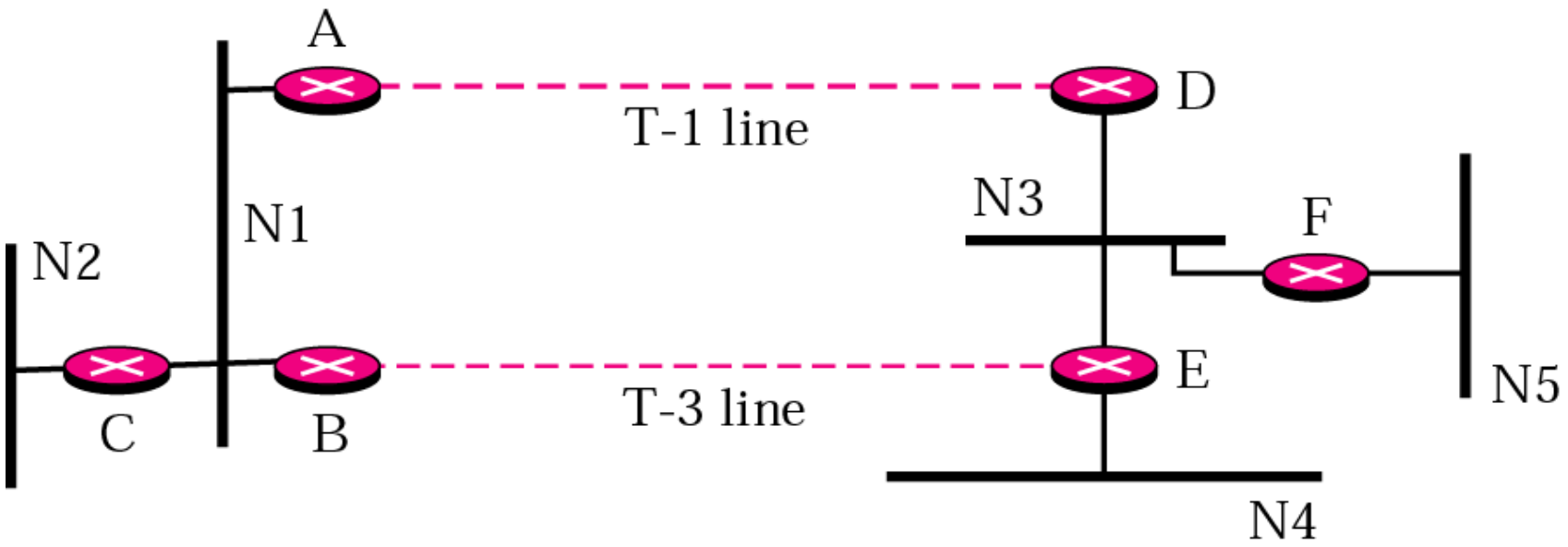


Figure 21.13 Graphical representation of an internet

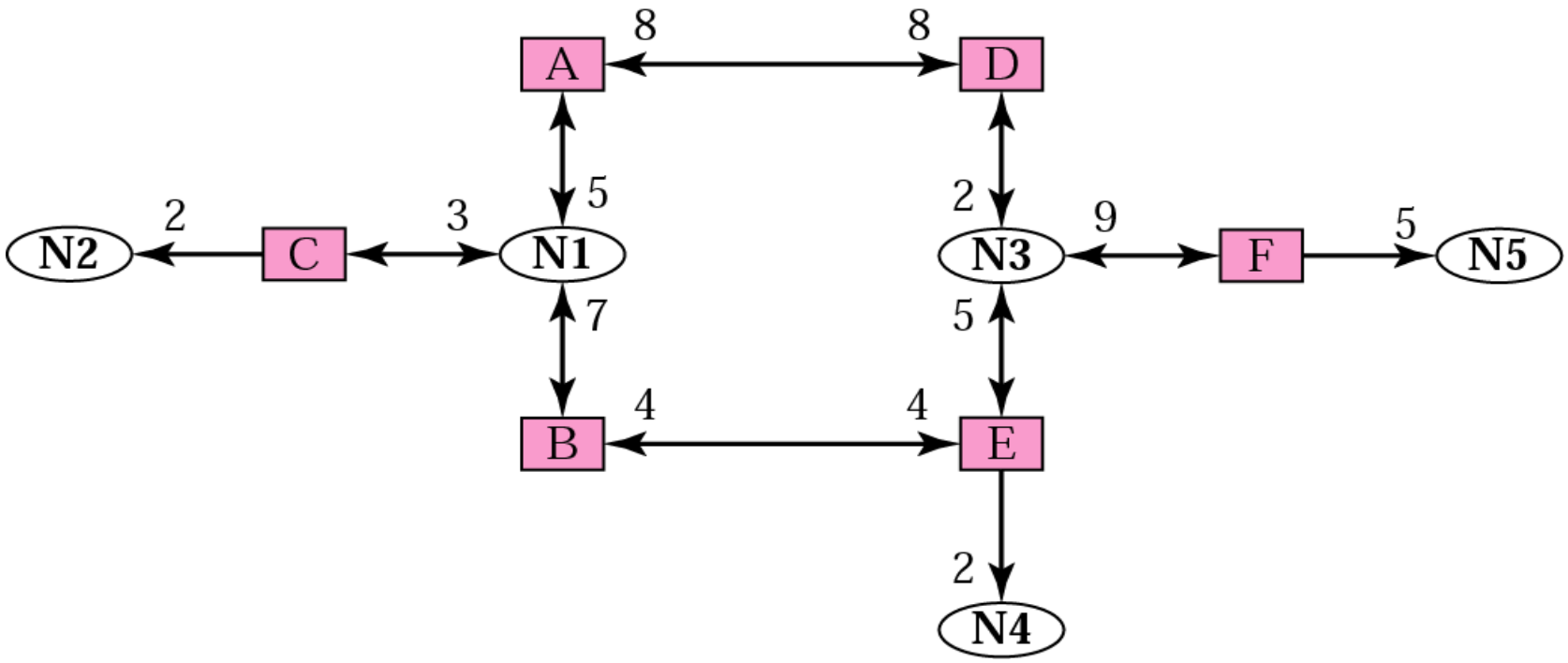


Figure 21.14 Types of LSAs

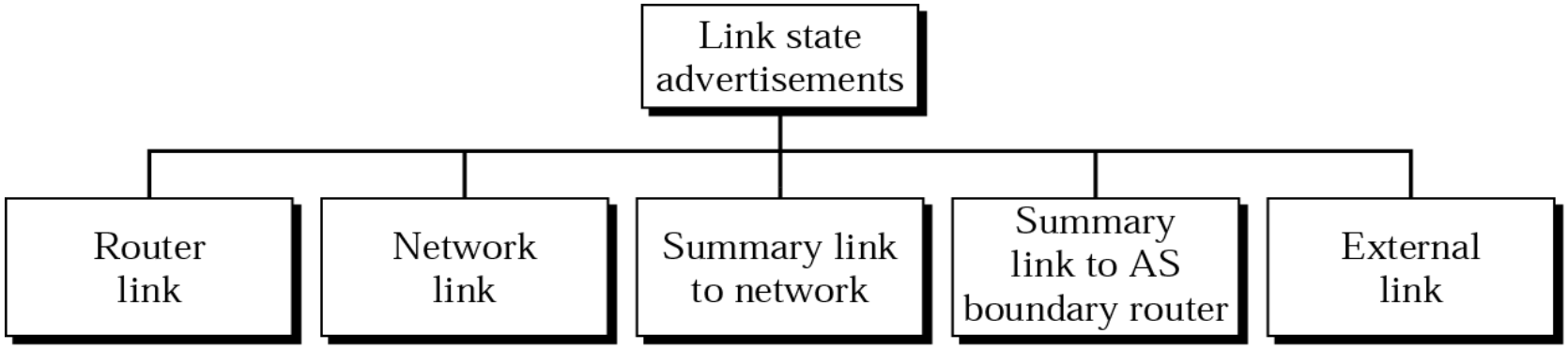


Figure 21.15 **Router link**

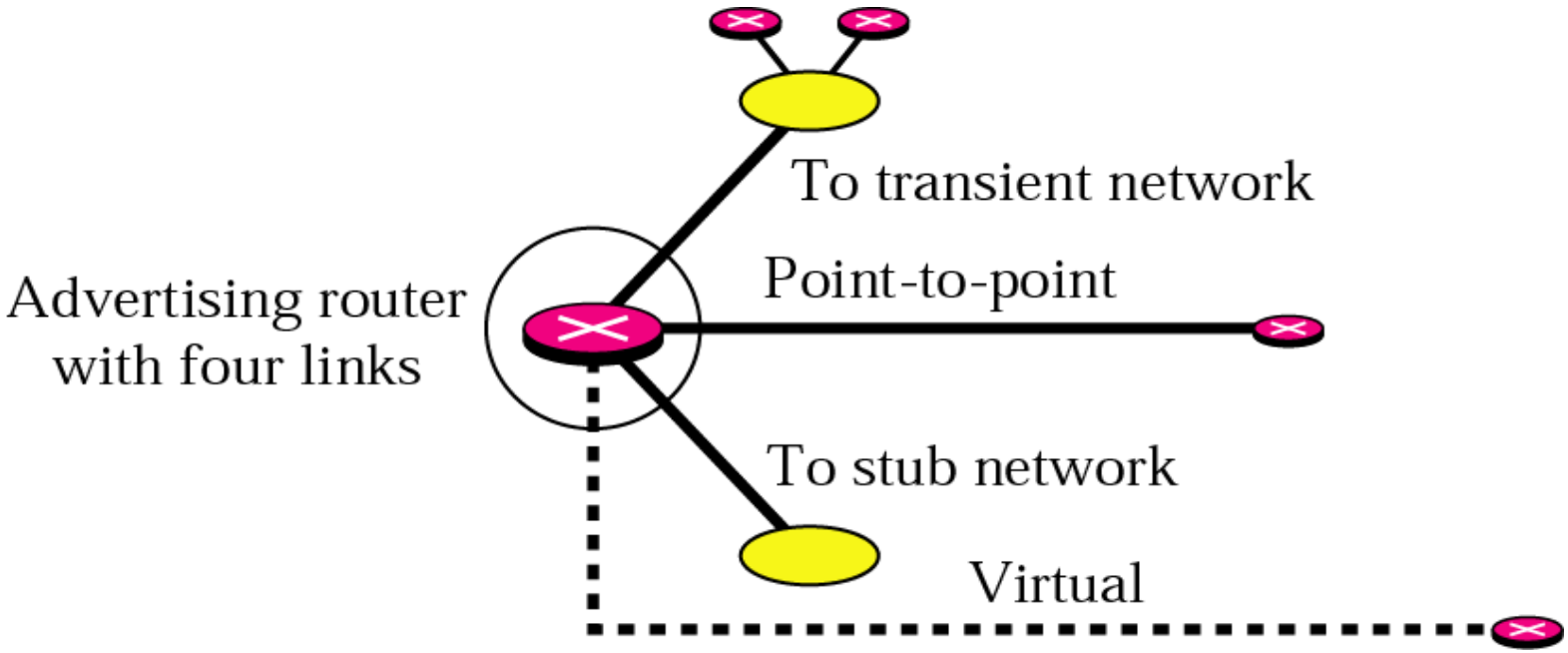


Figure 21.16 **Network link**

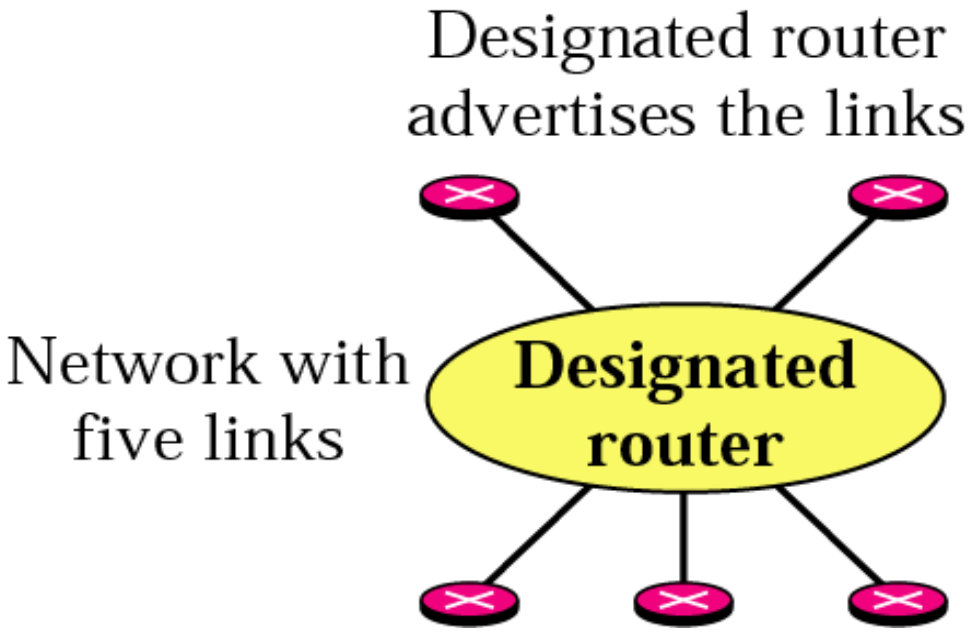


Figure 21.17 Summary link to network

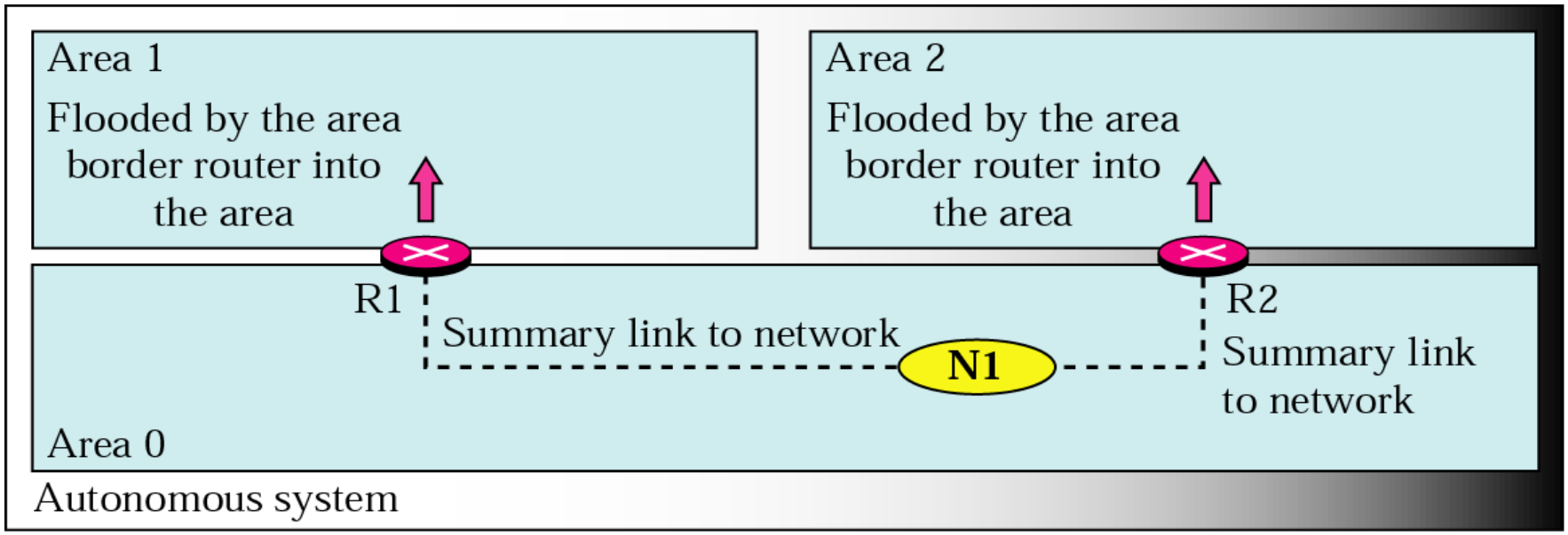


Figure 21.18 Summary link to AS boundary router

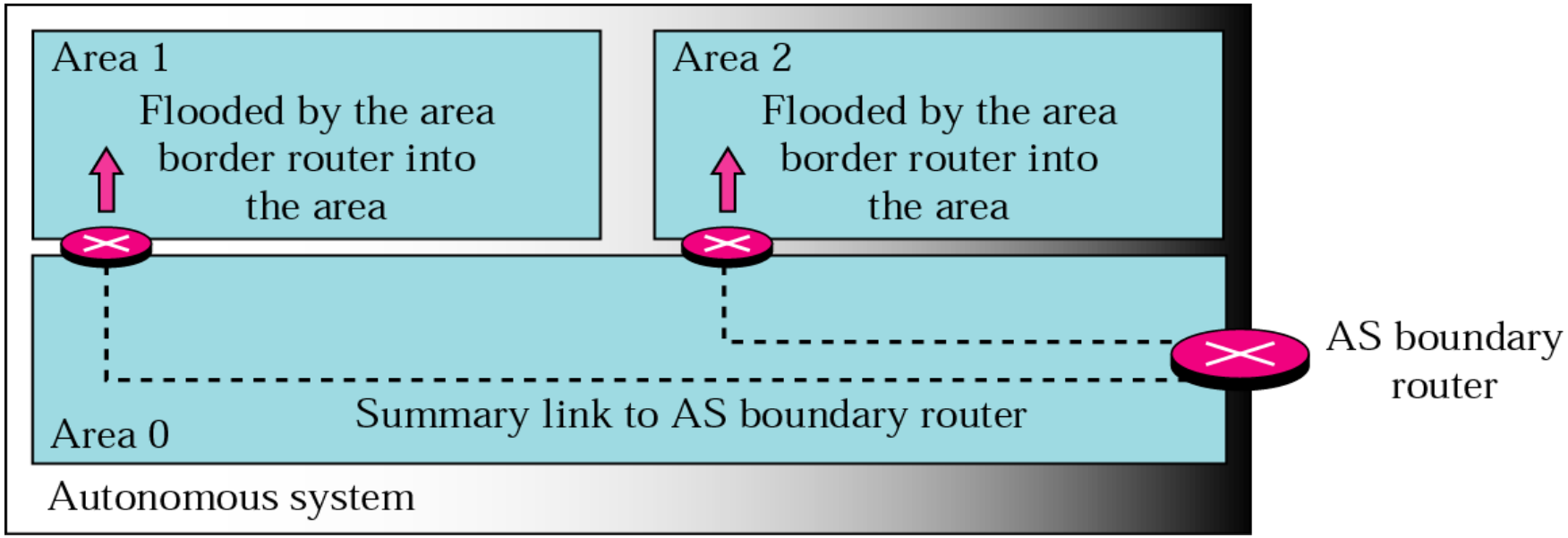
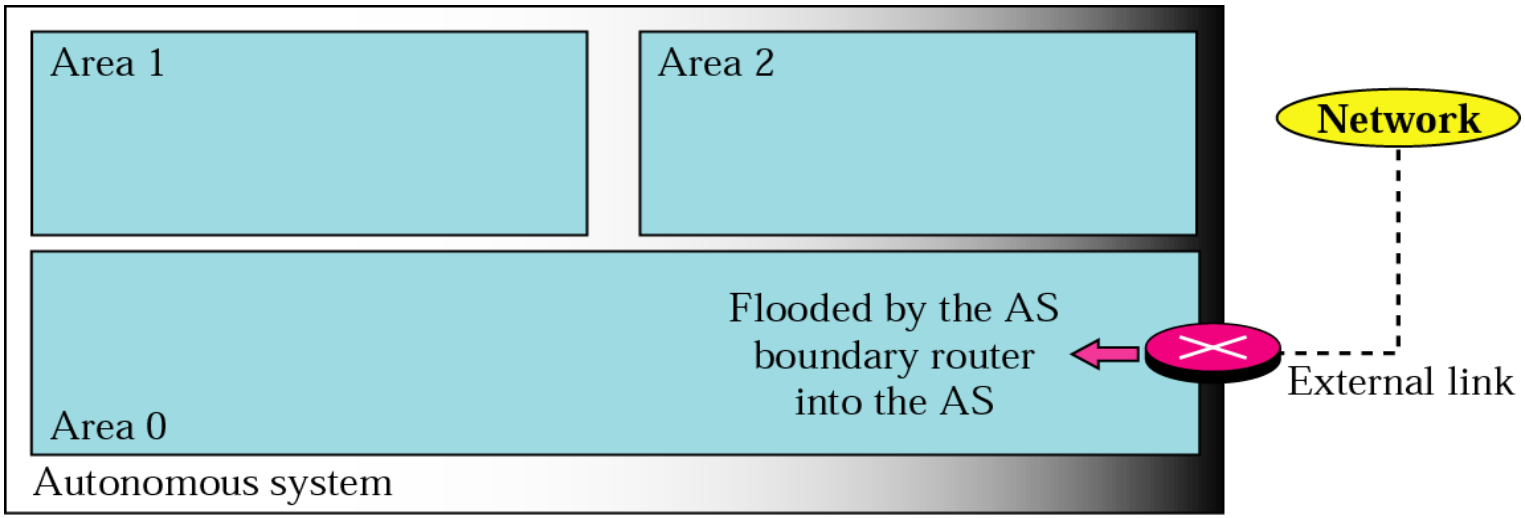
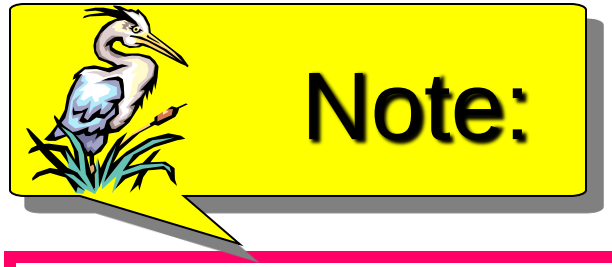


Figure 21.19 External link





In OSPF, all routers have the same link state database.

Dijkstra Algorithm

1. Start with the local node (router): the root of the tree.
2. Assign a cost of 0 to this node and make it the first permanent node.
3. Examine each neighbor node of the node that was the last permanent node.
4. Assign a cumulative cost to each node and make it tentative.
5. Among the list of tentative nodes
    1. Find the node with the smallest cumulative cost and make it permanent.
    2. If a node can be reached from more than one direction
        1. Select the direction with the shortest cumulative cost.
6. Repeat steps 3 to 5 until every node becomes permanent.

Figure 21.20 Shortest-path calculation

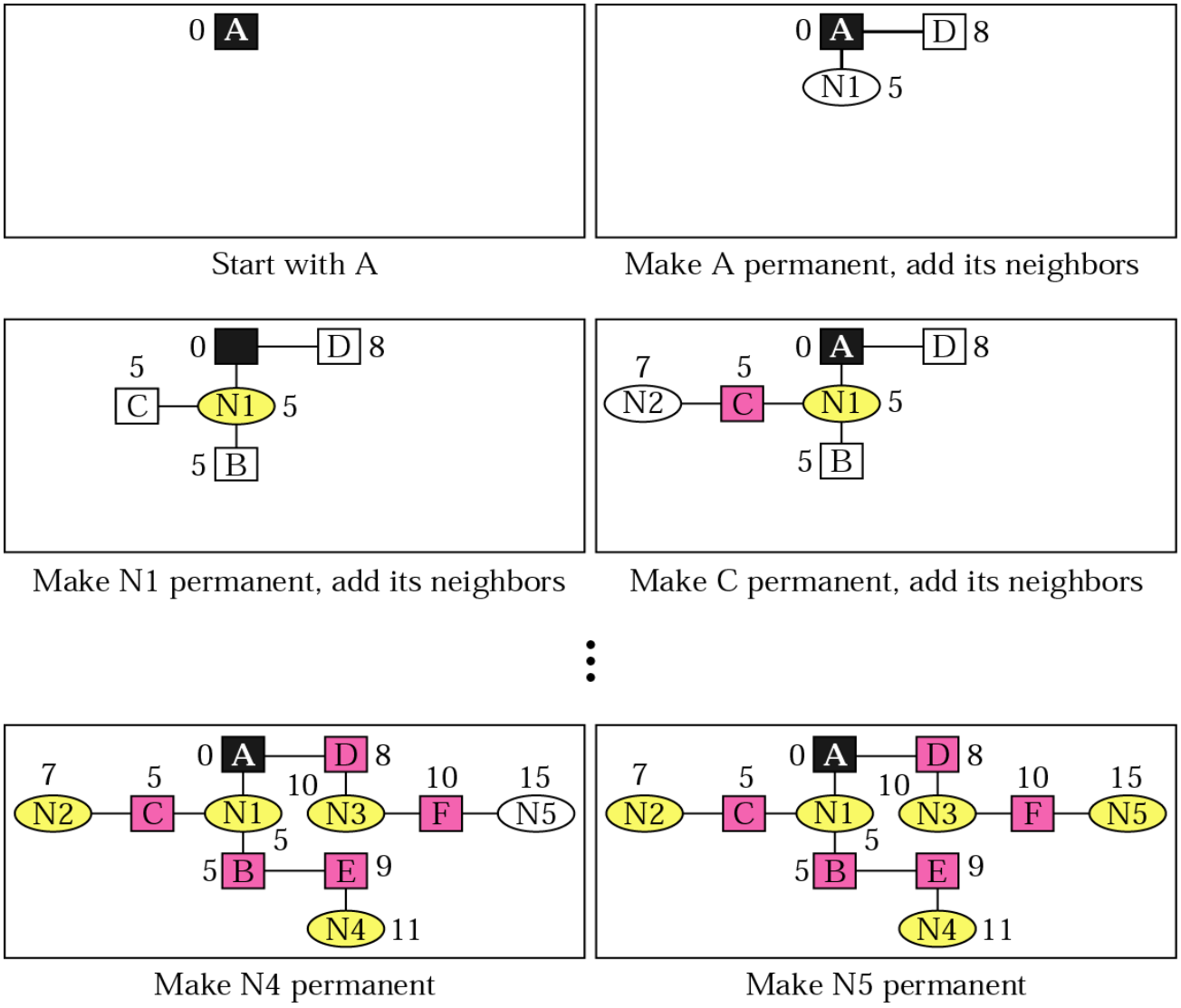


Table 21.2 Link state routing table for router A

Network	Cost	Next Router	Other Information
N1	5	C	
N2	7	D	
N3	10	B	
N4	11	D	
N5	15	C	

Table 21.3 Path vector routing table

Network	Next Router	Path
N01	R01	AS14, AS23, AS67
N02	R05	AS22, AS67, AS05, AS89
N03	R06	AS67, AS89, AS09, AS34
N04	R12	AS62, AS02, AS09

Figure 21.21 Path vector messages

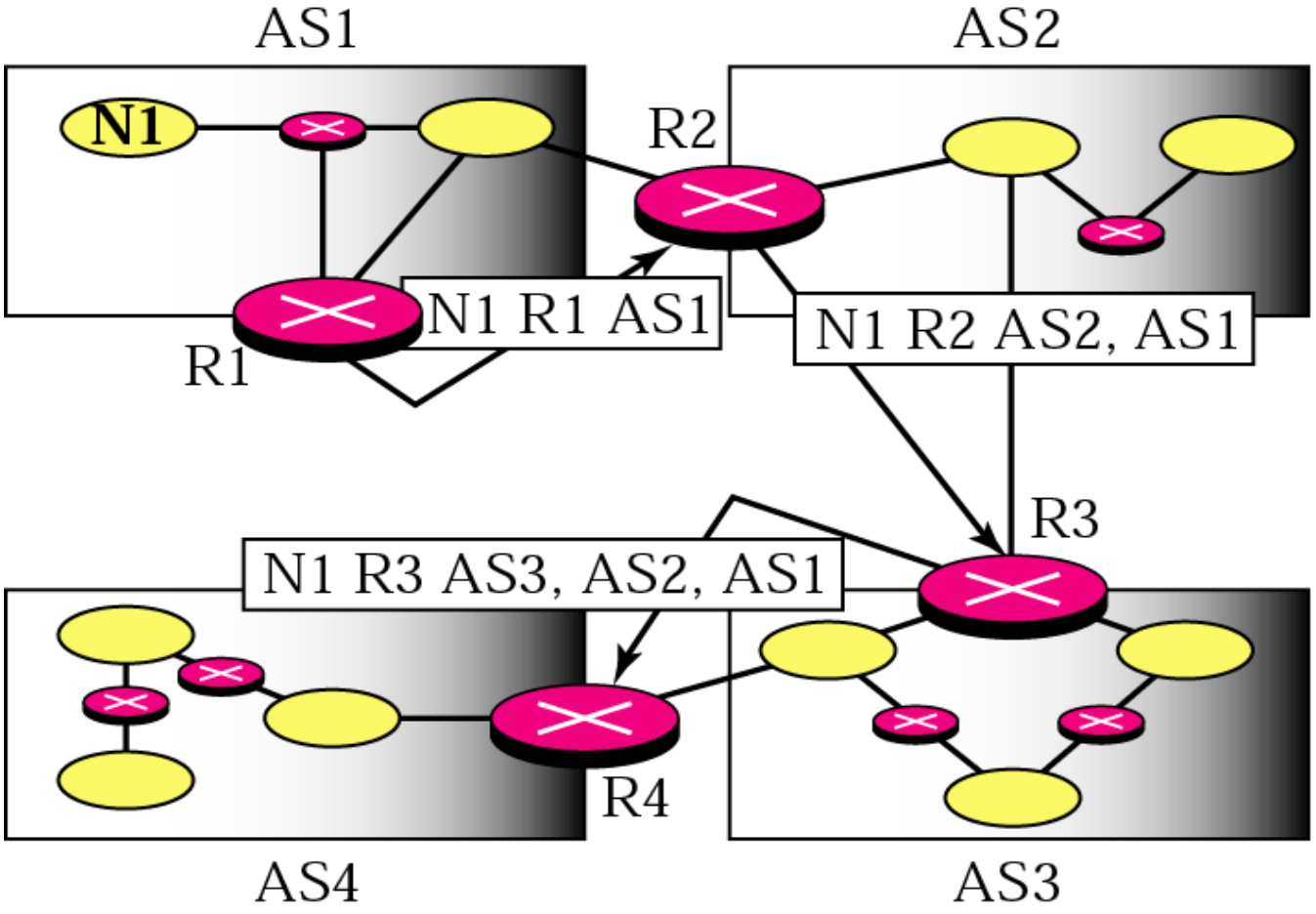
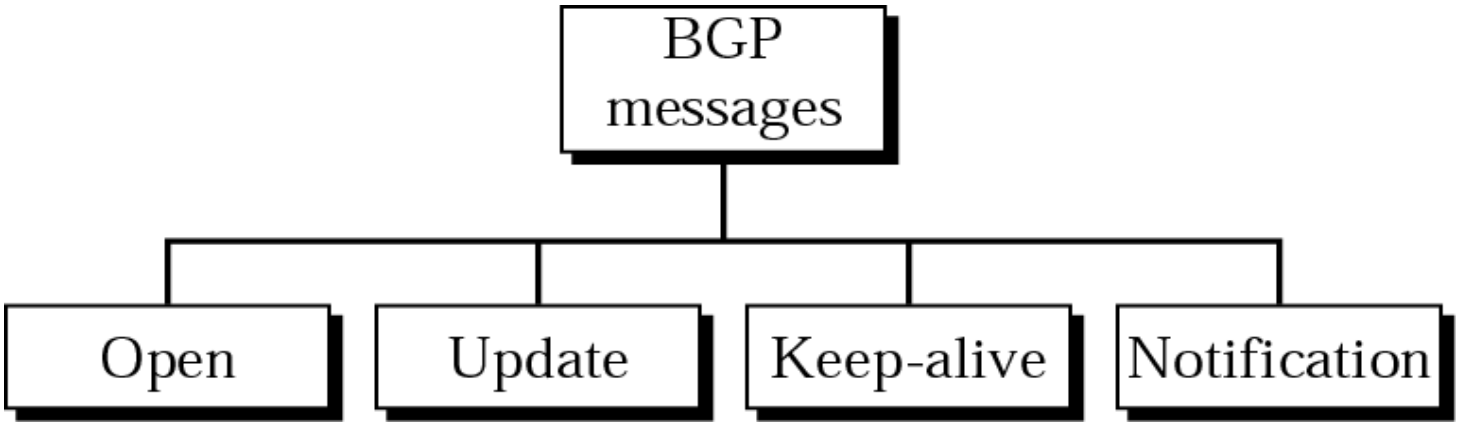


Figure 21.22 Types of BGP messages



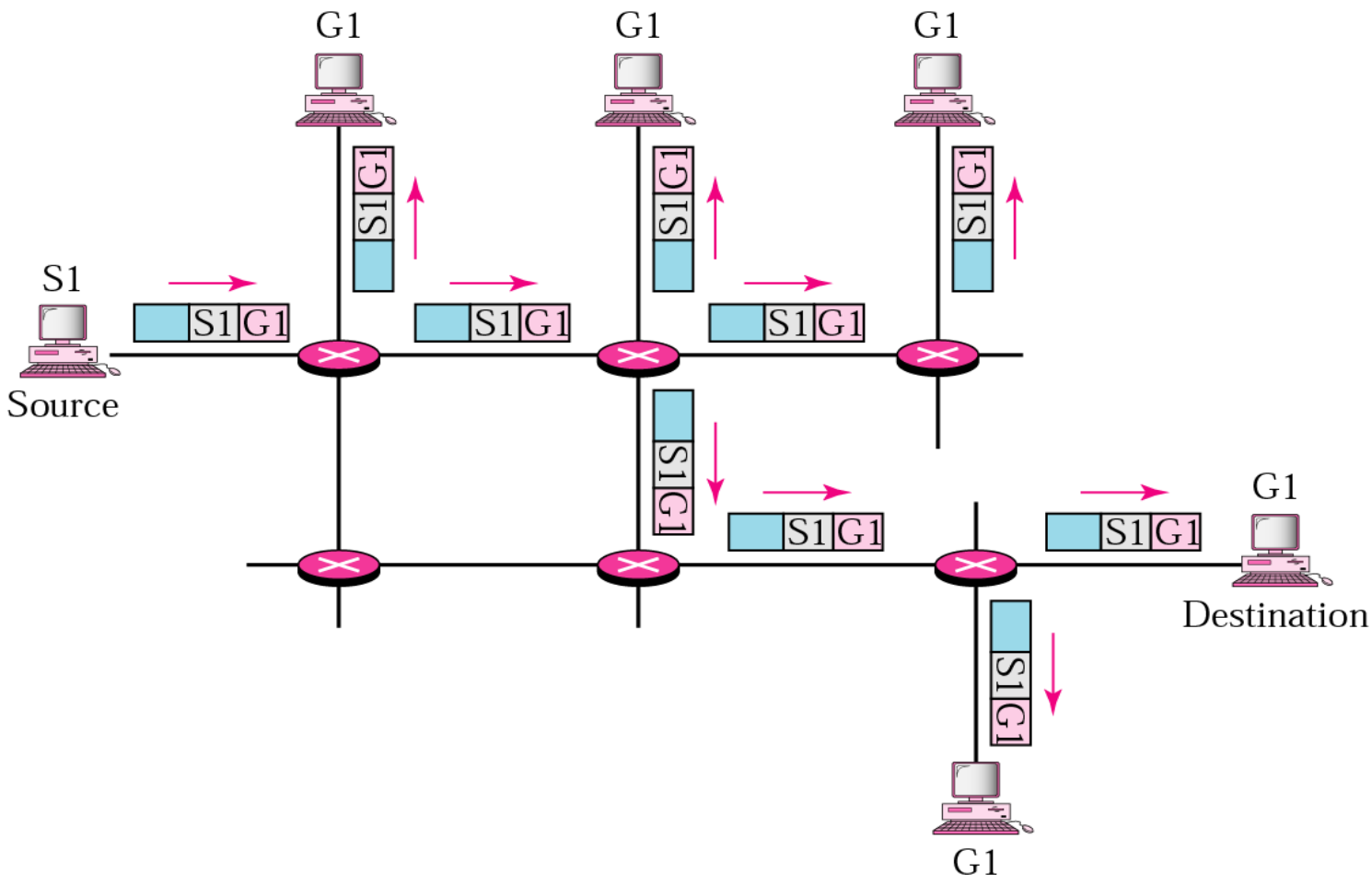
21.3 Multicast Routing

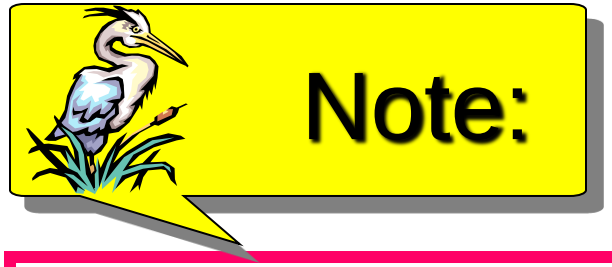
IGMP

Multicast Trees

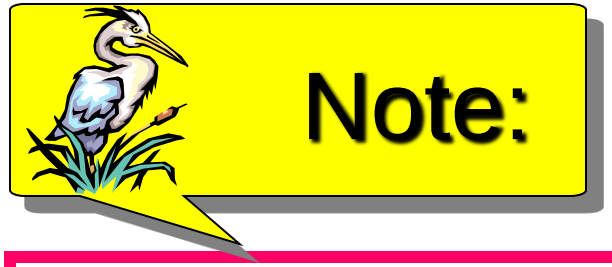
MBONE

Figure 21.23 Multicasting





In multicast routing, the router may forward the received packet through several of its ports.



IGMP is a group management protocol. It helps a multicast router create and update a list of loyal members related to each router interface.

Figure 21.24 **IGMP message types**

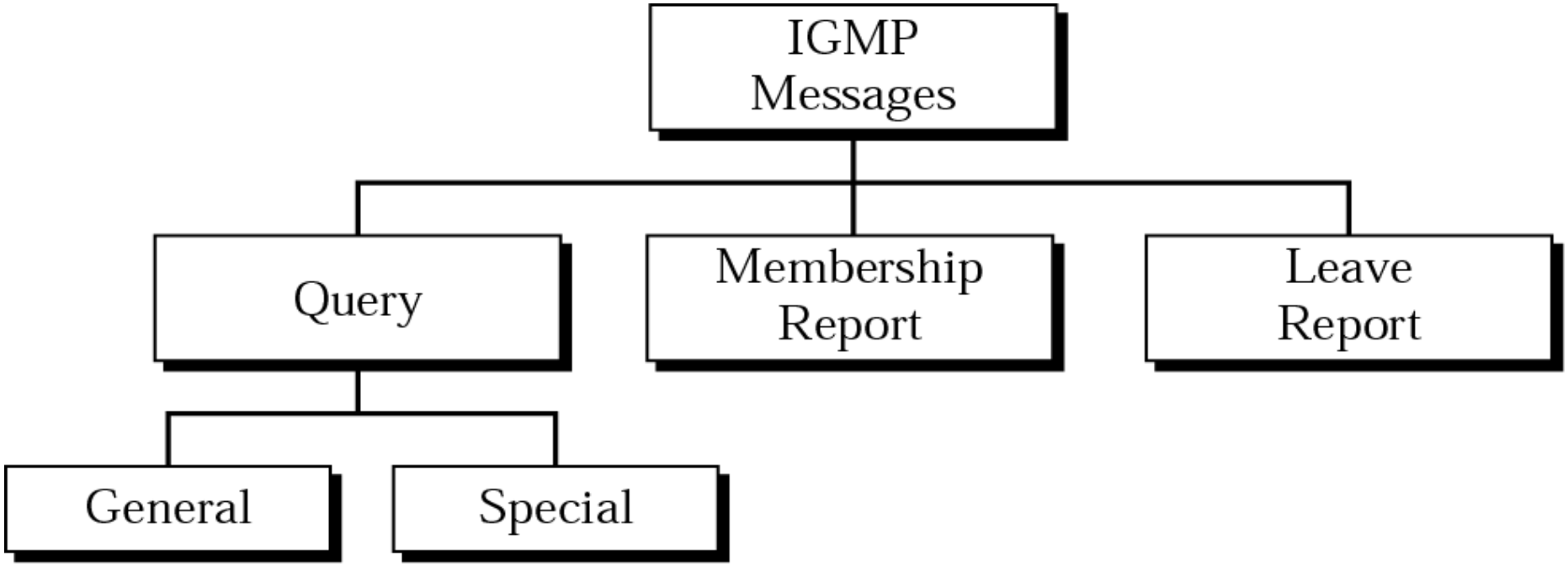


Figure 21.25 IGMP message format

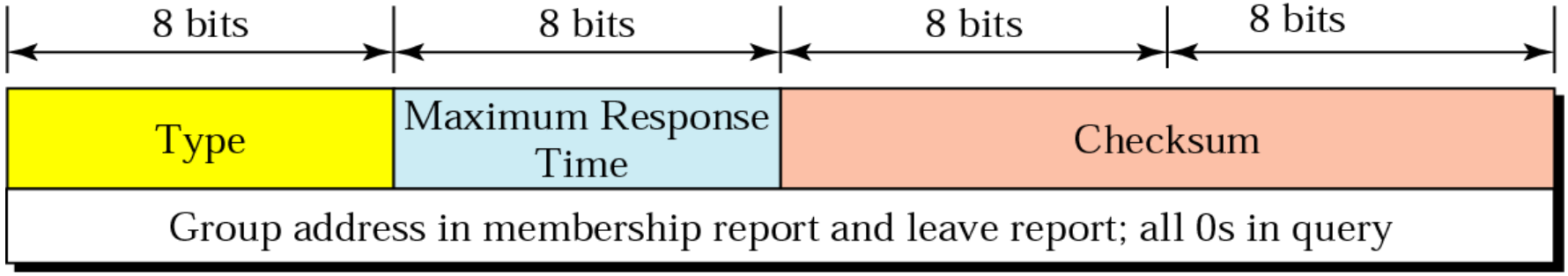


Table 21.4 IGMP type field

Type	Value
General or special query	0x11 or 00010001
Membership report	0x16 or 00010110
Leave report	0x17 or 00010111

Figure 21.26 IGMP operation

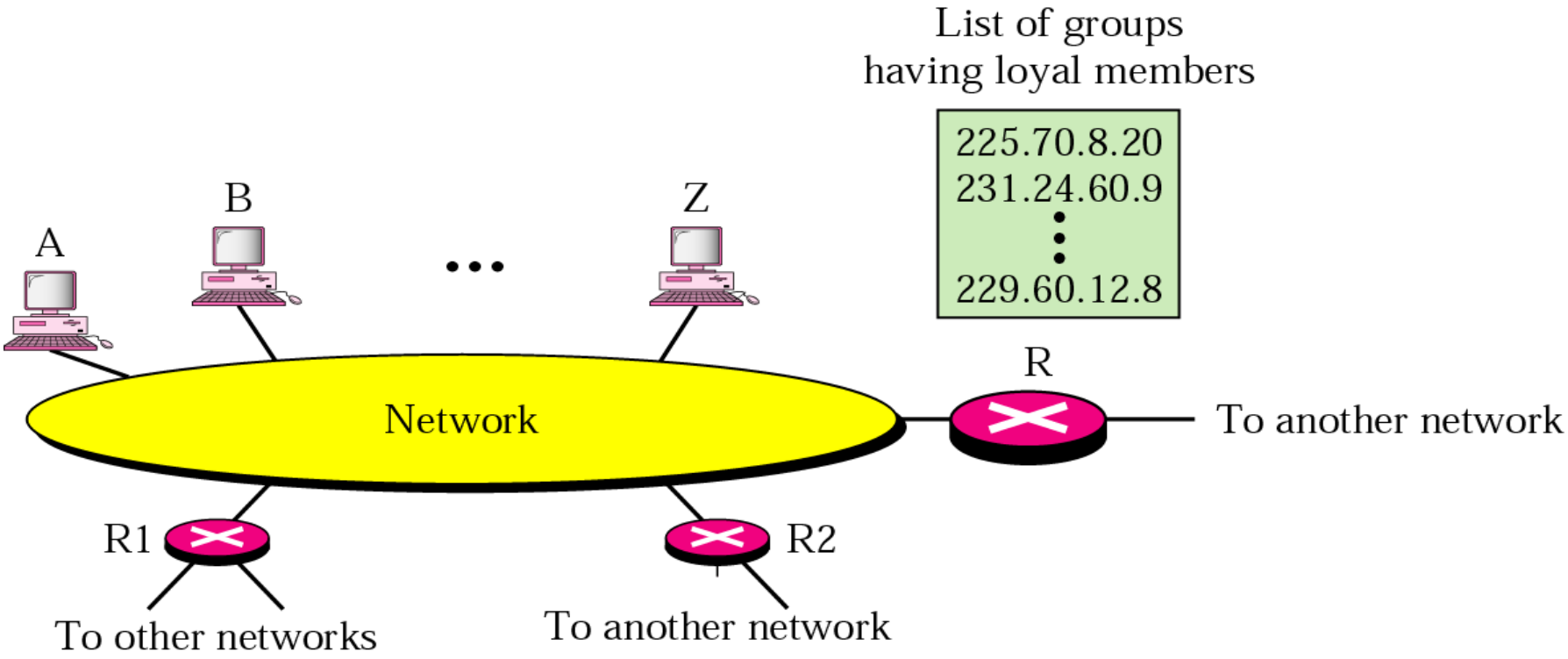
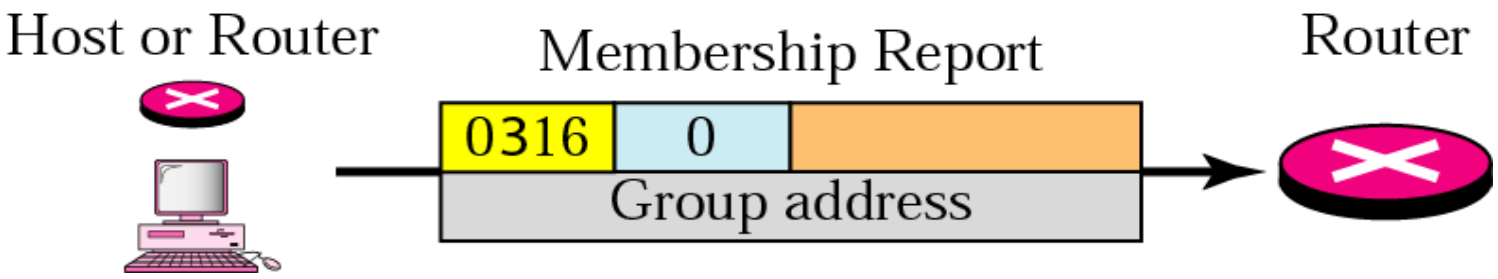
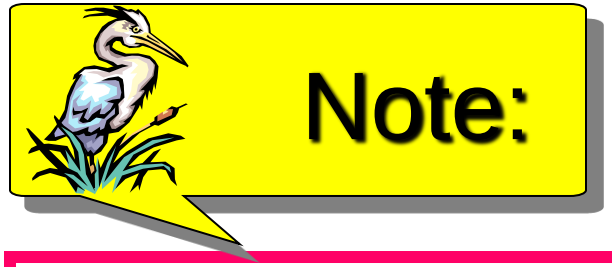


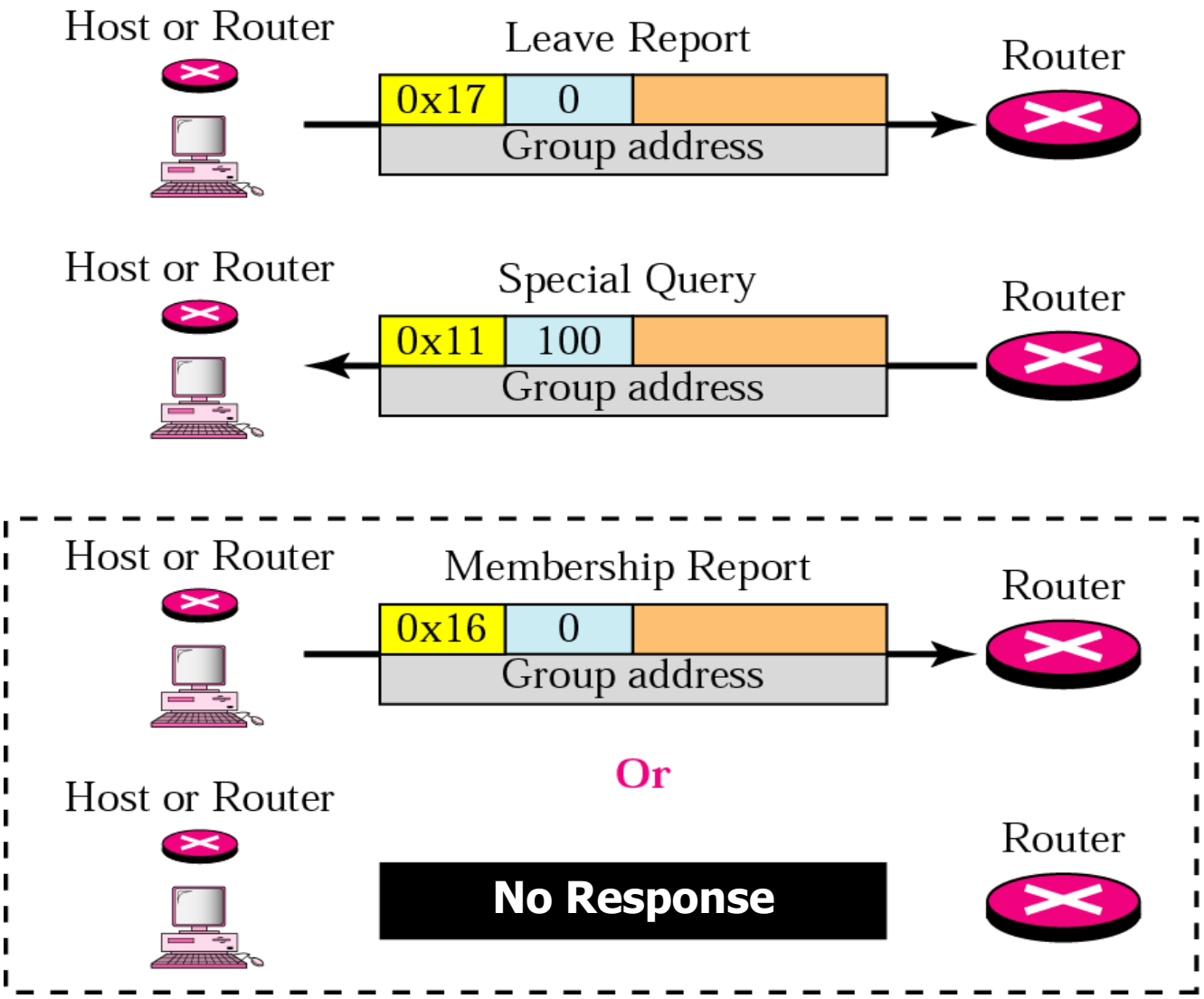
Figure 21.27 Membership report

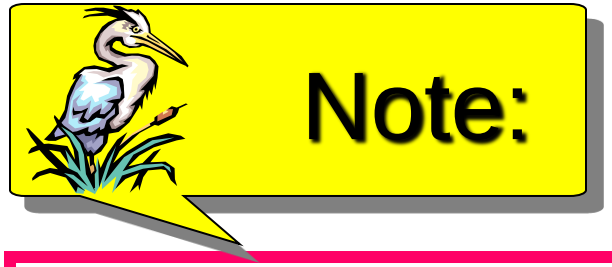




In IGMP, a membership report is sent twice, one after the other.

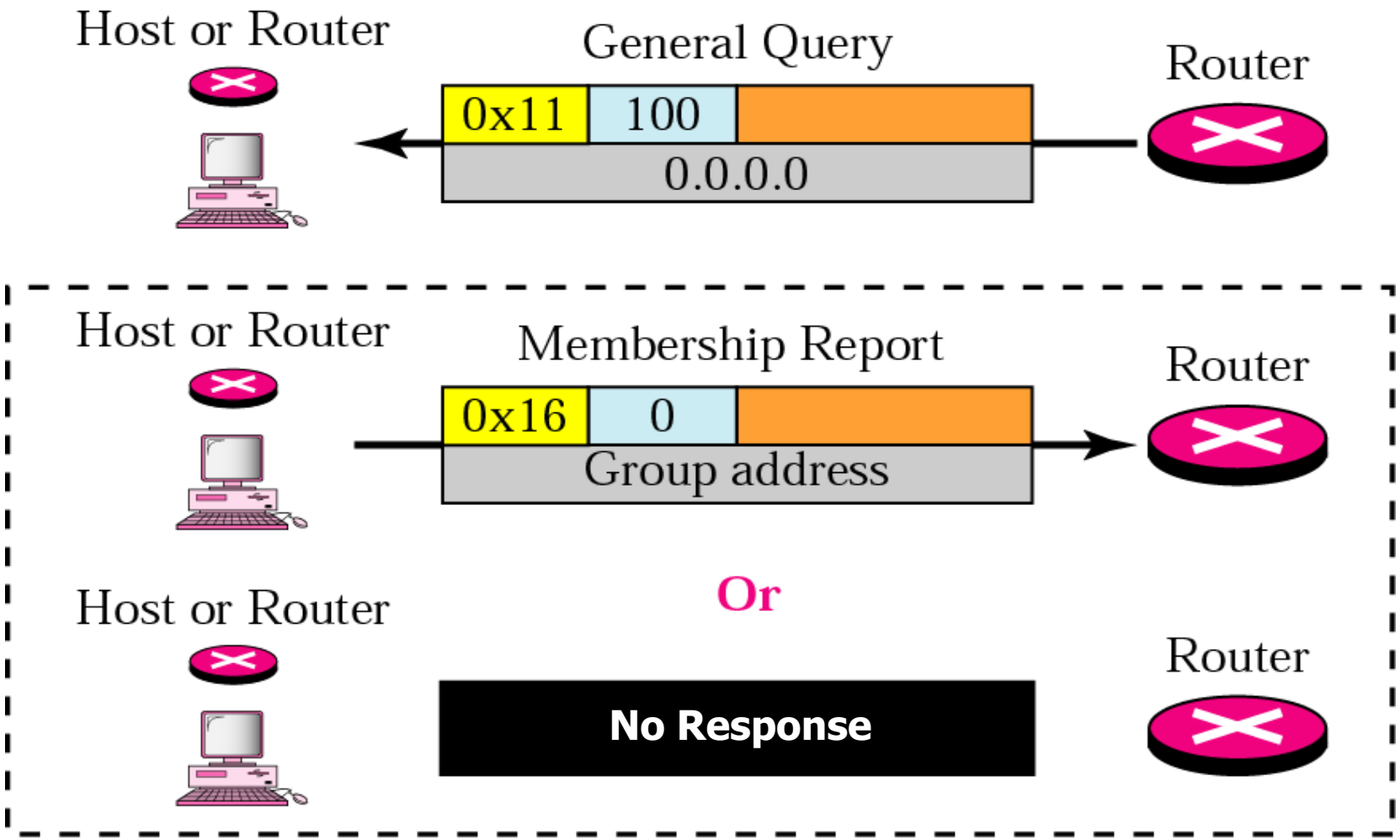
Figure 21.28 Leave report





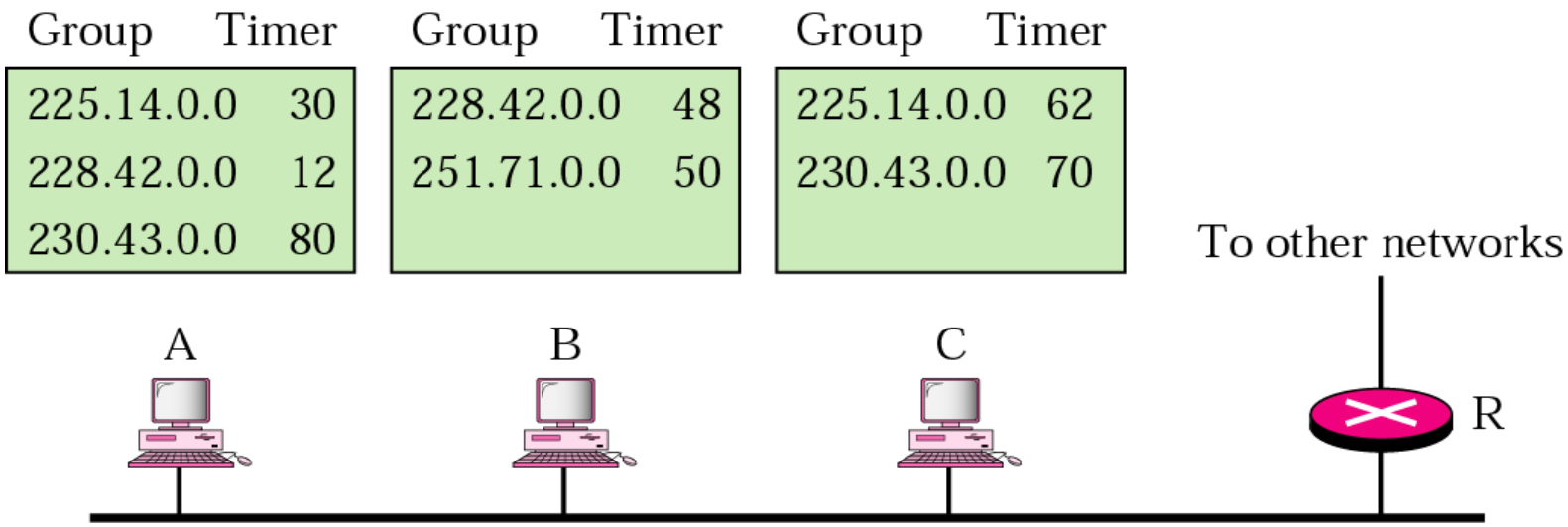
The general query message does not define a particular group.

Figure 21.29 General query message



Example 1

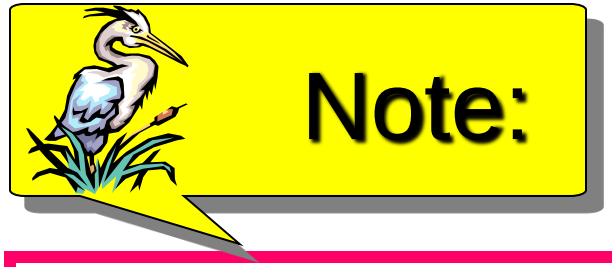
Imagine there are three hosts in a network, as shown in Figure 21.30 (below). A query message was received at time 0; the random delay time (in tenths of seconds) for each group is shown next to the group address. Show the sequence of report messages.



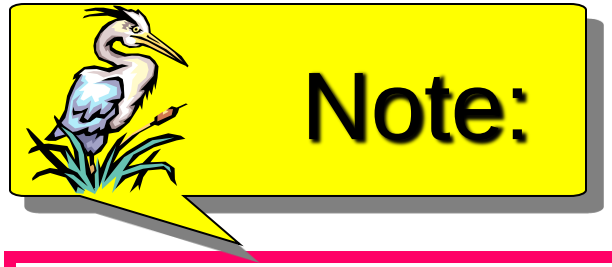
Solution

The events occur in this sequence:

1. Time 12. The timer for 228.42.0.0 in host A expires and a membership report is sent, which is received by the router and every host including host B which cancels its timer for 228.42.0.0.
2. Time 30. The timer for 225.14.0.0 in host A expires and a membership report is sent, which is received by the router and every host including host C which cancels its timer for 225.14.0.0.
3. Time 50. The timer for 251.71.0.0 in host B expires and a membership report is sent, which is received by the router and every host.
4. Time 70. The timer for 230.43.0.0 in host C expires and a membership report is sent, which is received by the router and every host including host A which cancels its timer for 230.43.0.0.



In a source-based tree approach, the combination of source and group determines the tree.



In the group-shared tree approach, the group determines the tree.

Figure 21.31 Logical tunneling

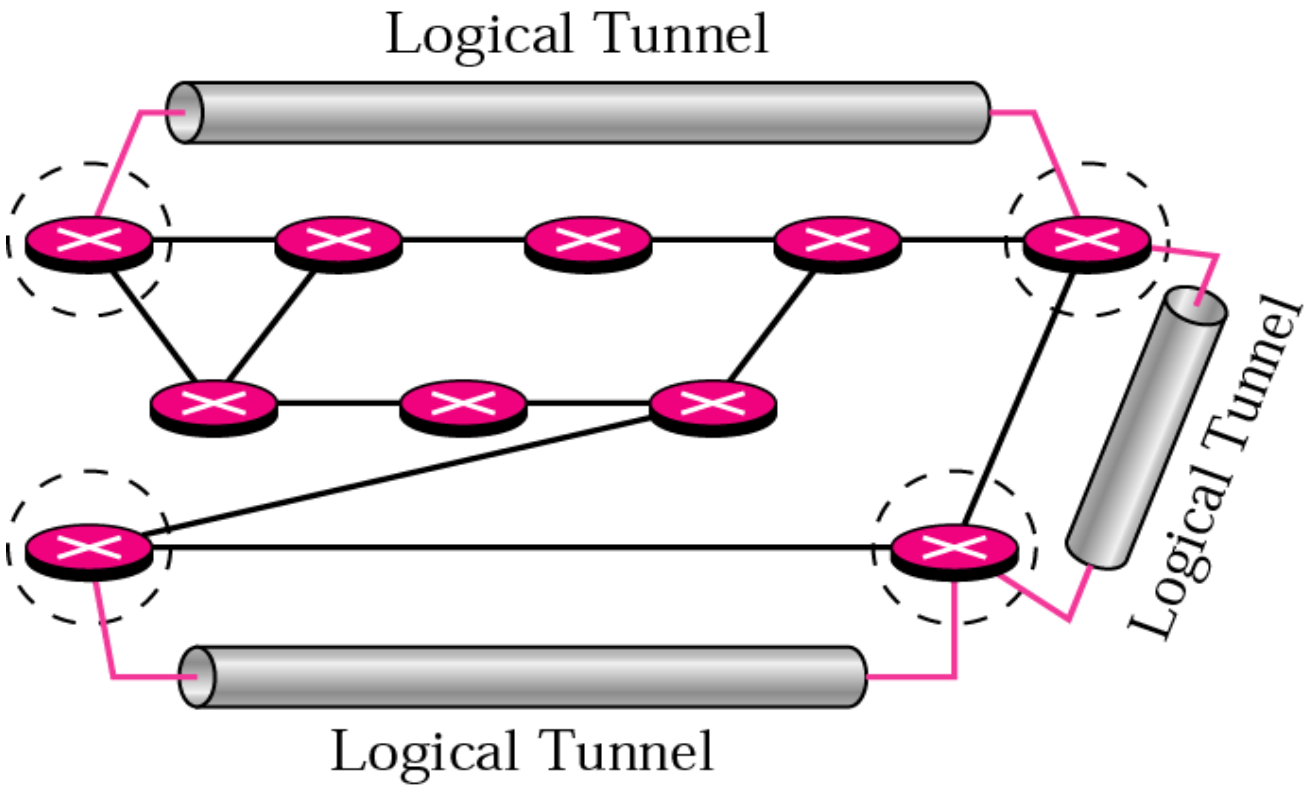
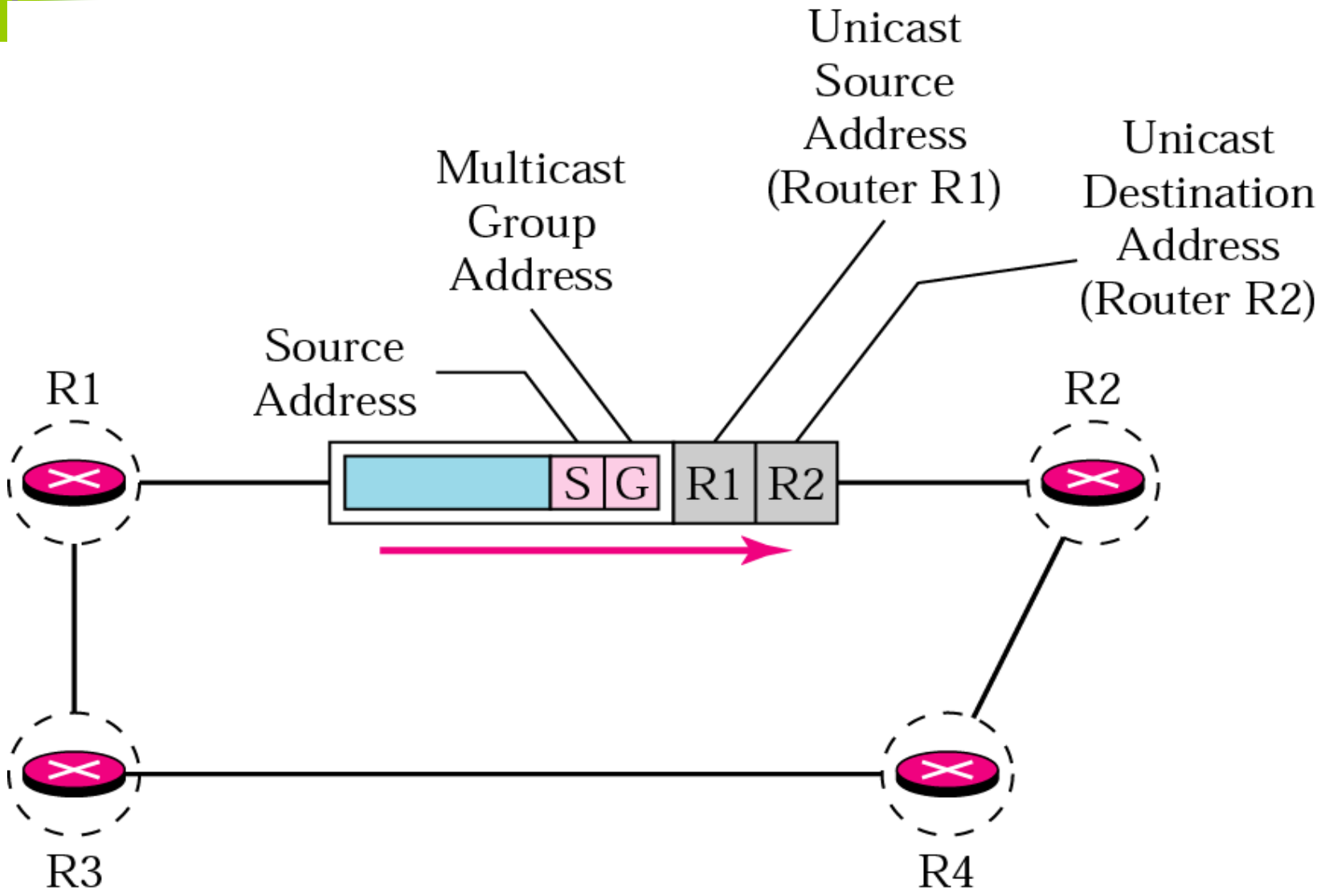


Figure 21.32 MBONE



21.4 Multicast Routing Protocols

DVMRP

MOSPF

CBT

PIM

Figure 21.33 Multicast routing protocols

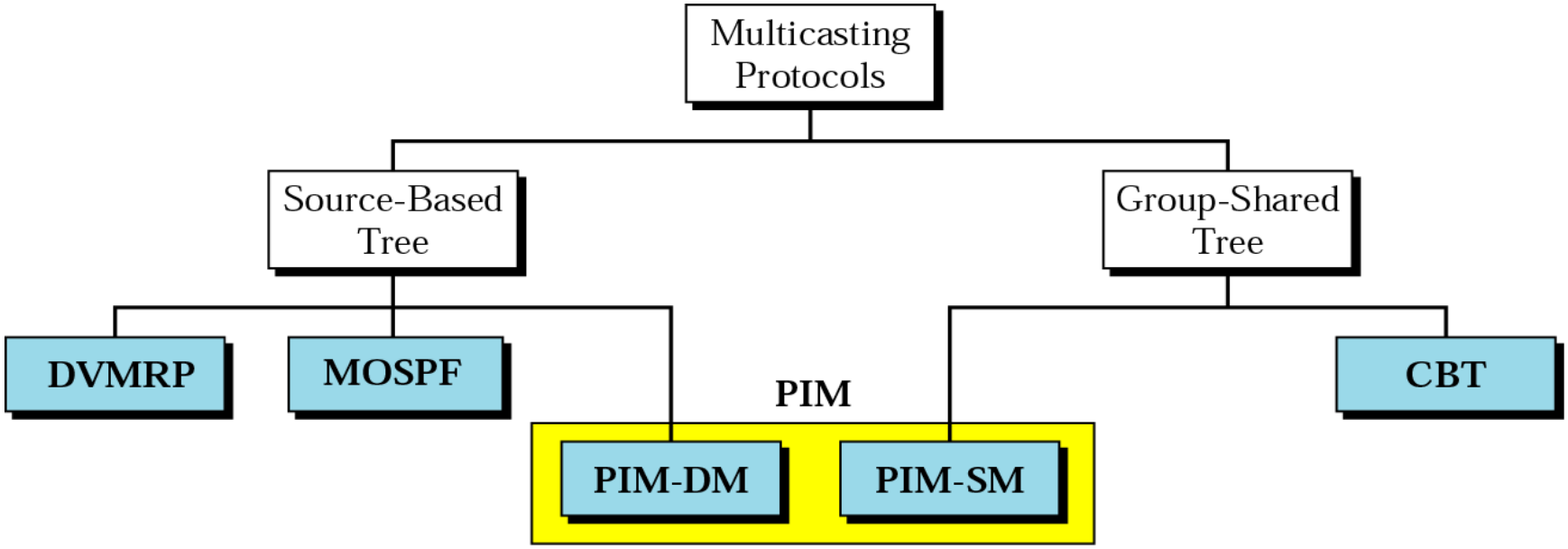
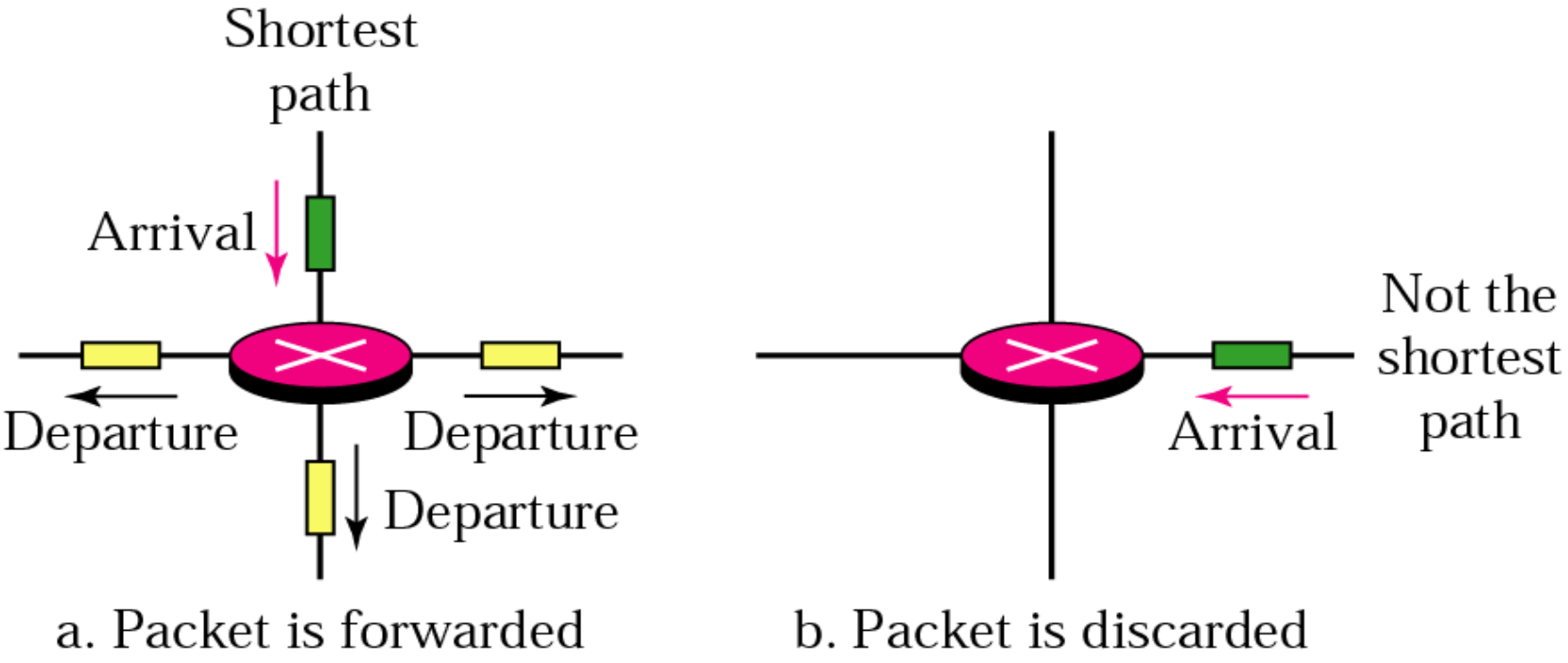
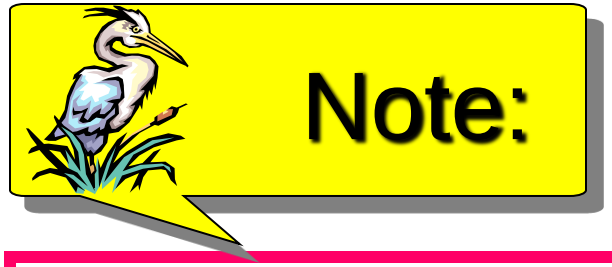


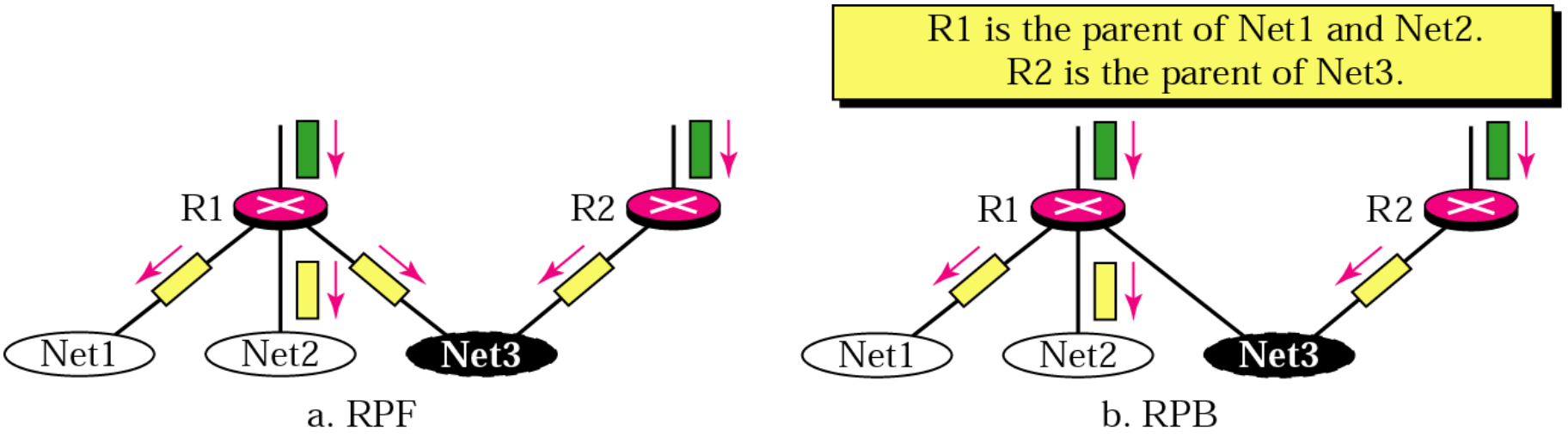
Figure 21.34 Reverse path forwarding

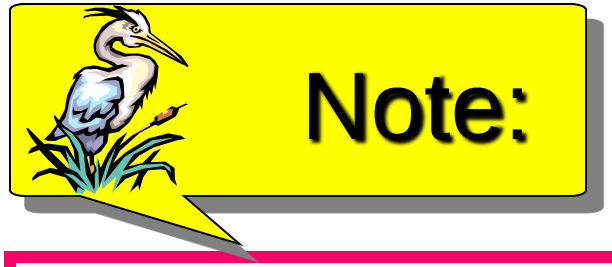




In reverse path forwarding, the router forwards only the packets that have traveled the shortest path from the source to the router; all other copies are discarded. RPF prevents the formation of loops.

Figure 21.35 RPF versus RPB

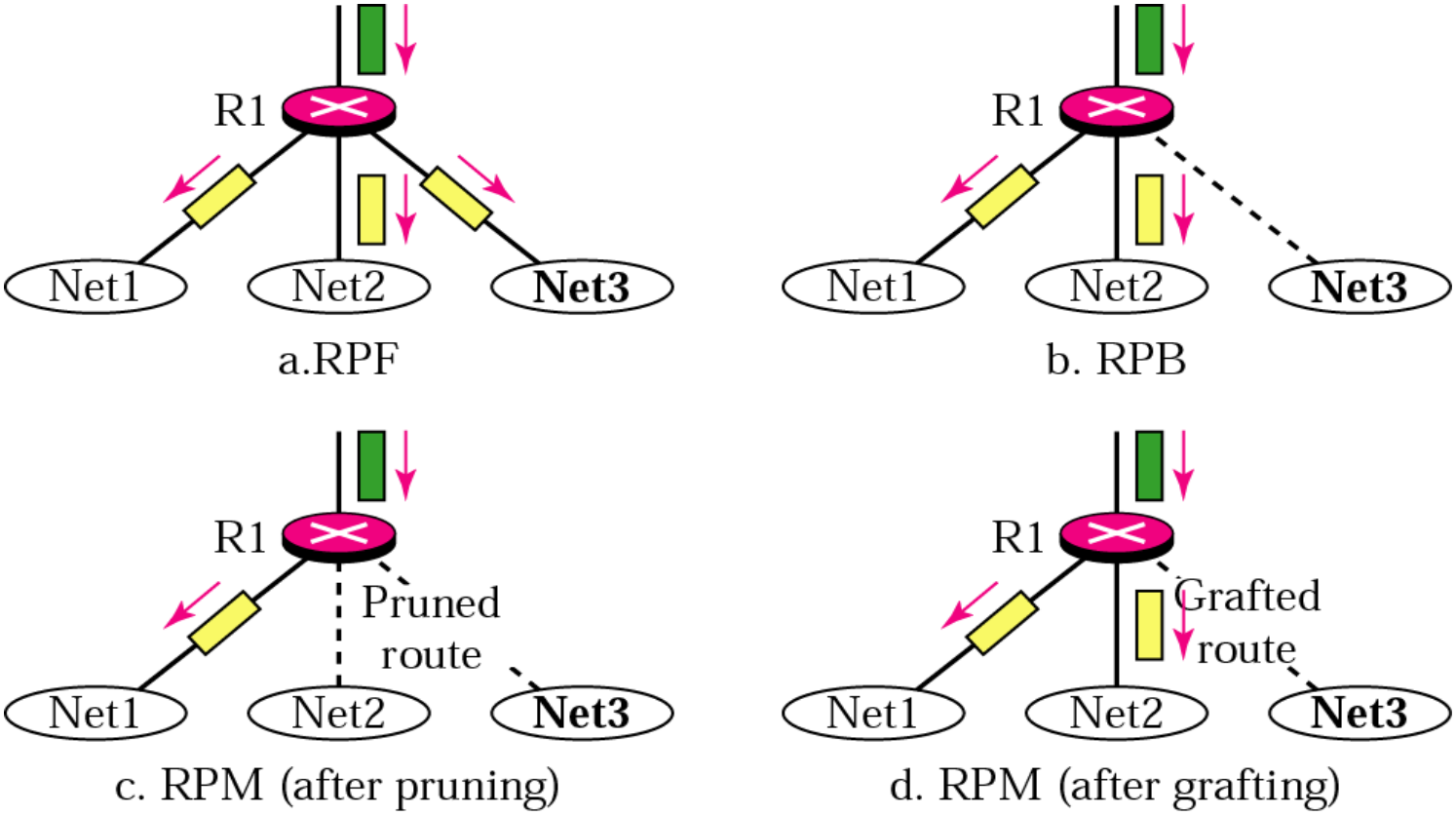


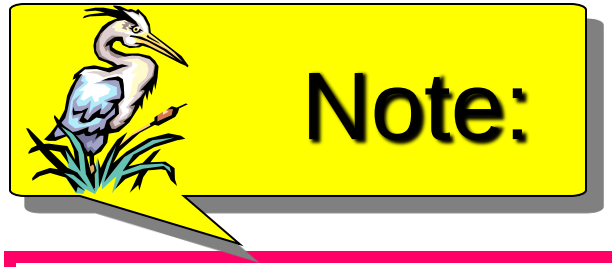


RPB creates a shortest-path broadcast tree from the source to each destination.

It guarantees that each destination receives one and only one copy of the packet.

Figure 21.36 RPF, RPB, and RPM





RPM adds pruning and grafting to RPB to create a multicast shortest-path tree that supports dynamic membership changes.

Figure 21.35 RPF versus RPB

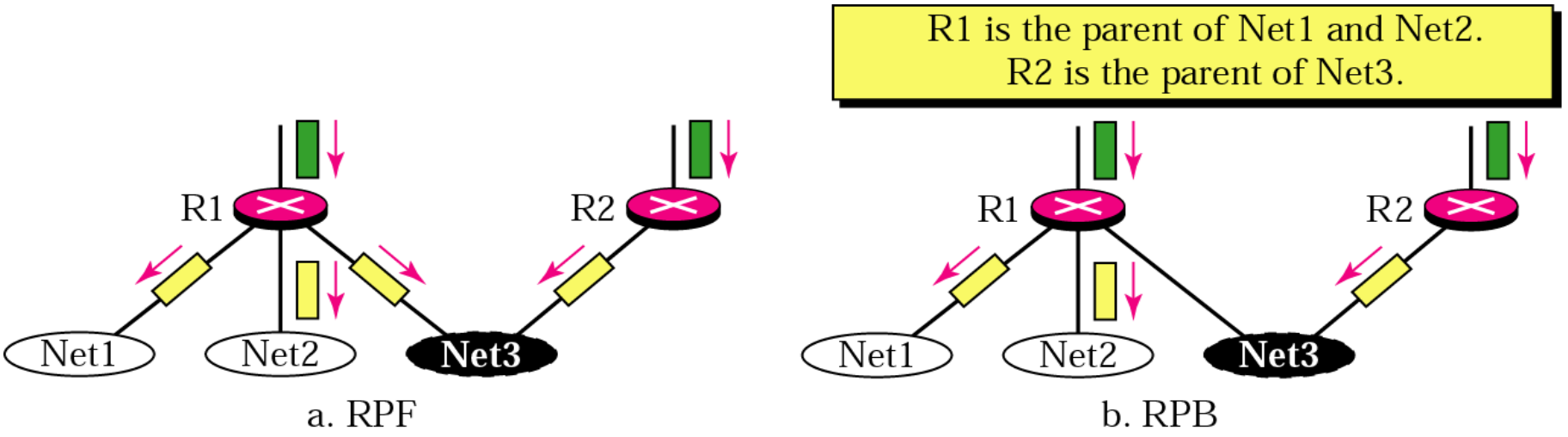
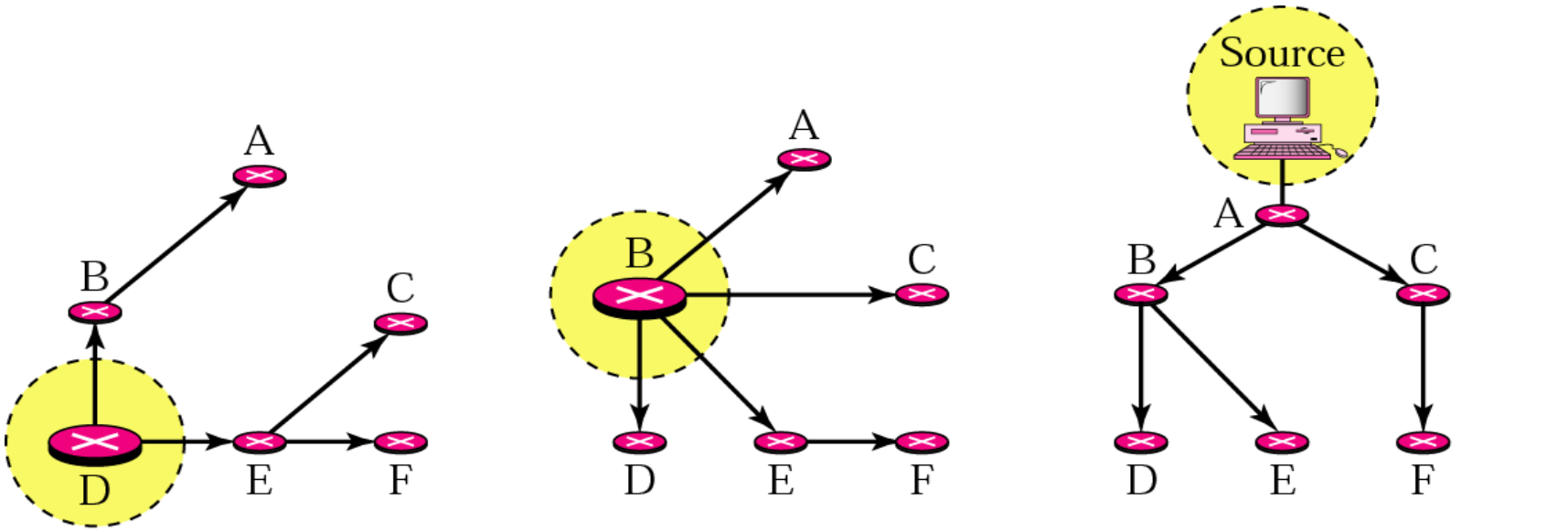


Figure 21.37 Unicast tree and multicast tree



a. Unicast tree for D

b. Unicast tree for B

c. Multicast tree for all routers

Figure 21.38 **Shared-group tree with rendezvous router**

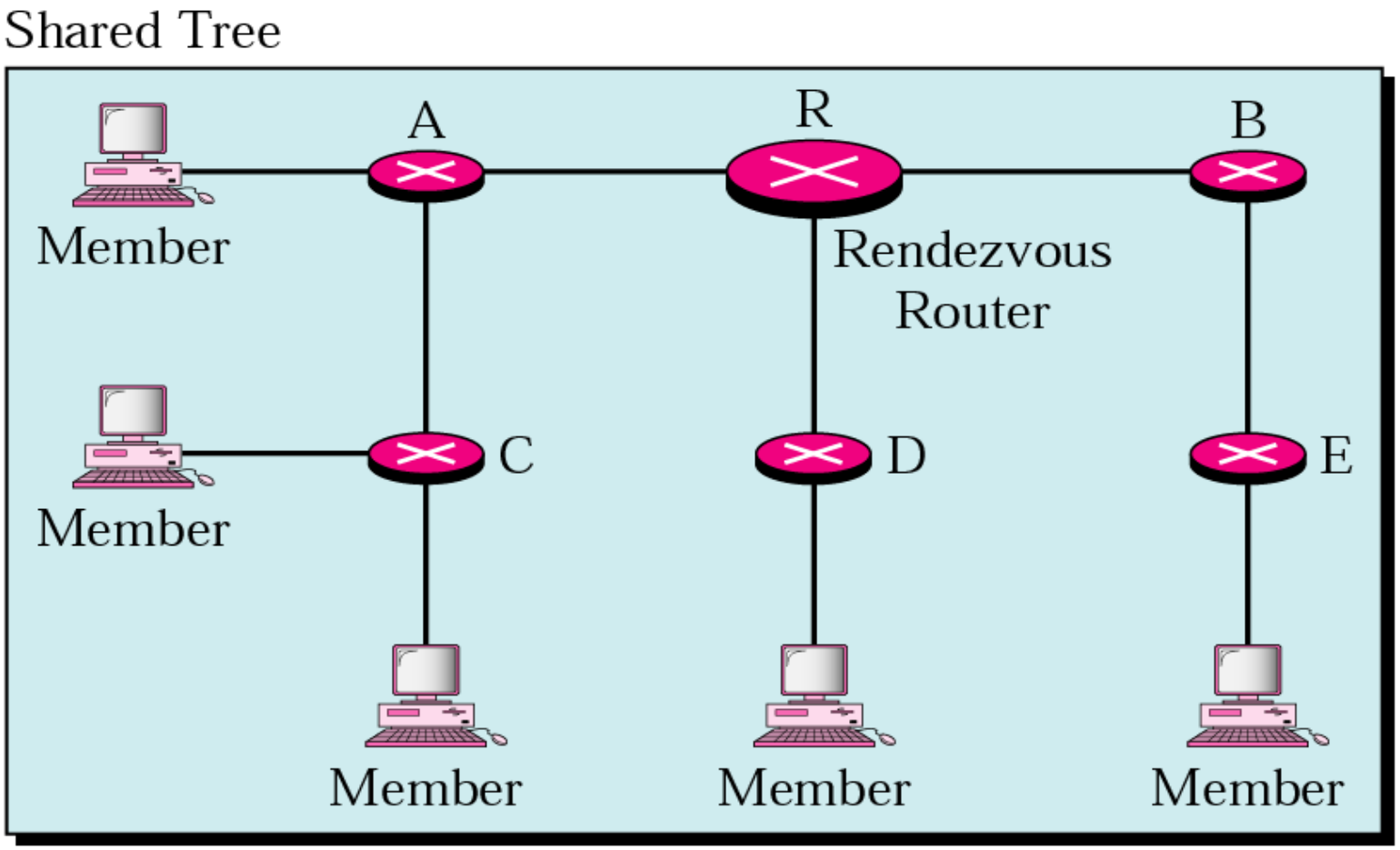
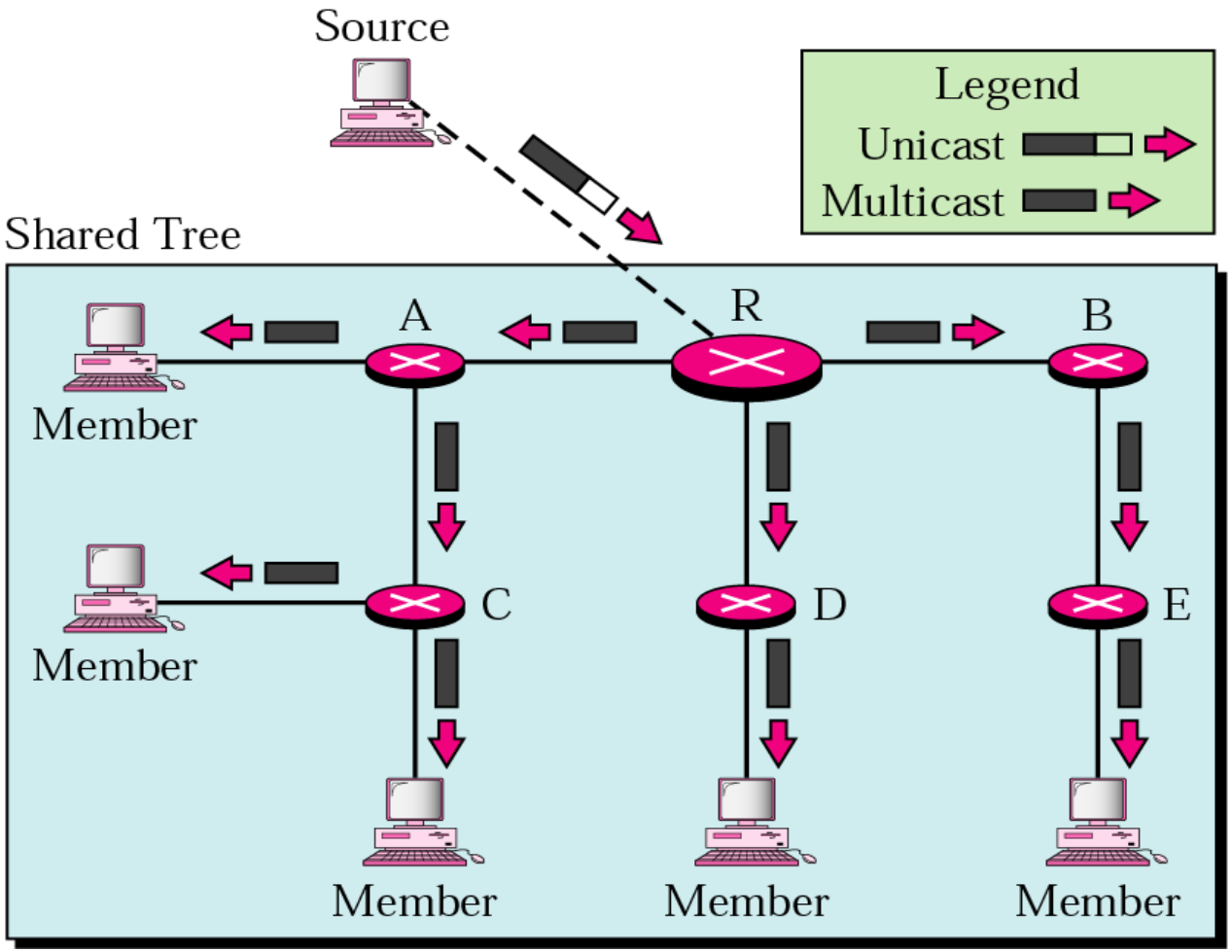
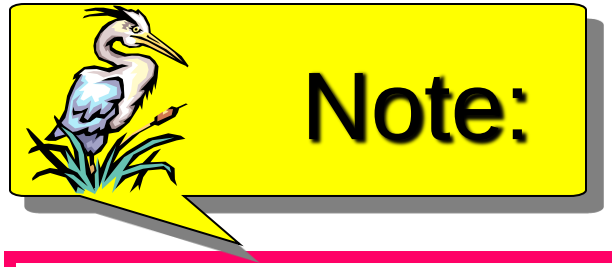
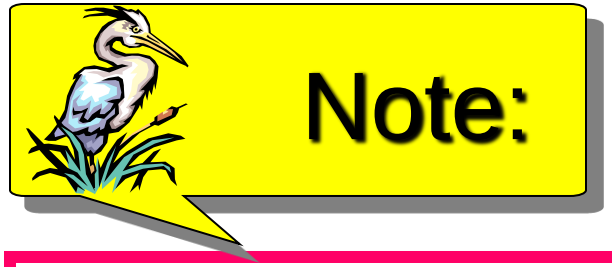


Figure 21.39 Sending a multicast packet to the rendezvous router

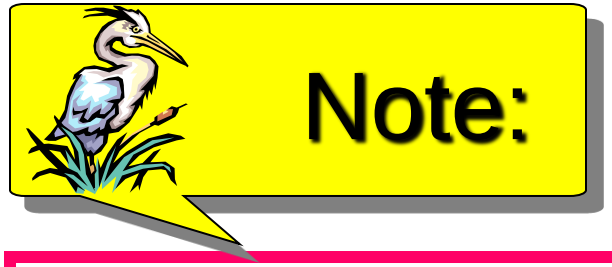




In CBT, the source sends the multicast packet to the core router. The core router decapsulates the packet and forwards it to all interested hosts.



PIM-DM uses RPF and pruning and grafting strategies to handle multicasting. However, it is independent of the underlying unicast protocol.



PIM-SM is similar to CBT but uses a simpler procedure.